MATHEMATICS

APPLICATION OF DERIVATIVE THEORY SHEET ENGLISH

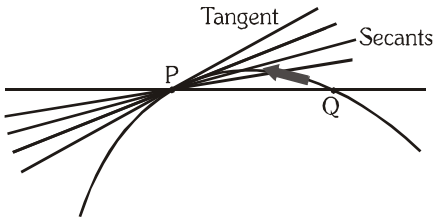
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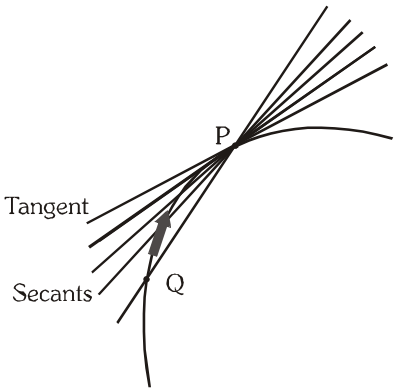
TANGENT & NORMAL

1.1 INTRODUCTION ABOUT TANGENT & NORMAL

Tangent to the curve at a point

The tangent to the curve at 'P' is the line through P whose slope is limit of the secant slopes as $Q \to P$ from either side.





Normal to the curve at a point

A line which is perpendicular to the tangent at the point of contact is called normal to the curve at that point.

(a) The value of the derivative at $P(x_1, y_1)$ gives the slope of the tangent to the curve at P.

Symbolically $f'(x_1) = \left.\frac{dy}{dx}\right]_{(x_1, y_1)}$ = Slope of tangent at $P(x_1, y_1)$ = m(say).

(b) Equation of tangent at (x_1, y_1) is ; $y - y_1 = \left.\frac{dy}{dx}\right]_{(x_1, y_1)} (x - x_1)$

(c) Equation of normal at (x_1, y_1) is; $y - y_1 = -\frac{1}{\left.\frac{dy}{dx}\right]_{(x_1, y_1)}} (x - x_1)$.

Note :

The point P (x_1, y_1) will satisfy the equation of the curve & the equation of tangent & normal line.

Questions

1 Find equation of tangent at (1,1) on the curve $\sqrt{x} + \sqrt{y} = a$?

Ans $T : x + y = 2, N : x = y$

Sol. $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\sqrt{\frac{y}{x}}$

$\left(\frac{dy}{dx}\right)_{(1,1)} = -1 = m_T$

EQT : $y - 1 = (-1)(x - 1)$

$x + y = 2$

EON : $y - 1 = 1(x - 1)$

$y = x$

2 Find equation of tangent and normal at (1,3) on the curve $y = x^2 + x + 1$?

Ans T : $y = 3x$, N : $x + 3y = 10$

Sol. $\frac{dy}{dx} = 2x + 1$

$\left(\frac{dy}{dx}\right)_{(1,3)} = 3 = m_T$

EOT : $y - 3 = 3(x - 1)$

EON : $y - 3 = -\frac{1}{3}(x - 1)$

3 Find equation of tangent at $x = 1$ for the curve $y = \int_{x^2}^{x^3} \frac{dt}{\sqrt{1 + t^2}}$

Ans. T : $\sqrt{2}y + 1 = x$, N : $y + \sqrt{2}x = \sqrt{2}$

Sol. At $x = 1$, $y = 0$

$\frac{dy}{dx} = \frac{3x^2}{\sqrt{1 + x^6}} - \frac{2x}{\sqrt{1 + x^4}}$

$\left(\frac{dy}{dx}\right)_{x=1} = \frac{3}{\sqrt{2}} - \frac{2}{\sqrt{2}} = \frac{1}{\sqrt{2}} = m_T$

EOT : $y - 0 = \frac{1}{\sqrt{2}}(x - 1)$

EON : $y - 0 = -\sqrt{2}(x - 1)$

4 Find the equation of the tangent to the curve $y = be^{-x/a}$ at the point where the curve crosses the y-axis.

Ans. $\frac{x}{a} + \frac{y}{b} = 1$

Sol. At y-axis : $x = 0$, $y = b$

$\frac{dy}{dx} = be^{-\frac{x}{a}}\left(\frac{-1}{a}\right)$

$\left(\frac{dy}{dx}\right)_{x=0} = \frac{-b}{a}$

EOt : $y - b = \frac{-b}{a}(x - 0)$

5. Find the tangent and normal for $x^{2/3} + y^{2/3} = 2$ at (1, 1).

Ans. T : $x + y = 2$ and N : $x - y = 0$

Sol. $\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx} = 0$

$\left(\frac{dy}{dx}\right)_{(1,1)} = -1 = m_T$

EOT : $y - 1 = (-1)(x - 1)$
EON : $y - 1 = (1)(x - 1)$

6. Find the equation of tangent and normal to the curve $f(x) = \begin{cases} x-2 & \text{if } x < 1 \\ x^2 - x - 1 & \text{if } x \geq 1 \end{cases}$ at $x = 1$ if it exists.

Ans. T : $x - y - 2 = 0$; $x + y = 0$
(f is continuous and differentiable at $x = 1$ and $f'(1) = 1$)

Sol. $f(1^+) = f(1^-) = f(1) = -1$
$f(x)$ is continuous at $x = 1$

$$f'(x) = \begin{cases} 1; & x < 1 \\ 2x - 1; & x > 1 \end{cases}$$

$f'(1^+) = f'(1^-) = 1$
$f(x)$ is differentiable at $x = 1$
At $x = 1$, $y = -1$
$f'(1) = 1 = m_T$
EOT : $y + 1 = 1(x - 1)$; EON : $y + 1 = (-1)(x - 1)$

7. A curve in the plane is defined by the parametric equations $x = e^{2t} + 2e^{-t}$ and $y = e^{2t} + e^t$. Equation for the line tangent to the curve at the point $t = \ln 2$ is.

Ans.

Sol. when $t = \ln 2$ $x = e^{\ln 4} + 2e^{-\ln 2} = 4 + 1 = 5$
and $y = e^{\ln 4} + e^{\ln 2} = 6$
$\therefore$ point $(5, 6)$

now $y = e^{2t} + e^t \Rightarrow \frac{dy}{dt} = 2e^{2t} + e^t$

$x = e^{2t} + 2e^{-t} \Rightarrow \frac{dx}{dt} = 2(e^{2t} - e^{-t})$

$$\frac{dy}{dx} = \frac{2e^{2t} + e^t}{2(e^{2t} - e^{-t})} = \frac{2(4) + 2}{2(4 - (1/2))} = \frac{10}{7}$$

$\therefore$ Tangent $y - 6 = \frac{10}{7}(x - 5)$; $7y - 42 = 10x - 50$; $10x - 7y = 8$ **Ans.**

8. Tangent to the curve $y = \sin^{-1}\frac{2x}{1 + x^2}$ at $x = \sqrt{3}$

Ans $x + 2y = \sqrt{3} + \frac{2\pi}{3}$

Sol. $\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\begin{cases} 2\tan^{-1}x & ; \ |x|\le 1 \\ \pi-2\tan^{-1}x & ; \ x>1 \\ -\pi-2\tan^{-1}x & ; \ x<-1 \end{cases}$

At $x=\sqrt{3}$; $y=\pi-2\tan^{-1}\sqrt{3}=\pi-\frac{2\pi}{3}=\frac{\pi}{3}$

$\frac{dy}{dx}=\frac{-2}{1+x^2}$

$\left(\frac{dy}{dx}\right)_{x=\sqrt{3}}=\frac{-2}{4}=\frac{-1}{2}=m_T$

EOT : $y-\frac{\pi}{3}=-\frac{1}{2}\left(x-\sqrt{3}\right)$

9. Tangent to the curve $y=\sin^{-1}\frac{2x}{1+x^2}$ at $x=\sqrt{3}$ is inclined at $\cot^{-1}(k)$ where $k=$ _____

Ans. -2

Sol. $m_T=\frac{-1}{2}$ (as solved in Question 8)

$\tan\theta=\frac{-1}{2}$

$\cot\theta=-2\Rightarrow\theta=\cot^{-1}(-2)$

10 For Curve $y^3=2x-e^{x/2}$. Find equation of tangent at $x=0$ and $y=-1$?

Ans. $2y-x+2=0$

Sol. $3y^2\frac{dy}{dx}=2-e^{\frac{x}{2}}\cdot\frac{1}{2}$

$\left(\frac{dy}{dx}\right)_{(0,-1)}=\frac{1}{3}\left(\frac{3}{2}\right)=\frac{1}{2}$

EOT : $y+1=\frac{1}{2}(x-0)$

11 Find equation of tangent to the curve $y=x^2+|x|$ at $x=-2$?

Ans. $5x+y+4=0$

Sol. At $x=-2$; $y=4+2=6$

$y=x^2-x$

$\left(\frac{dy}{dx}\right)_{x=-2}=-4-1=-5$

EOT : $y-6=-5(x+2)$

12 Find equation of tangent for curve $y\begin{cases}\sqrt{2-\cos x} & x \le \frac{\pi}{2}\\ 1+\sin x & x > \frac{\pi}{2}\end{cases}$ at $x = \frac{\pi}{2}$

Ans. tangent does not exist.

Sol. $\frac{dy}{dx} = \begin{cases}\sin x; & x < \frac{\pi}{2}\\ \cos x; & x > \frac{\pi}{2}\end{cases}$

LHD = 1, RHD = 0 at $x = \frac{\pi}{2}$

Unique tangent does not exist

13 Find equation of tangent and normal to the curve $x = 2t$, $y = t^2$ at $t = 2$.

Ans. T : $2x - y = 4$, N : $x + 2y = 12$

Sol. $\frac{dy}{dx} = \frac{2t}{2} = t$

$\left(\frac{dy}{dx}\right)_{t=2} = 2 = m_T$

At $t = 2 \Rightarrow x = 4, y = 4$

EOT : $y - 4 = 2(x - 4)$

EON : $y - 4 = \frac{-1}{2}(x - 4)$

14 Find equation of tangent and normal to the curve $x = 2(\theta + \sin\theta)$ $y = 2(1 + \cos\theta)$ at $\theta = \pi/2$ **[O]**

Ans. T : $x + y = 4 + \pi$, N : $y = x - \pi$

Sol. At $\theta = \frac{\pi}{2}; x = 2\left(\frac{\pi}{2} + 1\right); y = 2$

$\frac{dy}{dx} = \frac{-2\sin\theta}{2(1+\cos\theta)}$

$\left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{2}} = -1 = m_T$

EOT : $y - 2 = (-1)(x - (\pi + 2))$

EON : $y - 2 = (1)(x - (\pi + 2))$

15 Find tangent to $x = a\sin^3 t$ and $y = a\cos^3 t$ at $t = \pi/2$. **[O]**

Ans. $y = 0$

Sol. At $t = \frac{\pi}{2}; x = a, y = 0$

$$\frac{dy}{dx}=\frac{3a\cos^2 t(-\sin t)}{3a\sin^2 t\cos t}=-\cot t$$

$$\left(\frac{dy}{dx}\right)_{t=\frac{\pi}{2}}=0=m_T$$

EOT : $y-0=0(x-a)$

$\Rightarrow y=0$

16 Find the equation of a tangent and normal at x = 0 if they exist on the curve $y=x^{1/3}(1-\cos x)$ **[O]**

Ans. T : y = 0, N : x = 0

Sol. $f'(0)=\underset{h\to 0}{\text{Lim}}\frac{h^{1/3}(1-\cos h)}{h}=\underset{h\to 0}{\text{Lim}}\,h^{4/3}\frac{(1-\cos h)}{h^2}=0$

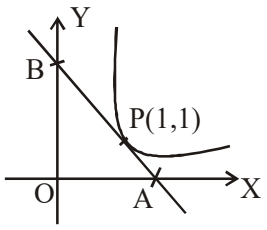
17 The triangle formed by the tangent to the curve $f(x)=x^2+bx-b$ at the point (1, 1) and the co-ordinate axes lies in the first quadrant. If its area is 2, then the value of b is : **[JEE – 2001]**

Ans. –3

Sol. $f'(1)=2+b$

EOT : $y-1=(2+b)(x-1)$

$$A\left(\frac{1+b}{2+b},0\right);B(0,-1-b)$$



Area of $\Delta AOB=\frac{1}{2}\times\left(\frac{1+b}{2+b}\right)(-1-b)=2$

$-1-b^2-2b=8+4b$

$b^2+6b+9=0$

$(b+3)^2=0$

$b=-3$

18. If the normal to the curve $y(x)=\int_0^x\left(2t^2-15t+10\right)dt$ at a point (a, b) is parallel to the line $x+3y=-5$, $a>1$, then the value of $|a+6b|$ is equal to _____. **[JEE(Main)Mar.2021]**

Ans. 406

Sol. $\frac{dy}{dx}=3$

$2x^2-15x+10=3$

$2x^2-15x+7=0$

$(x-7)(2x-1)=0$

$\Rightarrow x=7\ (\because\ x>1)\ \Rightarrow a=7$

$$b=\int_0^7\left(2t^2-15t+10\right)dt$$

$$b = \left(\frac{2t^3}{3} - \frac{15t^2}{2} + 10t\right)_0^7$$

$$= \frac{686}{3} - \frac{735}{2} + 70$$

$$= \frac{1372 - 2205 + 420}{6}$$

$$b = \frac{-413}{6}$$

$|a + 6b| = |7 - 413| = 406$

19. If the equation of the normal to the curve $y = \frac{x-a}{(x+b)(x-2)}$ at the point $(1,-3)$ is $x - 4y = 13$, then the value of $a + b$ is equal to _____ .

Ans. 4

Sol. $y = \frac{x-a}{(x+b)(x-2)}$

At point $(1, -3)$,

$$-3 = \frac{1-9}{(1+b)(1-2)}$$

$\Rightarrow 1 - a = 3(1 + b)$ ...(1)

Now, $y = \frac{x-a}{(x+b)(x-2)}$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+b)(x-2)\times(1) - (x-a)(2x+b-2)}{(x+b)^2(x-2)^2}$$

At $(1, -3)$ slope of normal is $\frac{1}{4}$ hence $\frac{dy}{dx} = -4$,

So, $-4 = \frac{(1+b)(-1) - (1-a)b}{(1+b)^2(-1)^2}$

Using equation (1)

$$\Rightarrow -4 = \frac{(1+b)(-1) - 3(b+1)b}{(1+b)^2}$$

$$\Rightarrow -4 = \left|\frac{(-1) - 3b}{(1+b)}\right| (b \neq -1)$$

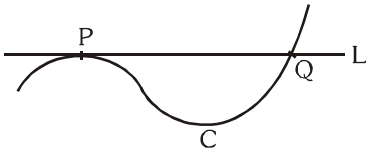
$\Rightarrow b = -3$

So, $a = 7$

Hence, a + b = 7 – 3 = 4

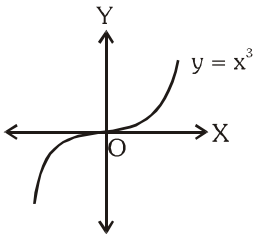
1.2 IMPORTANT POINTS

1. Tangent can intersect the curve.



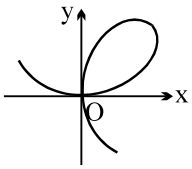
Here L is tangent to C at P, and cutting it again at Q.

2. Tangent can intersect the curve at point of tangency itself.



Here X-axis is tangent to $y = x^3$ at origin.

3. A line can have infinite point of tangency for a curve.
4. Straight line is tangent to itself.
5. A line can be tangent and normal to the same curve.
6. A line can be tangent and normal to the same curve at the same point.



$x^3 + y^3 - 3xy = 0$ (Folium of descartes)

1.3 OBJECTIVE CONCEPTS

1. Tangent to a second degree curve at (x_1, y_1) can be obtained by replacing

   $x^2 \to xx_1$; $y^2 \to yy_1$; $2xy \to yx_1 + xy_1$; $2x \to x + x_1$; $2y \to y + y_1$: $C \to C$ in equation of curve.

2. For an even function if tangent at x = 0 exists, then it will be horizontal.
3. For a polynomial curve passing through origin tangent at origin can directly be obtained by putting "lowest degree term in x and y" = 0

Ex. (i) $x^2 + y^2 - 4x + 3y = 0$ T : $-4x + 3y = 0$

(ii) $y^2 = 5x$ T : $5x = 0$

(iii) $x^3 + y^3 - 3xy = 0$ T : $-3xy = 0$

Some Points from Coordinate Geometry

1. If a line cuts off equal non zero intercepts on coordinate axis, its slope is –1.
2. If a line cuts off non zero intercepts equal in magnitude then slope is ±1.

3. If a line is equally inclined with coordinate axis then slope is ± 1.

1.4 VERTICAL TANGENT

Case-1	LHD $= \infty$	RHD $= \infty$	Vertical Tangent exists
Case-2	LHD $= \infty$	RHD $= -\infty$	Vertical Tangent does not exist
Case-3	LHD $= -\infty$	RHD $= -\infty$	Vertical Tangent exists

1 Which of the following cases the function $f(x)$ has a vertical tangent at $x = 0$.

(i) $f(x) = x^{1/3}$ (ii) $f(x) = \sqrt{|x|}$ (iii) $f(x) = x^{2/3}$

Sol. **Vertical Tangent:**

Concept: $y = f(x)$ has a vertical tangent at the point $x = x_0$ if

$$\lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h} = \infty \text{ or } -\infty \text{ but not both}$$

for example the functions

$f(x) = x^{1/3}$ and $f(x) = \text{sgn}\, x$ both have a vertical tangent at $x = 0$

but $f(x) = x^{2/3}$; $f(x) = \sqrt{|x|}$ and $f(x) = \begin{cases} 0 \text{ if } x < 0 \\ 1 \text{ if } x \geq 0 \end{cases}$ have no vertical tangent

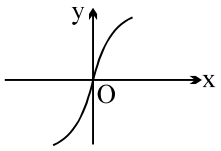
Explanation:

(i) $f(x) = x^{1/3}$

$$f'(0^+) = \lim_{h\to 0} \frac{h^{1/3}}{h} = \frac{1}{h^{2/3}} \to \infty$$

$$f'(0^-) = \lim_{h\to 0} \frac{(-h)^{1/3}}{-h} = \frac{1}{h^{2/3}} \to \infty$$

$\Rightarrow x = 0$ is a vertical tangent

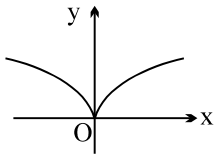


(ii) $f(x) = \sqrt{|x|} = \begin{cases} \sqrt{x} & \text{if } x \geq 0 \\ \sqrt{-x} & \text{if } x < 0 \end{cases}$

$$f'(0^+) = \lim_{h\to 0} \frac{\sqrt{h}-0}{h} \to \infty$$

$$f'(0^-) = \lim_{h\to 0} \frac{\sqrt{h}}{-h} \to -\infty$$

no vertical tangent at $x = 0$

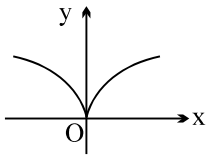


(iii) $f(x) = x^{2/3}$

$$f'(0^+) = \lim_{h\to 0} \frac{h^{2/3}}{h} \to \infty$$

$$f'(0^-) = \lim_{h\to 0} \frac{(-h)^{2/3}}{-h} = -\frac{1}{h^{2/3}} \to -\infty$$

no vertical tangent at $x = 0$



2 SOME IMPORTANT PARAMETRIC COORDINATES

	Curve	Parametric coordinates
(1)	$x^2 + y^2 = a^2$	$(a\cos\theta, a\sin\theta)$
(2)	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$(a\cos\theta, b\sin\theta)$

(3) $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $(a\sec\theta, b\tan\theta)$

(4) $\frac{x^n}{a^n}+\frac{y^n}{b^n}=1$ $(a\cos^{\frac{2}{n}}\theta, b\sin^{\frac{2}{n}}\theta)$

(5) $xy = c^2$ $(ct, \frac{c}{t})$

(6) $y^2 = 4ax$ $(at^2, 2at)$

(7) $x^2 = 4ay$ $(2at, at^2)$

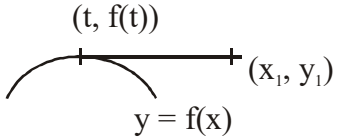
(8) $y = mx + c$ $(t, mt + c)$

(9) $y = f(x)$ $(t, f(t))$

3.1 ULTIMATE CONCEPT

Always equate the slope of tangent.

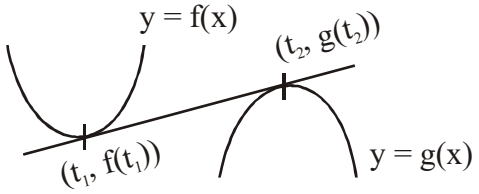
Ex.1 Find equation of tangent from (x_1, y_1) to the curve $y = f(x)$.



$$f'(t) = \frac{f(t)-y_1}{t-x_1}$$

solve and get t.

Ex.2 Find equation to common tangent to the curves $y = f(x)$ and $y = g(x)$.



$$f'(t_1) = g'(t_2) = \frac{g(t_2)-f(t_1)}{t_2-t_1}$$

1. Tangent at point P(2,8) on the curve $y = x^3$ interect the curve again at Q. Find coordinates of Q.

Ans. (–4, –64)

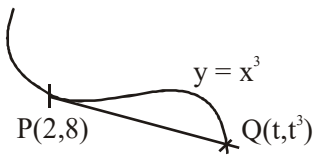
Sol. $m_{PQ} = \left(\frac{dy}{dx}\right)_P$

$$\frac{t^3-8}{t-2} = \left(3x^2\right)_{x=2}$$

$t^2 + 4 + 2t = 12$

$t^2 + 2t - 8 = 0$

$t = -4, 2$

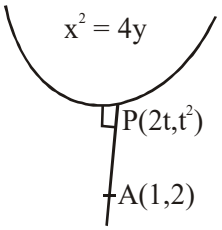


Q(–4, –64)

2. Find equation of the normal to the curve $x^2 = 4y$ which passes through (1, 2).

Ans. $x + y = 3$

Sol. $m_{AP} = \left(\dfrac{-1}{\dfrac{dy}{dx}}\right)_P$



$\dfrac{t^2-2}{2t-1} = -\dfrac{1}{t}$

$t^3 - 2t = -2t + 1$

$t = 1$

$P(2, 1)$

EON : $y - 1 = (-1)(x - 2)$

3. Find abscisca of all points on curve $y = x^3 - 3x$ such that tangent on those points passes through (2, –2)

Ans. $1, 1 \pm \sqrt{3}$

Sol. Let $P(t, t^3 - 3t)$

$\dfrac{dy}{dx} = \left(3x^2 - 3\right)_{x=t} = \dfrac{\left(t^3 - 3t\right) - (-2)}{t - 2}$

$(3t^2 - 3)(t - 2) = t^3 - 3t + 2$

$3(t - 1)(t + 1)(t - 2) = (t - 1)(t^2 + t - 2)$

$t = 1$ or $3(t^2 - t - 2) = t^2 + t - 2$

$2t^2 - 4t - 4 = 0$

$t^2 - 2t - 2 = 0$

$t = 1 \pm \sqrt{3}$

4. Find equation of normals to the curve $x^2 = 4y$ which passes through (4, – 1).

Ans. $x + y = 3$

Sol. Similar as question no. 2

5. A line is drawn touching the curve $y - \dfrac{2}{3-x} = 0$. Find the line if its slope/gradient is 2.

Ans. $y - 2x + 2 = 0$; $y - 2x + 10 = 0$

Sol. $\dfrac{dy}{dx} = \dfrac{2}{(3-x)^2} = 2$

$\Rightarrow x = 4$ or 2

$P(4, -2)$ or $P(2, 2)$

EOT : $y + 2 = 2(x - 4)$ OR EOT : $y - 2 = 2(x - 2)$

6. Find point on curve $y = 2x^3 - 15x^2 + 34x - 20$ such that tangent on that point is parallel to line $y + 2x = 0$

Ans. (2, 4) and (3, 1)

Sol. $m_T = -2$

$\dfrac{dy}{dx} = 6x^2 - 30x + 34 = 2$

$x^2 - 5x + 6 = 0$

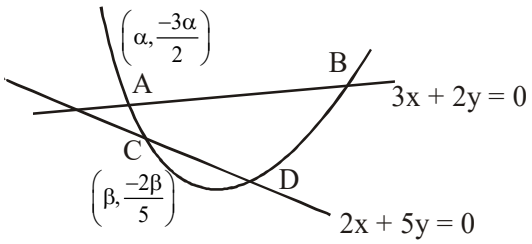
$x = 2, 3$

At x = 3 & at x = 2

y = 1 y = 4

7. Show that tangents to the curve $3x^2 + 4xy + 5y^2 - 4 = 0$ at the points, in which it is intersected by the lines $3x + 2y = 0$ and $2x + 5y = 0$, is parallel to co-ordinate axis

Sol.



$$6x + 4x\frac{dy}{dx} + 4y + 10y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-(3x + 2y)}{2x + 5y}$$

$$\left(\frac{dy}{dx}\right)_{A \text{ or } B} = 0 \qquad \& \qquad \left(\frac{dy}{dx}\right)_{C \text{ or } D} = \infty$$

Tangents at A & B are parallel to x-axis & tangents at C & D are parallel to y-axis.

8. Prove that all points on the curve $y^2 = 4a\left(x + a\sin\frac{x}{a}\right)$ at which the tangent is parallel to the x-axis lie on a parabola. **[O]**

Sol. Let P(h, k)

$$2y\frac{dy}{dx} = 4a\left(1 + a\cos\left(\frac{x}{a}\right)\cdot\left(\frac{1}{a}\right)\right)$$

$$\left(\frac{dy}{dx}\right)_P = \frac{2a}{k}\left(1 + \cos\left(\frac{h}{a}\right)\right) = 0$$

$$\cos\frac{h}{a} = -1$$

$$\Rightarrow \sin\frac{h}{a} = 0$$

$$\Rightarrow k^2 = 4ah \qquad \{(h, k) \text{ lie on } y^2 = 4a\left(x + a\sin\left(\frac{x}{a}\right)\right)\}$$

$y^2 = 4ax$ which is parabola

9. Tangents are drawn from the origin to the curve $y = \sin x$. Prove that their point of contacts lie on the curve $x^2y^2 = x^2 - y^2$. **[O]**

Sol. Let P(h, k) lie on $y = \sin x$

$$\frac{dy}{dx} = (\cos x)_P = \frac{k - 0}{h - 0}$$

$\cosh = \frac{k}{h}$

$\sin h = k$ {(h, k) lie on $y = \sin x$}

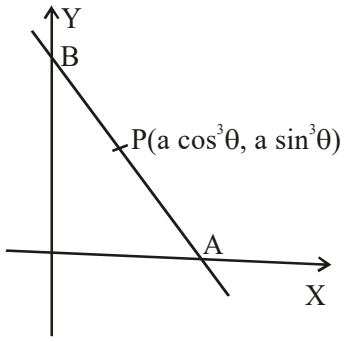
$\cos^2 h + \sin^2 h = 1$

$\frac{k^2}{h^2} + k^2 = 1$

$y^2 + x^2y^2 = x^2$

10. Show that the portion of the tangent to the curve $x = a\cos^3\theta$ and $y = a\sin^3\theta$ intercepted between the coordinate axes is constant.

Sol $l = a$



$\frac{dy}{dx} = \frac{3a\sin^2\theta\cos\theta}{3a\cos^2\theta(-\sin\theta)} = -\tan\theta$

EOT :

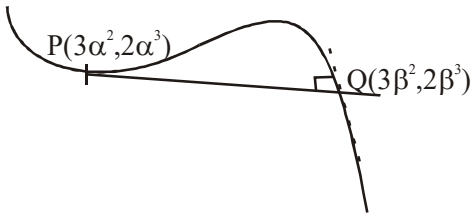
$y - a\sin^3\theta = -\tan\theta\,(x - a\cos^3\theta)$

$B(0, a\sin\theta)$; $A(a\cos\theta, 0)$

$|AB| = a$ = constant

11. Find the equation of the straight line which is tangent at one point and normal at another point of the curve, $x = 3t^2$, $y = 2t^3$. **[O]**

Ans. $\sqrt{2}\,x + y - 2\sqrt{2} = 0$ or $\sqrt{2}\,x - y - 2\sqrt{2} = 0$



Sol.

$\frac{dy}{dx} = \frac{6t^2}{6t} = t$

$m_{PQ} = (m_T)_P = (m_n)_Q$

$$\frac{2\left(\beta^3-\alpha^3\right)}{3\left(\beta^2-\alpha^2\right)}=\alpha=\frac{-1}{\beta}$$

$\alpha\beta = -1$ & $2(\beta^2 + \alpha^2 + \alpha\beta) = 3\alpha(\beta + \alpha)$

$2\beta^2 + 2\alpha^2 - 2 = -3 + 3\alpha^2$

$\alpha^2 - 2\beta^2 - 1 = 0$

$$\alpha^2-\frac{2}{\alpha^2}-1=0$$

$\alpha^4 - \alpha^2 - 2 = 0$

$\alpha^2 = 2, -1$

$$\alpha=\pm\sqrt{2}$$

Equation of PQ : $y-4\sqrt{2}=\sqrt{2}\left(x-6\right)$

or $y+4\sqrt{2}=-\sqrt{2}\left(x-6\right)$

12. If p_1 and p_2 are the lengths of the perpendiculars from origin on the tangent and normal drawn to the curve $x^{2/3} + y^{2/3} = 6^{2/3}$ respectively. Find the value of $\sqrt{4p_1^2 + p_2^2}$. **[O]**

Ans. **6**

Sol. Let $x = 6\cos^3\theta$ $y = 6\sin^3\theta$

$$\frac{dy}{dx}=\frac{6(3\sin^2\theta\cos\theta)}{-6(3\cos^2\theta\sin\theta)}=-\tan\theta$$

Equ. of tangent

$y - 6\sin^3\theta = -\tan\theta\ (x - 6\cos^3\theta) \Rightarrow p_1 = 6\sin\theta\cos\theta$

Equ. of normal

$y - 6\sin^3\theta = \cot\theta\ (x - 6\cos^3\theta) \Rightarrow p_2 = 6(\cos^2\theta - \sin^2\theta)$

$$\sqrt{4p_1^2+p_2^2}=6\sqrt{4\sin^2\theta\cos^2\theta+\cos^4\theta+\sin^4\theta-2\sin^2\theta\cos^2\theta}=6$$

13. For the curve $xy = c^2$, prove that the portion of the tangent intercepted between the coordinate axes is bisected at the point of contact. **[O]**

Sol: Let the point at which tangent is drawn be (α, β) on the curve $xy = c^2$.

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{(\alpha,\beta)}=-\frac{\beta}{\alpha}$$

Thus, the equation of tangent is,

$$y-\beta=-\frac{\beta}{\alpha}(x-\alpha)$$

$$\Rightarrow \quad y\alpha-\alpha\beta=-x\beta+\alpha\beta$$

$$\Rightarrow \quad x\beta+y\alpha=2\alpha\beta$$

$$\Rightarrow \quad \frac{x}{2\alpha}+\frac{y}{2\beta}=1$$

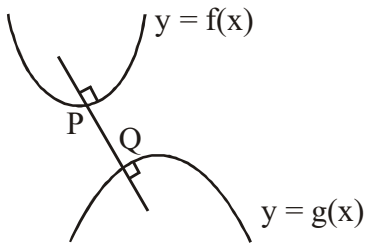
It is clear that the tangent line cuts x and y-axis at A(2α , 0) and B(0, 2β) and the point (α, β) bisects AB.

3.2 MINIMUM DISTANCE BETWEEN TWO CURVES

Minimum distance between two curves will be :

(i) Zero if the curves intersect each other.

(ii) Along their common normal if they do not have any common point.



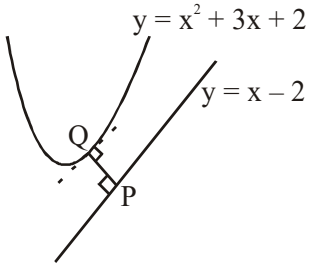
Note : Given a fixed point A(a, b) and a moving point P(x, f (x)) on the curve y = f (x). Then

AP will be maximum or minimum if it is normal to the curve at P.

1. Find minimum distance between curves $y = x^2 + 3x + 2$ and $y = x - 2$.

Ans. $\frac{3}{\sqrt{2}}$

Sol.



$Q\ (t, t^2 + 3t + 2)$

$m_{PQ} = -1$

$$\frac{-1}{2t+3} = -1$$

$2t + 3 = 1$

$t = -1$

$Q(-1, 0)$

$$|PQ| = \frac{|-1-0-2|}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

2. Find the minimum distance between $x^2 + y^2 = 2$ and $xy = 9$.

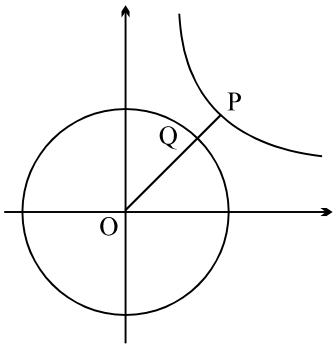
Ans. $2\sqrt{2}$

Sol. $xy = 9 \left\langle \begin{matrix} x = 3t \\ y = 3/t \end{matrix} \right.$

$\frac{dy}{dx} = -\frac{9}{x^2}$

slope of the normal at the point P(t)

$\frac{x^2}{9} = t^2$



equation of normal at P is

$y - \frac{3}{t} = t^2(x - 3t)$ ....(1)

minimum distance = PQ i.e. along common normal

$\therefore$ (1) must pass through origin

$\therefore \quad -\frac{3}{t} = -3t^2 \quad \Rightarrow \quad t^4 = 1 \quad \Rightarrow \quad t = 1 \text{ or } -1$

($t = -1$ corresponds to a point $xy = 9$ in the 3rd quadrant)

$\therefore \quad t = 1$

hence P is (3, 3) $\Rightarrow \quad m_{OP} = 1$

equation of OP is $y = x \Rightarrow$ Q(1, 1)

hence $d_{min.} = \sqrt{4+4} = 2\sqrt{2}$ Ans.

3. Find the minimum value of $y = (x_1 - x_2)^2 + \left(\sqrt{2 - x_1^2} - \frac{9}{x_2}\right)^2$, $x_1 \in (0, \sqrt{2})$; $x_2 \in R^+$;

Ans. 8

Sol. Put $\sqrt{2 - x_1^2} = y_1$

& $\frac{9}{x_2} = y_2$

Now, $y = (x_1 - x_2)^2 + (y_1 - y_2)^2$

which denotes distance between two points (x_1, y_1), (x_2, y_2) which lies on $x^2 + y^2 = 2$ & $xy = a$ respectively.

Now, same as question 2.

4. A straight line l passes through the points (3, 0) and (0, 4). The point A lies on the parabola $y = 2x - x^2$. Find the distance p from point A to the straight line and indicate the coordinates of the point A (x_0, y_0) on the parabola for which the distance from the parabola to the straight line is the least. **[O]**

Ans. 11/15

Sol. $\ell : \frac{x}{3} + \frac{y}{4} = 1; m_\ell = \frac{-\left(\frac{1}{3}\right)}{\frac{1}{4}} = \frac{-4}{3}$

$A(t, 2t - t^2)$

$m_N = \frac{3}{4}$

$\frac{-1}{2-2t} = \frac{3}{4} \Rightarrow -4 = 6 - 6t$

$$t = \frac{5}{3}$$

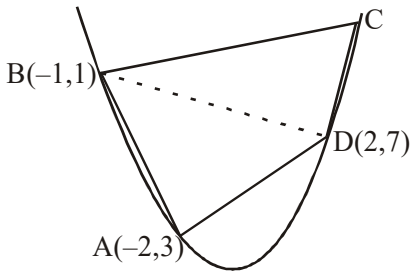
$$A\left(\frac{5}{3}, \frac{5}{9}\right)$$

$$P = \frac{\left|\frac{5}{9} + \frac{5}{36} - 1\right|}{\sqrt{\frac{1}{9} + \frac{1}{16}}} = \frac{\frac{11}{36}}{\frac{5}{12}} = \frac{11}{15}$$

5. Four points A, B, C, D lie in that order on the parabola $y = ax^2 + bx + c$. The coordinates of A, B, and D are known A (–2, 3), B (–1, 1), D (2, 7). Find the coordinates of C for which the area of the quadrilateral ABCD is the greatest. **[O]**

Ans. a = b = c = 1, (1/2, 7/4)]

Sol.



$3 = 4a - 2b + c$
$7 = 4a + 2b + c$
$1 = a - b + c$

$a = b = c = 1$
$y = x^2 + x + 1$
Let $C(t, t^2 + t + 1)$
$ar(\text{quad ABCD})_{max}$ when $ar(\Delta BCD)_{max}$

$$\Rightarrow ar(\Delta BCD) = \frac{1}{2}\begin{vmatrix} t & t^2+t+1 & 1 \\ -1 & 1 & 1 \\ 2 & 7 & 1 \end{vmatrix}$$

$$A = \frac{1}{2}\left(3t^2 - 3t - 6\right)$$

$$\frac{dA}{dt} = \frac{1}{2}(6t - 3) = 0 \Rightarrow t = \frac{1}{2}$$

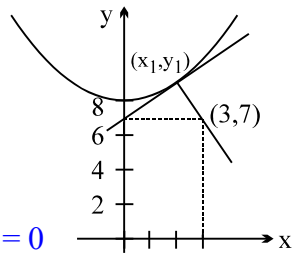
$$C\left(\frac{1}{2}, \frac{7}{4}\right)$$

6. A aeroplane of enemy is flying along the curve given by $y = x^2 + 7$. A soldier placed at (3, 7) wants to shoot down the helicopter when it is nearest to him. Find the nearest distance.

Ans. $\sqrt{5}$

[Hint: $\left.\frac{dy}{dx}\right]_{x_1,y_1} = 2x_1$

$\therefore \quad (2x_1)\left(\frac{7-y_1}{3-x_1}\right) = -1$



$2x_1(7-y_1) = x_1 - 3$

$14x_1 - 2x_1y_1 = x_1 - 3$

$13x_1 + 3 = 2x_1(x_1^2 + 7)$

$2x_1^3 + x_1 - 3 = 0 \quad \Rightarrow \quad 2x_1^3 + 3x_1 - 3 = 0$

$2x_1^2(x_1 - 1) + 2x_1(x_1 - 1) + 3(x_1 - 1)$

$(x_1 - 1)(2x_1^2 + 2x_1 + 3)$

$x_1 = 1$

7. Find the shortest distance between the curves , $9x^2 + 9y^2 - 30y + 16 = 0$ and $y^2 = x^3$. **[O]**

Ans $\frac{\sqrt{13}}{3} - 1$

Sol. $9x^2 + 9y^2 - 30y + 16 = 0$ can be rewritten as , $x^2 + \left(y - \frac{5}{3}\right)^2 = 1$

Any point on the curve $y^2 = x^3$ can be taken as (t^2, t^3).

Let L be the distance between the centre of the given circle and the point (t^2, t^3) , then

$$K = L^2 = t^4 + \left(t^3 - \frac{5}{3}\right)^2$$

Now , we calculate the minimum value of L . Required distance = L – radius of given circle.

Now , $\frac{dK}{dt} = 4t^3 + 2\left(t^3 - \frac{5}{3}\right)3t^2 = 0$

for maximum or minimum , $t = 0$ or 1

Now , $\frac{d^2K}{dt^2} = 12t^2 + 30t^4 - 20t$ $\qquad \left.\frac{d^2K}{dt^2}\right|_{t=0} = 0$ $\qquad$ But , $\left.\frac{d^3K}{dt^3}\right|_{t=0} \neq 0$

$\Rightarrow$ There is neither maxima nor minima at $t = 0$.

Also , $\frac{d^2K}{dt^2} > 0$ at $t = 1$ $\quad \Rightarrow \quad$ L^2 is minimum at $t = 1$ i.e., L is minimum at $t = 1$

So , shortest distance = (value of L at $t = 1$) – (Radius of the circle) = $\frac{\sqrt{13}}{3} - 1$.

8. Minimum distance between the curves $y^2 = 4x$ & $x^2 + y^2 - 12x + 31 = 0$ is α , then find $[\alpha]$ **[O]**
(Where $[.] \rightarrow$ G.I.F.)

Ans. 2

Sol.

Let normal be $y + tx = 2t + t^3$

for shortest distance it must pass through (6, 0)

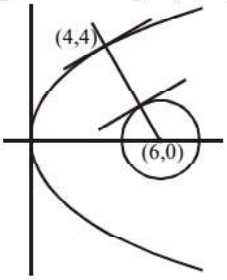
$\Rightarrow \quad 6t = 2t + t^3.$

$t = 2$

shortest distance

$= \sqrt{4+16} - \sqrt{5}$

$= \sqrt{20} - \sqrt{5}$



4.1 ANGLE OF INTERSECTION BETWEEN TWO CURVES

Let $y = f(x)$ and $y = g(x)$ be two given intersecting curves. Angle of intersection of these curves is defined as the acute angle between the tangents that can be drawn to the given curves at the point of intersection .

Let (x_1 , y_1) be the point of intersection. $\Rightarrow y_1 = f(x_1) = g(x_1)$

Slope of the tangent drawn to the curve $y = f(x)$ at (x_1 , y_1) i.e., $m_1 = \left(\frac{df}{dx}\right)_{(x_1 , y_1)}$

Similarly slope of the tangent drawn to the curve $y = g(x)$ at (x_1, y_1) i.e., $m_2 = \left(\frac{dg}{dx}\right)_{(x_1 , y_1)}$

If θ be the angle (acute) of intersection, then

$\tan\theta \rightarrow \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$ m_1 and m_2 are finite

$\left|\frac{1}{m_1}\right|$ $m_2 \to \infty$

1. Find the acute angle between the curves

(i) $y = \sin x$ & $y = \cos x$

Ans. $\tan^{-1}(2\sqrt{2})$

Sol. $m_1 = \cos x, m_2 = -\sin x$

Point of Intersection at $x = \frac{\pi}{4}$

$m_1 = \frac{1}{\sqrt{2}}, m_2 = \frac{-1}{\sqrt{2}}$

$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = 2\sqrt{2}$

(ii) $y = x^2$ and $6y = 7 - x^3$ at (a, a).

Ans. $\frac{\pi}{2}$

Sol. Point of intersection (1, 1)

$C_1 : \frac{dy}{dx} = 2x \Rightarrow m_1 = 2$

$C_2 : \frac{6dy}{dx} = -3x^2 \Rightarrow m_2 = -\frac{1}{2}$

$m_1 m_2 = -1$

$\theta = \frac{\pi}{2}$

2. Find the angle between the curve $2y^2 = x^3$ and $y^2 = 32x$

Ans. at (0, 0) $\theta = 90^\circ$; at (8, 16) & (8, –16) $\theta = \tan^{-1}(1/2)$

Sol. Solving $2y^2 = x^3$ and $y^2 = 32x$

we get (0, 0) ; (8, 16) and (8, – 16)

for $\sqrt{2}\, y = x\sqrt{x}$ or $\sqrt{2}\, y = -x\sqrt{x}$

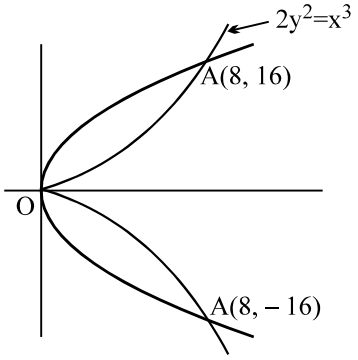
at (0, 0) $\left.\frac{dy}{dx}\right|_{(0,0)} = 0$

for I

at (0, 0) $\left.\frac{dy}{dx}\right|_{(0,0)} = \infty$

for II

hence angle = 90°



now $\left.\frac{dy}{dx}\right|_I = \frac{3x^2}{4y} = \frac{3}{4}\cdot\frac{64}{16} = 3$

$\left.\frac{dy}{dx}\right|_{II} = \frac{32}{2y} = \frac{16}{16} = 1 \qquad \therefore \qquad \tan\theta = \frac{3-1}{1+3} = \frac{2}{4} = \frac{1}{2}$

3. Find angle betweeen the curves $x^3 + y^3 = 3xy$ and $y^2 = x$ at point other than origin

Ans. $\tan^{-1}\left(16^{1/3}\right)$

Sol. For point of intersection

$y^6 + y^3 = 3y^3$

$y = 0,\ y = 2^{\frac{1}{3}}$

$x = 0,\ x = 2^{\frac{2}{3}}$

$$P\left(2^{\frac{2}{3}}, 2^{\frac{1}{3}}\right)$$

$$C_1 : 3x^2 + 3y^2\frac{dy}{dx} = 3\left(x\frac{dy}{dx} + y\right)$$

$$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x} \Rightarrow m_1 = \infty$$

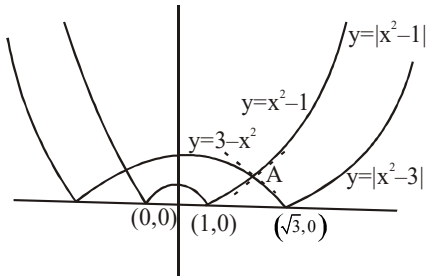
$$C_2 : 2y\frac{dy}{dx} = 1 \Rightarrow m_2 = \frac{1}{2^{\frac{4}{3}}}$$

$$\tan\theta = \left|\frac{1}{m_2}\right| = 16^{\frac{1}{3}}$$

4. Find the acute angles between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their point of intersection.

Ans. $\theta = \tan^{-1}\left|\frac{4\sqrt{2}}{7}\right|$

Sol.



$A : x^2 - 1 = 3 - x^2$

$x^2 = 2$

$x = \sqrt{2}$

$y = 1$

$m_1 = 2\sqrt{2}, m_2 = -2\sqrt{2}$

$$\tan\theta = \left|\frac{4\sqrt{2}}{7}\right|$$

5. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical, is (are) **[O]**

(A) $\left(\pm\frac{4}{\sqrt{3}}, -2\right)$ (B) $\left(\pm\frac{\sqrt{11}}{3}, 1\right)$ (C) (0, 0) (D) $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$

Ans. D

Sol. $3y^2\frac{dy}{dx} + 6x = 12\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{-2x}{y^2 - 4}$$

$y = 2$ or $y = -2$

$8 + 3x^2 = 24$ $\quad -8 + 3x^2 = -24$

$x^2 = \frac{16}{3}$ No real value of x.

(D) option correct.

4.2 ORTHOGONAL CURVES

Two curves are said to be orthogonal if angle of intersection is 90º at every common point.

Sometimes we do not need to find common point to check orthogonality.

1. Check whether curves are orthogonal.

(i) $y^2 = 4ax$ & $y = e^{-x/2a}$

Ans. Yes

Sol. Let common point be (x_1, y_1)

$2y\frac{dy}{dx} = 4a$ $\qquad \frac{dy}{dx} = e^{-\frac{x}{2a}}\left(-\frac{1}{2a}\right)$

$m_1 = \frac{2a}{y_1}$ $\qquad m_2 = \frac{-y_1}{2a}$

$m_1m_2 = -1$

Curves are orthogonal.

(ii) $xy = a^2$ & $x^2 - y^2 = b^2$

Ans. Yes

Sol. Let common point bhe (x_1, y_1)

$y + x\frac{dy}{dx} = 0$ $\qquad 2x - 2y\frac{dy}{dx} = 0$

$m_1 = \frac{-y_1}{x_1}$ $\qquad m_2 = \frac{x_1}{y_1}$

$m_1m_2 = -1$

curves are orthogonal

(iii) $y = ax$ & $x^2 + y^2 = c^2$

Ans. Yes

Sol. Let common point be (x_1, y_1)

$m_1 = a;\ m_2 = \frac{-x_1}{y_1}$

$m_1m_2 = \frac{-ax_1}{y_1} = \frac{-y_1}{y_1} = -1$ $\qquad \{(x_1, y_1)$ lie on $y = ax\}$

Curves are orthogonal.

2. If the curves, $x^3 + pxy^2 = -2$ and $3x^2y - y^3 = 2$ are orthogonal. Find p.

Ans. $p = -3$

Sol. Let common point be (x_1, y_1)

$3x_1^2 + py_1^2 + 2px_1y_1\frac{dy}{dx} = 0$ $\qquad$ $3x_1^2\frac{dy}{dx} + 6x_1y_1 - 3y_1^2\frac{dy}{dx} = 0$

$m_1 = -\frac{3x_1^2 + py_1^2}{2px_1y_1}$ $\qquad$ $m_2 = \frac{2x_1y_1}{y_1^2 - x_1^2}$

$\Rightarrow m_1m_2 = -1$

$-\left(\frac{3x_1^2 + py_1^2}{2px_1y_1}\right)\left(\frac{2x_1y_1}{y_1^2 - x_1^2}\right) = -1$

$\Rightarrow 3x_1^2 + py_1^2 = py_1^2 - px_1^2$

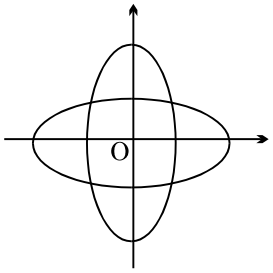
$\Rightarrow p = -3$

3. Find the condition for the two concentric conics $a_1x^2 + b_1y^2 = 1$ and $a_2x^2 + b_2y^2 = 1$ to intersect orthogonally.

Ans. $\frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{b_2} - \frac{1}{b_1}$

Sol. $a_1x^2 + b_1y^2 = 1$ & $a_2x^2 + b_2y^2 = 1$

differentiating, $2a_1x + 2b_1y \cdot \frac{dy}{dx} = 0$



$\frac{dy}{dx} = -\frac{a_1x}{b_1y}$

$\left.\frac{dy}{dx}\right|^{I}_{x_1,y_1} = -\frac{a_1x_1}{b_1y_1}$ ....(1)

|||ly $\left.\frac{dy}{dx}\right|^{II}_{x_1,y_1} = -\frac{a_2x_1}{b_2y_1}$ ....(2)

If the two curves are orthogonal

$\frac{a_1x_1}{b_1y_1} \cdot \frac{a_2x_1}{b_2y_1} = -1$

$\frac{a_1a_2}{b_1b_2}\frac{x_1^2}{y_1^2} = -1$ ....(3)

now $a_1x_1^2 + b_1y_1^2 = 1$ & $a_2x_1^2 + b_2y_1^2 = 1$ (as x_1y_1 satisfy the given equations)

subtracting $(a_1 - a_2)x_1^2 + (b_1 - b_2)y_1^2 = 0$

$\frac{x_1^2}{y_1^2} = -\frac{b_1 - b_2}{a_1 - a_2}$

putting in (3) $\dfrac{a_1a_2}{b_1b_2}\dfrac{b_1-b_2}{a_1-a_2}=1$

$$\frac{a_1-a_2}{a_1a_2}=\frac{b_1-b_2}{b_1b_2} \quad \Rightarrow \quad \frac{1}{a_2}-\frac{1}{a_1}=\frac{1}{b_2}-\frac{1}{b_1}$$

4. Find the condition for the line $y=mx$ to cut at right angles the conic $ax^2+2hxy+by^2=1$. Hence find the direction of the axes of the conic. **[O]**

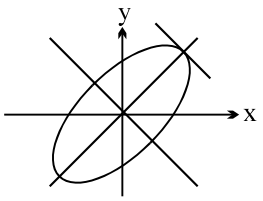
Ans. $m^2h+m(a-b)-h=0$

Sol. $ax^2+2hxy+by^2=1$ ....(1)

$$2ax+2h\left[x\frac{dy}{dx}+y\right]+2by\frac{dy}{dx}=0$$

$$(xh+by)\frac{dy}{dx}=-(ax+hy)$$

$$\frac{dy}{dx}=-\frac{ax+hy}{hx+by}$$



$$\therefore \quad +m\left(\frac{ax+hy}{hx+by}\right)=+1$$

$$m\left[\frac{ax+h\cdot mx}{hx+b\cdot mx}\right]=1$$

$m(a+hm)=h+bm$

$m^2h+(a-b)m-h=0$

5. The curve $y=ax^3+bx^2+cx+5$ touches the x-axis at $(-2, 0)$ and cuts the y-axis at a point Q where its gradient is 3. Find a, b, c.

Ans. $a=-½, b=-3/4, c=3$

Sol. $0=-8a+4b-2c+5$ ...(1)

$$\frac{dy}{dx}=3ax^2+2bx+c$$

$$\left(\frac{dy}{dx}\right)_{(-2,0)}=12a-4b+c=0 \qquad ...(2)$$

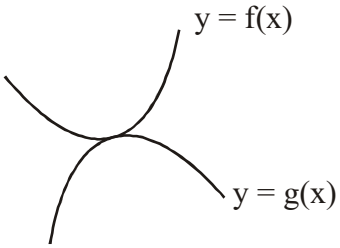
$$\left(\frac{dy}{dx}\right)_{x=0}=c=3 \qquad ...(3)$$

from (1), (2) & (3) :

$$a=-\frac{1}{2}, b=-\frac{3}{4}, c=3$$

4.3 CASE WHEN TWO CURVES TOUCH EACH OTHER

If f(x) and g(x) touch each other then their slopes at their common point must be same.



$f(x_1) = g(x_1)$

$f'(x_1) = g'(x_1)$

Q.1 Check whether curves $x^2 + y^2 = 8$ and $xy = 4$ touch each other at (2, 2)

Ans Yes

Sol. $2x + 2y\dfrac{dy}{dx} = 0$ $\qquad$ $x\dfrac{dy}{dx} + y = 0$

$\left(\dfrac{dy}{dx}\right)_{(2,2)} = -1$ $\qquad$ $\left(\dfrac{dy}{dx}\right)_{(2,2)} = -1$

$m_1 = m_2$

Q.2 If $y = e^x$ and $y = kx$ touch each other, find k.

Ans e

Sol. Let common point be (x_1, y_1)

$\dfrac{dy}{dx} = e^{x_1}$ $\qquad$ $\dfrac{dy}{dx} = k$

$e^{x_1} = k$

$y_1 = kx_1 = k$

$\Rightarrow x_1 = 1, y_1 = e$

$k = e$

Q.3 If $y = \ln x$, $y = kx$ touch each other, find k

Ans 1/e

Sol. Let common point be (x_1, y_1)

$\dfrac{1}{x_1} = k$

$x_1 = \dfrac{1}{k}$ & $y_1 = kx_1$

$y_1 = 1, x_1 = e$

$k = \dfrac{1}{e}$

Q.4 $y = \ln x$, $y = kx^2$ touch each other, find k.

Ans 1/2e

Sol. Let common point be (x_1, y_1)

$$\frac{1}{x_1} = 2kx_1$$

$$x_1^2 = \frac{1}{2k} \ \& \ y_1 = kx_1^2$$

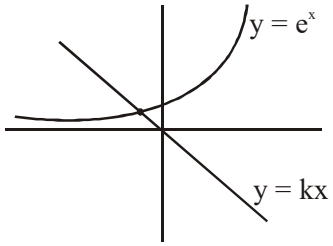
$$\Rightarrow y_1 = \frac{1}{2}, x = \sqrt{e}$$

$$\& \ k = \frac{1}{2e}$$

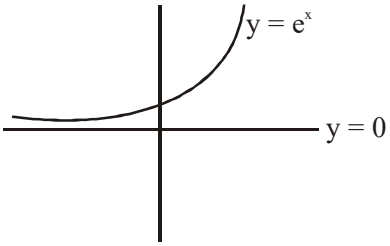
Q.5 Analyze equation $e^x = kx$ for number of solutions

Ans. If $k < 0$ then one solution, , If $0 \le k < e$ no solution, If $k = e$ exactly one solution, If $k > e$ two solution

Sol. (i) $k < 0$ (ii) $k = 0$

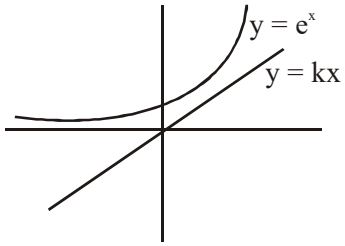


One Point of intersection
One solution

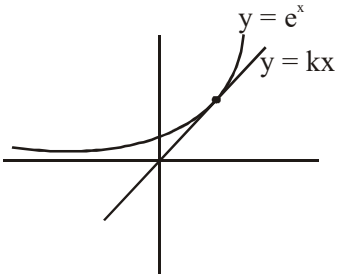


No point of intersection
No solution

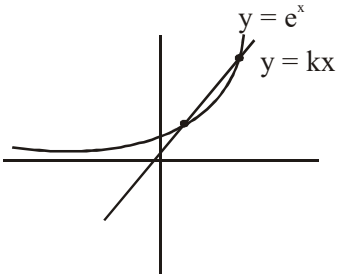
(iii) $k > 0$



$0 < k < e$
no point of intersection
no solution



$k = e$
1 point of intersection
1 solution

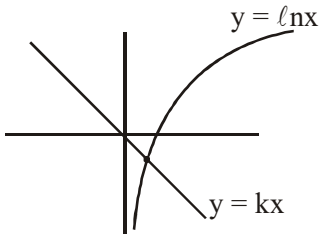


$k > e$
2 point of intersection
2 solution

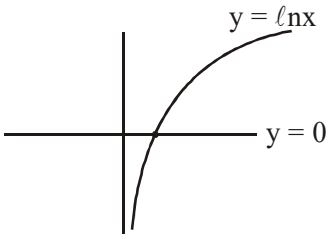
Q.6 Analyze equation $\ln x = kx$ for number of solutions

Ans. If $k > 1/e$ no solution, If $k = 1/e$ exactly one solution, If $0 < k < 1/e$ two solution, if $k \le 0$ then one solution

Sol. (i) $k < 0$ (ii) $k = 0$

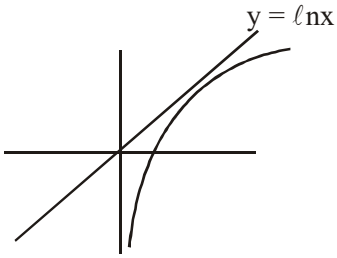


1 point of intersection
1 solution



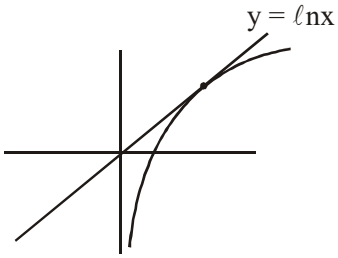
1 point of intersection
1 solution

(iii) $k > 0$



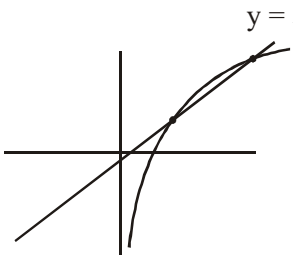
$k > \frac{1}{e}$

No point of intersection
No solution



$k = \frac{1}{e}$

1 point of intersection
1 solution



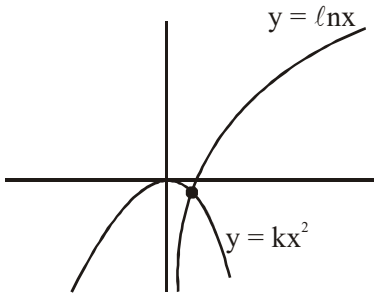
$0 < k < \frac{1}{e}$

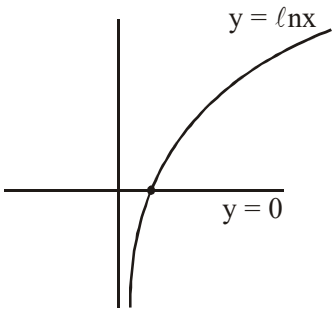
2 Point of intersection
2 Solution

Q.7 Analyze equation $\ln x = kx^2$ for number of solution

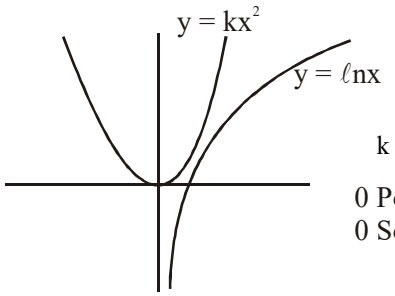
Ans. If $k > 1/2e$ no solution, If $k = 1/2e$ exactly one solution, If $0 < k < 1/2e$ two solution, if $k \leq 0$ then one solution

Sol. (i) $k < 0$ (ii) $k = 0$

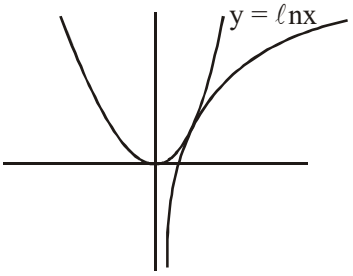




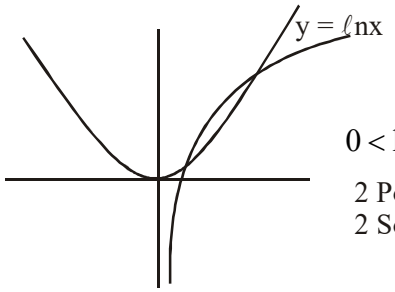
(iii) $k > 0$



$k > \frac{1}{2e}$
0 Point of intersection
0 Solution



$k = \frac{1}{2e}$
1 Point of intersection
1 Solution



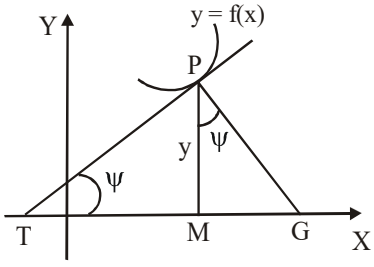
$0 < k < \frac{1}{2e}$
2 Point of intersection
2 Solution

5 SOME IMPORTANT LENGTHS

Let the tangent and the normal at any point (x , y) of the curve y = f(x) meet the x–axis at T and G respectively . Draw the ordinate PM . Then the lengths TM , MG are called the sub–tangent and sub–normal respectively . The lengths PT , PG are sometimes referred to as the lengths of the tangent and the normal respectively .

Clearly $\angle MPG = \psi$

Also , $\tan\psi = \frac{dy}{dx}$



From the figure , we have :

(i) Length of Tangent :

$= TP = MP\ |\operatorname{cosec}\psi| = |y|\ \sqrt{(1+\cot^2\psi)}$

$= |y|\ \sqrt{\left[1+\left(\frac{dx}{dy}\right)^2\right]} = \left|y\frac{\sqrt{1+m^2}}{m}\right|$

(ii) Length of Sub–tangent :

$= TM = MP\ |\cot\psi| = \left|y\frac{dx}{dy}\right| = \left|\frac{y}{m}\right|$

(iii) Length of Normal :

$= GP = MP\ |\sec\psi| = |y|\ \sqrt{(1+\tan^2\psi)} = |y|\ \sqrt{\left[1+\left(\frac{dy}{dx}\right)^2\right]} = \left|y\sqrt{1+m^2}\right|$

(iv) Length of Sub–normal :

$= MG = MP\ |\tan\psi| = \left|y\frac{dy}{dx}\right| = |ym|$

Another method

equation of PT $y - y_1 = m(x - x_1)$

equation of PG $y - y_1 = \frac{-1}{m}(x - x_1)$

$T \equiv (x_1 - \frac{y_1}{m}, 0)$ $G \equiv (x_1 + my_1, 0)$

Apply distance formula to get all four lengths.

1. Find the length of tangent, subtangent, normal and subnormal at the point (2,4) of the curve $y^2 = 8x$.

Ans. $4\sqrt{2}, 4, 4\sqrt{2}, 4$

Sol. Differentiating the equation of the curve w.r.t. x, we get

$2y\frac{dy}{dx} = 8 \Rightarrow dy/dx = 8/2y = 4/y$

$\therefore \left(\frac{dy}{dx}\right)_{(2,4)} = 4/4 = 1$

Therefore at the point (2,4):

length of tangent $= \frac{4\sqrt{1+(1)^2}}{1} = 4\sqrt{2}$

length of sub tangent $= 4/1 = 4$

length of normal $= 4\sqrt{1+(1)^2} = 4\sqrt{2}$

length of sub normal $= 4.1 = 4$

2. Find all four lengths for $\sqrt{x} + \sqrt{y} = 3$ at (1,4)

Ans. $2\sqrt{5}, 2, 4\sqrt{5}, 8$

Sol. $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0$

$$\left(\frac{dy}{dx}\right)_{(1,4)} = -2$$

$\therefore$ Length of tangent $= 4\sqrt{1+\frac{1}{4}} = 2\sqrt{5}$

Length of subtangent $= \left|\frac{4}{(-2)}\right| = 2$

Length of normal $= 4\sqrt{1+4} = 4\sqrt{5}$

Length of subnormal $= |4(-2)| = 8$

3. $y = ax^n$. If length of subnormal is always constant find n $(n > 0)$

Ans. $\frac{1}{2}$

Sol. $\frac{dy}{dx} = nax^{n-1}$

Length of subnormal : $y_1\left(nax_1^{n-1}\right)$

$\Rightarrow na^2x_1^{2n-1} = \text{constant}$

if $2n - 1 = 0 \Rightarrow n = \frac{1}{2}$

4. $y = be^{x/a}$. Prove that length of subtangent is constant.

Sol. $\frac{dy}{dx} = \frac{b}{a}e^{\frac{x}{a}}$

Length of subtangent $\Rightarrow \frac{y_1}{\frac{b}{a}e^{\frac{x_1}{a}}} = a = \text{constant}$

5. Show that for the curve $by^2 = (x + a)^3$ the square of the subtangent varies as the subnormal.

Sol. TPT $\left(\frac{y}{(dy/dx)}\right)^2 = k \cdot y\left(\frac{dy}{dx}\right)$ or $y = k\left(\frac{dy}{dx}\right)^3$

now $2by\frac{dy}{dx} = 3(x + a)^2 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{3(x+a)^2}{2by}$

hence $\frac{y}{(dy/dx)^3} = \frac{y\,8b^3y^3}{2y(x+a)^6} = \frac{8b^3}{2y}\left(\frac{y^2}{(x+a)^3}\right)^2 = \frac{8b^3}{27}\cdot\frac{1}{b^2} = \frac{8b}{27}$ Ans.

6. Show that at any point on the hyperbola $xy = c^2$, the subtangent varies as the abscissa and the subnormal varies as

the cube of the ordinate of the point of contact. **[O]**

Sol. $x\dfrac{dy}{dx}+y=0$

$\dfrac{dy}{dx}=\dfrac{-y_1}{x_1}$

Subtangent : $\left|\dfrac{y_1}{-\dfrac{y_1}{x_1}}\right|=|-x_1|$

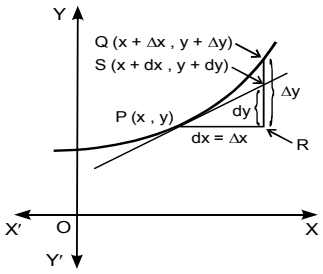
Subnormal : $\left|y_1\left(\dfrac{-y_1}{x_1}\right)\right|=\left|\dfrac{-y_1^3}{c^2}\right|$

6.1 APPROXIMATION

In this section, we will use differentials to approximate values of certain quantities. Let $f: D \to R$, $D \subset R$, be a given function and let $y = f(x)$. Let Δx denote a small increment in 'x'. Recall that the increment in y corresponding to the increment in x, denoted by Δy is given by $\Delta y = f(x + \Delta x) - f(x)$. We define the following :

(i) The differential of 'x', denoted by dx is defined by $dx = \Delta x$.
(ii) The differential of 'y', denoted by dy is defined by

$dy = f'(x)\,dx$ or $dy = \left(\dfrac{dy}{dx}\right)\Delta x$.



In case $dx = \Delta x$ is relatively small when compared with x, dy is a good approximation of Δy and we denote it by, $dy = \Delta y$.
For Geometrical meaning of Δx, Δx, dx and dy refer to the figure given above.

Q.1 Use differential to approximate, $\sqrt{36.6}$.

Ans. 6.05

Sol. Take $y = \sqrt{x}$. Let $x = 36$ and let $\Delta x = 0.6$. Then :

$\Delta y = \sqrt{x + \Delta x} - \sqrt{x} = \sqrt{36.6} - \sqrt{36} = \sqrt{36.6} - 6$

or $\sqrt{36.6} = 6 + \Delta y$

Now, dy is approximately equal to Δy and is given by,

$dy = \left(\dfrac{dy}{dx}\right)\Delta x = \dfrac{1}{2\sqrt{x}}(0.6) = \dfrac{1}{2\sqrt{36}}(0.6) = 0.05 \quad \left(\text{as } y = \sqrt{x}\right)$

Thus, the approximate value of $\sqrt{36.6}$ is $6 + 0.05 = 6.05$.

Q.2 Find the approximate value of square root of 25.2.

Ans 5.02

Sol. Let $f(x) = \sqrt{x}$

Now, $f(x + \Delta x) - f(x) = f'(x) \cdot \Delta x = \frac{\Delta x}{2\sqrt{x}}$

we may write, $25.2 = 25 + 0.2$

Taking $x = 25$ and $\Delta x = 0.2$, we have

$$f(25.2) - f(25) = \frac{0.2}{2\sqrt{25}}$$

or $f(25.2) - \sqrt{25} = \frac{0.2}{10} = 0.02 \quad \Rightarrow \quad f(25.2) = 5.02$

or $\sqrt{(25.2)} = 5.02$

6.2 RATE MEASURE

In various fields of applied mathematics one has the quest to know the rate at which one variable is changing, with respect to other. The rate of change naturally refers to time. But we can have rate of change with respect to other variables also.

An economist may want to study how the investment changes with respect to variations in interest rates.

A physician may want to know, how small changes in dosage can affect the body's response to a drug.

A physicist may want to know the rate of charge of distance with respect to time.

All questions of the above type can be interpreted and represented using derivatives.

Definition : The average rate of change of a function f(x) with respect to x over an interval [a, a + h] is defined as $\frac{f(a+h)-f(a)}{h}$.

Definition : The instantaneous rate of change of f(x) with respect to x is defined as $f'(x) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$, provided the limit exists.

Note : To use the word 'instantaneous', x may not be representing time. We usually use the word 'rate of change' to mean 'instantaneous rate of change'.

Q.1 How fast the area of a circle increases when its radius is 5cm;
(i) with respect to radius (ii) with respect to diameter

Ans. (i) 10π cm^2/cm (ii) 5π cm^2/cm

Sol. (i) $A = \pi r^2$, $\frac{dA}{dr} = 2\pi r$

$$\therefore \left.\frac{dA}{dr}\right]_{r=5} = 10\pi \text{ cm}^2/\text{cm}.$$

(ii) $A = \frac{\pi}{4}D^2$, $\frac{dA}{dD} = \frac{\pi}{2}D$

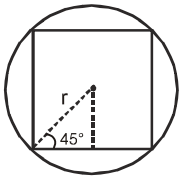
$$\therefore \left.\frac{dA}{dD}\right]_{D=10} = \frac{\pi}{2} \cdot 10 = 5\pi \text{ cm}^2/\text{cm}.$$

Q.2 If area of circle increases at a rate of 2cm^2/sec, then find the rate at which area of the inscribed square

increases.

Ans. $\frac{4}{\pi}$

Sol. Area of circle, $A_1 = \pi r^2$. Area of square, $A_2 = 2r^2$ (see figure)



$$\frac{dA_1}{dt} = 2\pi r \frac{dr}{dt}, \qquad \frac{dA_2}{dt} = 4r \cdot \frac{dr}{dt}$$

$$\because \quad 2 = 2\pi r \cdot \frac{dr}{dt} \quad \Rightarrow \quad r\frac{dr}{dt} = \frac{1}{\pi}$$

$$\therefore \quad \frac{dA_2}{dt} = 4 \cdot \frac{1}{\pi} = \frac{4}{\pi} \text{ cm}^2/\text{sec}$$

$\therefore$ Area of square increases at the rate $\frac{4}{\pi}$ cm^2/sec.

Q.3 The volume of a cube is increasing at a rate of 7 cm^3/sec. How fast is the surface area increasing when the length of an edge is 4 cm?

Ans. 7

Sol. Let at some time t, the length of edge is x cm.

$$v = x^3 \quad \Rightarrow \quad \frac{dv}{dt} = 3x^2 \frac{dx}{dt} \qquad \left(\text{but } \frac{dv}{dt} = 7\right)$$

$$\Rightarrow \quad \frac{dx}{dt} = \frac{7}{3x^2} \text{ cm/sec.}$$

Now $S = 6x^2$

$$\frac{dS}{dt} = 12x \frac{dx}{dt} \quad \Rightarrow \quad \frac{dS}{dt} = 12x \cdot \frac{7}{3x^2} = \frac{28}{x}$$

when x = 4 cm, $\frac{dS}{dt} = 7 \text{ cm}^2/\text{sec}$

Q.4 Sand is pouring from pipe at the rate of 12 cm^3/s. The falling sand forms a cone on the ground in such a way that the height of the cone is always one - sixth of radius of base. How fast is the height of the sand cone increasing when height is 4 cm?

Ans $\frac{1}{48\pi}$

Sol. $V = \frac{1}{3}\pi r^2 h$

but $h = \frac{r}{6}$

$$\Rightarrow \quad V = \frac{1}{3}\pi (6h)^2 h$$

$$\Rightarrow \quad V = 12\pi h^3$$

$$\frac{dV}{dt} = 36\pi h^2 \cdot \frac{dh}{dt}$$

when, $\frac{dV}{dt} = 12 \text{ cm}^3/\text{s}$ and $h = 4$ cm

$$\frac{dh}{dt} = \frac{12}{36\pi.(4)^2} = \frac{1}{48\pi} \text{ cm/sec.}$$

Q.5 If the area of circle increases at a uniform rate, show that the rate of increase of the perimeter varies inversely at its radius. **[O]**

Ans.

Sol. $A = \pi r^2 \quad \Rightarrow \quad \frac{dA}{dt} = 2\pi r \frac{dr}{dt} = c$

$P = 2\pi r \quad \Rightarrow \quad \frac{dP}{dt} = 2\pi \frac{dr}{dt}$

$\therefore \quad \frac{dA}{dt} = r \cdot \frac{dP}{dt} = c; \quad \frac{dP}{dt} = \frac{c}{r}$

Q.6 If the side of an equilateral triangle increases at the rate of $\sqrt{3}$ cm/sec and area increase at the rate 12 cm^2/sec then the side of the equilateral triangle is _____.

Ans. 8

Sol. $\frac{dA}{dt} = 12; \ \frac{dx}{dt} = \sqrt{3}$ cm/sec; x = ?

now $A = \frac{\sqrt{3}}{4}x^2$; $\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2x \cdot \frac{dx}{dt}$; $12 = \frac{\sqrt{3}}{2} \cdot \sqrt{3} \cdot x = \frac{3x}{2} \Rightarrow x = 8$ Ans.

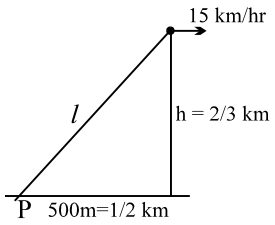
Q.7 An aeroplane is flying horizontally at a height of $\frac{2}{3}$ km with a velocity of 15 km/hr. Find the rate at which it is moving from a fixed point on the ground which it passed over 2 minutes ago.

Ans. 9 km/hr

[Sol. $l^2 = h^2 + x^2$

$2l \frac{dl}{dt} = 0 + 2x \frac{dx}{dt} \quad \Rightarrow \quad \frac{dl}{dt} = \frac{x}{l} \cdot \frac{dx}{dt}$

where $x = \frac{1}{2}$ km, $h = \frac{2}{3}$ km then $l = \frac{1}{4} + \frac{4}{9} = \frac{5}{6}$ km



$\therefore \quad \frac{dl}{dt} = \frac{1}{2} \cdot \frac{6}{5} \cdot 15 = 9$ km/ hr

Q.8 The height *h* of a right circular cone is 20 cm and is decreasing at the rate of 4 cm/sec. At the same time, the radius r is 10 cm and is increasing at the rate of 2 cm/sec. Find the rate of change of the volume in cm^3/sec.

Ans. $\frac{400\pi}{3}$

[Hint: $V = \frac{1}{3}\pi r^2 h$. differentiating w.r.t. time

$$\frac{dV}{dt} = \frac{\pi}{3}\left(r^2\frac{dh}{dt} + 2rh\frac{dr}{dt}\right);\ \text{when } h = 20,\ \frac{dh}{dt} = -4\ ;\ r = 10 \text{ and } \frac{dr}{dt} = 2; \text{ to find } \frac{dV}{dt}$$

$$= \frac{\pi}{3}[100\,(-4) + 2(10)(20)2] = \frac{\pi}{3}[400]$$

Q.9 On the curve $x^3 = 12y$, find the interval of values of x for which the abscissa changes at a faster rate than the ordinate? **[O]**

Ans. $x \in (-2, 2) - \{0\}$

Sol: Given $x^3 = 12y$, differentiating with respect to y

$$3x^2\frac{dx}{dy} = 12$$

$$\therefore \quad \frac{dx}{dy} = \frac{12}{3x^2}$$

The interval in which the abscissa changes at a faster rate than the ordinate, we must have

$$\Rightarrow \quad \left|\frac{dx}{dy}\right| > 1 \text{ or } \left|\frac{12}{3x^2}\right| > 1$$

$$\text{or} \quad \frac{4}{x^2} > 1 \Rightarrow \frac{4 - x^2}{x^2} > 0$$

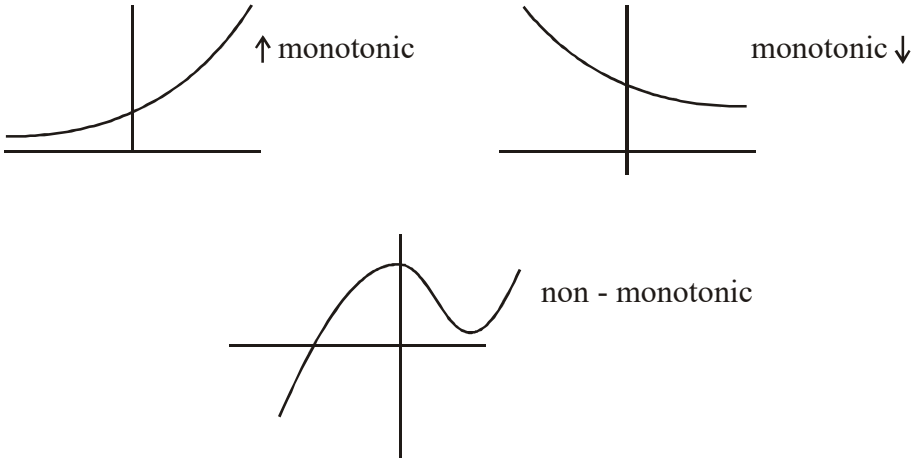
$$\Rightarrow x \in (-2, 2) - \{0\}.$$

Thus $x \in (-2, 2) - \{0\}$ is the required interval in which abscissa changes at a faster rate than the ordinate.

MONOTONOCITY

1.1 MONOTONIC FUNCTION

A function which is always increasing/decreasing in its domain.

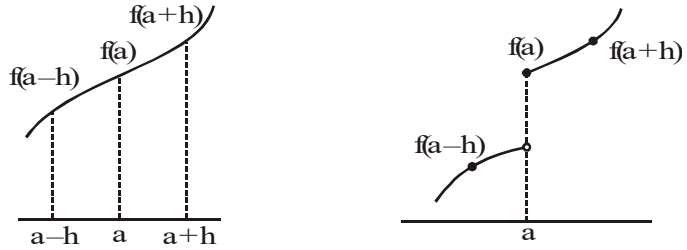


→ Increasing function : On moving left to right if graph of function moves upwards then it is called increasing function.

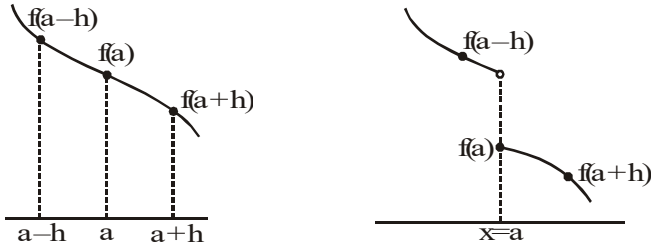
→ Decreasing function : On moving left to right if graph of function moves downwards then it is called decreasing function.

1. 2 MONOTONOCITY AT A POINT

(a) A function f(x) is called increasing function at point x = a, if in a sufficiently small neighbourhood of x = a ; $f(a-h) < f(a) < f(a+h)$



(b) A function f(x) is called a decreasing function at point x = a, if in a sufficiently small neighbourhood of x= a ; $f(a-h) > f(a) > f(a+h)$



Note :

(i) Monotonicity is defined in domain only.

(ii) To define monotonicity function need not to be continuous.

1.3 MONOTONOCITY IN AN INTERVAL

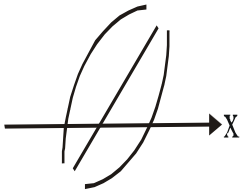
1. Increasing Function

$f(x)$ is said to be increasing in D_1 if for every $x_1, x_2 \in D_1$,

$x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$

It means that there is a certain increase in the value of $f(x)$ with an increase in the value of x .

[Refer to the adjacent figure]



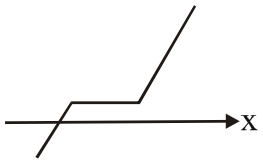
Increasing function

2. Non–Decreasing Function

$f(x)$ is said to be non–decreasing in D_1 if for every $x_1, x_2 \in D_1$,

$x_1 > x_2 \Rightarrow f(x_1) \geq f(x_2)$. It means that the value of $f(x)$ would never decrease with an increase in the value of x

[Refer to the adjacent figure]



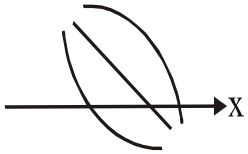
Non-decreasing function

3. Decreasing Function

$f(x)$ is said to be decreasing in D_1

if for every $x_1, x_2 \in D_1$, $x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$ it means that there is a certain decrease in the value of $f(x)$ with an increase in the value of x

[Refer to the adjacent figure]



Decreasing function

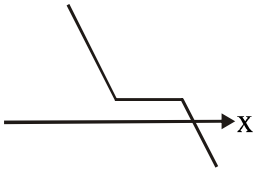
4. Non–increasing Function

$f(x)$ is said to be non–increasing in D_1

if for every $x_1 . x_2 \in D_1, x_1 > x_2 \Rightarrow f(x_1) \leq f(x_2)$.

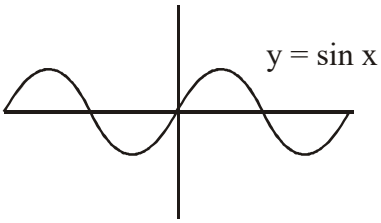
It means that the value of $f(x)$ would never increase with an increase in the value of x

[Refer to the adjacent figure]



Non-Increasing function

Example



(i) $\sin x$ is $\uparrow$ in $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$, True.

(ii) In $[-\pi, \pi]$

$\sin x$ is $\uparrow$ in $[-\pi, \pi]$, False.

2.1 DERIVATIVE APPROACH FOR MONOTONOCITY

1. Function f(x) is $\uparrow$ in interval I, then $\frac{dy}{dx} \geq 0 \ \forall \ x \in I$ {equality does not hold for an interval}

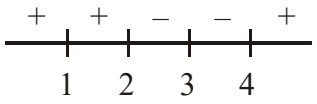
2. Function f(x) is $\downarrow$ in interval I, then $\frac{dy}{dx} \leq 0 \ \forall \ x \in I$ {equality does not hold for an interval}

$\Rightarrow$ To find intervals of monotonicity

$y = f(x)$

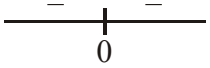
$\frac{dy}{dx} = f'(x)$ factorise.

Ex.: $f'(x) = (x-1)^2 (x-2) (x-3)^2 (x-4)$



$\uparrow$ in $(-\infty, 1] \cup [1, 2) \cup (4, \infty)$ $\downarrow$ in $(2, 3] \cup [3, 4)$

Ex. : $f(x) = \frac{1}{x}$



$f'(x) = \frac{-1}{x^2}$

$\downarrow$ in $(-\infty, 0) \cup (0, \infty)$ '0' is not in domain.

2.2 CRITICAL POINTS AND STATIONARY POINTS

Critical Points:

The points in domain of f(x) for which $f'(x)$ is equal to zero or doesn't exist are called critical points.

Q.1 Find critical points of following functions.

(i) $y = \sin x$ (ii) $y = \cos x$ (iii) $y = [x]$ (iv) $y = \{x\}$

(v) $y = \text{sgn}(x)$ (vi) $y = |x|$ (vii) $y = \tan x$ (viii) $y = e^x$

Ans. (i) $x=(2n+1)\frac{\pi}{2}$ (ii) $x=n\pi$ (iii) R (iv) I (v) R

(vi) $x=0$ (vii) ϕ (viii) ϕ

Sol. (i) $y=\sin x$

$$\frac{dy}{dx}=\cos x$$

as $\frac{dy}{dx}$ defined all over the number line so only $\frac{dy}{dx}=0$ will be checked.

$\Rightarrow \cos x = 0$

$$\Rightarrow x=(2n+1)\frac{\pi}{2}\forall n\in I$$

(ii) $y=\cos x$

$$\frac{dy}{dx}=-\sin x$$

as $\frac{dy}{dx}$ defined all over the number line so only $\frac{dy}{dx}=0$ will be checked.

$\Rightarrow -\sin x = 0 \Rightarrow \sin x = 0$

$\Rightarrow x = n\pi \ \forall \ n \in I$

(iii) $y=[x]$

as output of [x] is always integers (constants)

So, $\frac{dy}{dx}=0\forall x\in R-I$

also, $\forall x\in I$, $\frac{dy}{dx}=$ not defined

$\therefore$ Every $x \in R$ will be critical point.

(iv) $y=\{x\}$

$\forall \ x \notin I$ (for non integers)

$$\frac{dy}{dx}\neq 0$$

but $\forall x\in I$, $\frac{dy}{dx}=$ not defined

So, only integers will become critical points.

(v) $y=\text{sgn}(x)$

$$y=\begin{cases}-1, & x<0\\ 0; & x=0\\ 1; & x>0\end{cases}$$

$\frac{dy}{dx} = 0 \forall x \in R - \{0\}$

at $x = 0$, $\frac{dy}{dx}$ = not defined

$\therefore \forall\ x \in R$ will be critical point for sgn (x).

(vi) $y = |x|$

only at $x = 0$, $\frac{dy}{dx}$ = not defined.

$\therefore$ critical point is only $x = 0$.

(vii) $y = \tan x$

for function tan x domain $\in R - \left\{(2n+1)\frac{\pi}{2} \forall n \in I\right\}$

Now,

$\frac{dy}{dx} = \sec^2 x$

$\frac{dy}{dx} \neq 0 \forall x \in R$

Also, $x \in (2n+1)\frac{\pi}{2} \forall n \in I$ is not in domain.

$\therefore$ No critical point exist for $y = \tan x$

$\Rightarrow x \in \phi$

(viii) $\frac{dy}{dx} = e^x \neq 0 \forall x \in R$

No critical point exist.

$x \in \phi$

Stationary Points

The stationary points are the points of the domain where $f'(x) = 0$.

Questions

Q.2 Find Interval of Monotonocity

(i) $y = x^2 (x-2)^2$ (ii) $y = x^2 e^x$ (iii) $y = x \ln x$ (iv) $y = \frac{x}{\ln x}$ (v) $y = x + \ln (1 - 4x)$

(vi) $y = \sin x$ in $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (vii) $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x - 3$ in $[0, \pi]$

Ans

(i) increasing in $(0,1) \cup (2,\infty)$, decreasing in $(-\infty, 0) \cup (1,2)$

(ii) increasing in $(-\infty,-2) \cup (0,\infty)$, decreasing in $(-2, 0)$

(iii) increasing in $(1/e, \infty)$, decreasing in $(0, 1/e)$

(iv) increasing in (e, ∞), decreasing in $(0, 1) \cup (1,e)$

(v) increasing in $(-\infty, -3/4)$, decreasing in $(-3/4, 1/4)$

(vi) increasing in $[-\pi/2, \pi/2)$, decreasing in $(\pi/2, 3\pi/2]$

(vii) decreasing in $[0, \pi/2) \cup (2\pi/3, \pi]$ increasing in $(\pi/2, 2\pi/3)$

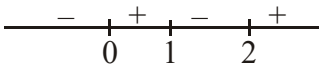
Sol. (i) $y = x^2 (x-2)^2$

$$\frac{dy}{dx} = 2x(x-2)^2 + x^2\, 2(x-2)$$

$$= 2x(x-2)[x-2+x]$$

$$= 2x(x-2)\,2(x-1) = 4\cdot x\cdot(x-1)(x-2)$$

on number line



I.O.D. $x \in (-\infty, 0) \cup (1, 2)$

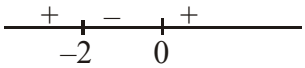
I.O.I. $x \in (0, 1) \cup (2, \infty)$

(ii) $y = x^2 \cdot e^x$

$$\frac{dy}{dx} = 2x\cdot e^x + x^2\cdot e^x = e^x \cdot x(x+2)$$

$$\frac{dy}{dx} = 0 \text{ at } x = 0, -2$$

on number line



I.O.D. $x \in (-2, 0)$

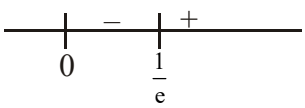
I.O.I. $x \in (-\infty, -2) \cup (0, \infty)$

(iii) $y = x\,\ell nx$

$$\frac{dy}{dx} = x\cdot\frac{1}{x} + \ell nx = 1 + \ell nx$$

$$\frac{dy}{dx} = 0 \text{ at } x = \frac{1}{e}$$

on number line



I.O.D. $x \in \left(0, \frac{1}{e}\right)$

I.O.I. $x \in \left(\frac{1}{e}, \infty\right)$

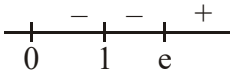
(iv) $y = \frac{x}{\ell nx}$

$$\frac{dy}{dx}=\frac{(\ell nx)\cdot(1)-x\cdot\left(\frac{1}{x}\right)}{(\ell nx)^2}$$

$$\frac{dy}{dx}=\frac{\ell nx-1}{(\ell nx)^2}$$

$\frac{dy}{dx}=0$ at x = e and not defined at x = 1.

on number line



I.O.D. $x \in (0, 1) \cup (1, e)$

I.O.I. $x \in (e, \infty)$

(v) $y = x + \ell n\,(1 - 4x)$

$1 - 4x > 0$

$4x - 1 < 0$

$$x < \frac{1}{4}$$

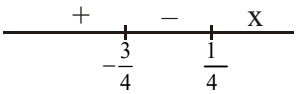
$$D_f \in \left(-\infty, \frac{1}{4}\right)$$

$$\frac{dy}{dx}=1+\frac{1}{(1-4x)}\times(-4)=1-\frac{4}{(1-4x)}$$

$$\frac{dy}{dx}=\frac{1-4x-4}{(1-4x)}=\frac{-(4x+3)}{(1-4x)}=\frac{4x+3}{4x-1}$$

$\frac{dy}{dx}=0$ at $x=-\frac{3}{4}$ and not defined at $x=\frac{1}{4}$

on number line



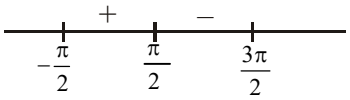
I.O.D. $x \in \left(-\frac{3}{4}, \frac{1}{4}\right)$

I.O.I. $x \in \left(-\infty, -\frac{3}{4}\right)$

(vi) $y = \sin x$ in $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$

$$\frac{dy}{dx} = \cos x; x \in \left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$$\frac{dy}{dx} = 0 \text{ at } x = -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$$



I.O.D. $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right]$

I.O. I. $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(vii) $f(x) = 3\cdot\cos^4 x + 10\cdot\cos^3 x + 6\cdot\cos^2 x - 3$ in $[0, \pi]$

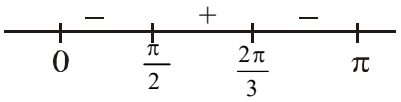
$f'(x) = 12\cos^3 x(-\sin x) + 30\cos^2 x(-\sin x) + 12\cos x(-\sin x)$

$$f'(x) = -12\sin x\cos x\left[\cos^2 x + \frac{5}{2}\cos x + 1\right]$$

$f'(x) = -3\sin 2x(2\cos x + 1)(\cos x + 2)$

$f'(x) = 0 \Rightarrow \sin 2x = 0$ or $\cos x = -\frac{1}{2}$ or $\cos x = -2$ (Not possible)

on number line



I.O.D. $x \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \pi\right]$

I.O.I. $x \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

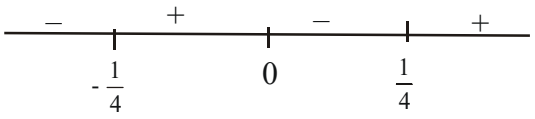
3. If S is the set, such that $f(x) = 8x^2 - \ln|x|$ increases in S, then S is

Ans $x \in \left(-\frac{1}{4}, 0\right) \cup \left(\frac{1}{4}, \infty\right)$

Sol $f'(x) = 16x - \frac{1}{x} = \frac{16}{x}\left(x^2 - \frac{1}{16}\right)$

For an increasing function. $f'(x) \geq 0 \Rightarrow \frac{1}{x}\left(x - \frac{1}{4}\right)\left(x + \frac{1}{4}\right) \geq 0$

Sign scheme for $f'(x)$:



$$\Rightarrow x \in \left[-\frac{1}{4}, 0\right) \cup \left[\frac{1}{4}, \infty\right)$$

4. Let $f(x) = 2x - \tan^{-1}x - \ln\left(x + \sqrt{1+x^2}\right)$; $x \in R$. Then find interval of monotonocity

Ans $x \in R$

Sol $f'(x) = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}} \geq 0 \; \forall x \in R$

and equality holds at $x = 0$ only.
So, f(x) is increasing in $(-\infty, \infty)$

5. Discuss the monotonicity of g(x), where $g(x) = 2f\left(\frac{x^2}{2}\right) + f(6 - x^2)$ for all $x \in R$. It is given that $f''(x) > 0$ for all $x \in R$.

Ans. $\uparrow (-2, 0) \cup (2, \infty)$, $\downarrow (-\infty, -2) \cup (0, 2)$

Sol. Given : $f''(x) > 0 \; \forall \; x \in R$ ...(1)
$\Rightarrow f'(x)$ is increasing $\forall \; x \in R$
and,

$$g(x) = 2f\left(\frac{x^2}{2}\right) + f\left(6 - x^2\right) \forall x \in R \quad ...(2)$$

$$g'(x) = 2 \cdot f'\left(\frac{x^2}{2}\right) \cdot \frac{2x}{2} + f'\left(6 - x^2\right)(-2x)$$

$$g'(x) = 2x\left[f'\left(\frac{x^2}{2}\right) - f'\left(6 - x^2\right)\right]$$

for I.O.I

$$2x\left[f'\left(\frac{x^2}{2}\right) - f'\left(6 - x^2\right)\right] > 0$$

C-I : $x > 0$ ...(3)

and $f'\left(\frac{x^2}{2}\right) - f'\left(6 - x^2\right) > 0$

$$\Rightarrow f'\left(\frac{x^2}{2}\right) > f'\left(6 - x^2\right)$$

$\frac{x^2}{2} > 6 - x^2$ (as $f'(x)$ is increasing)

$x^2 > 12 - 2x^2$
$3x^2 - 12 > 0$

$x^2 - 4 > 0$

$(x - 2)(x + 2) > 0$

$x \in (-\infty, -2) \infty (2, \infty)$ ...(4)

from (3) & (4)

$x \in (2, \infty)$

C-II : $x < 0$ ...(5)

and $f'\left(\frac{x^2}{2}\right) - f'\left(6 - x^2\right) < 0$

$\Rightarrow f'\left(\frac{x^2}{2}\right) < f'\left(6 - x^2\right)$

$3x^2 - 12 < 0$

$x^2 - 4 < 0$

$(x - 2)(x + 2) < 0$

$\Rightarrow x \in (-2, 2)$ ...(6)

from (5) & (6)

$x \in (-2, 0)$

$\therefore$ I.O.I. $x \in (-2, 0) \cup (2, \infty)$

for I.O.D

$2x\left[f'\left(\frac{x^2}{2}\right) - f'\left(6 - x^2\right)\right] < 0$

C–I : $x > 0$ ...(7)

and $f'\left(\frac{x^2}{2}\right) - f'\left(6 - x^2\right) < 0$

$\frac{x^2}{2} < 6 - x^2$

$3x^2 - 12 < 0$

$x^2 - 4 < 0$

$\Rightarrow x \in (-2, 2)$ ...(8)

from (7) & (8)

$x \in (0, 2)$

C–II : $x < 0$ ...(9)

and $f'\left(\frac{x^2}{2}\right) > f'\left(6 - x^2\right)$

$\Rightarrow 3x^2 - 12 > 0$

$x^2 - 4 > 0$

$\Rightarrow x \in (-\infty, -2) \cup (2, \infty)$ ...(10)

from (9) & (10)

$x \in (-\infty, -2)$

$\therefore$ I.O.D. $x \in (-\infty, -2) \cup (0, 2)$

6. Let $f : R \to R$ be defined as

$$f(x)=\begin{cases}-55x, & \text{if } x<-5\\ 2x^3-3x^2-120x, & \text{if } -5\le x\le 4\\ 2x^3-3x^2-36x-336, & \text{if } x>4,\end{cases}$$

Let $A=\{x\in R: f \text{ is increasing}\}$. Then A is equal to : **[JEE(Main)Feb.2021]**

Ans. $(-5,-4)\cup(4,\infty)$

Sol. $f(x)$

$\rightarrow -55x \qquad x<-5$

$\rightarrow 2x^3-3x^2-120x \qquad -5\le x\le 4$

$\rightarrow 2x^3-3x^2-36x-336 \qquad x>4$

$f'(x)$

$\rightarrow -55 \qquad x<-5$

$\rightarrow 6x^2-6x-120 \qquad -5<x<+4$

$\rightarrow 6x^2-6x-36 \qquad x>4$

C – I $\qquad x<-5$

$f'(x)<0$

$\Rightarrow f$ is decreasing

C – II $\qquad -5<x<4$

$f'(x)>0 \Rightarrow x\in(-\infty,-4)\cup(5,\infty)$

$\Rightarrow x\in(-5,-4)\Rightarrow f$ is increasing

C – III $\qquad x>4$

$f'(x)>0$

$6(x^2-x-6)>0\Rightarrow x\in(-\infty,-2)\cup(3,\infty)$

$\Rightarrow x\in(4,\infty)\Rightarrow f$ is increasing

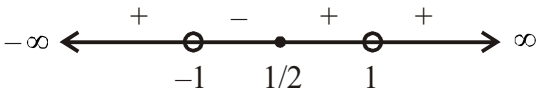
7. Let *f* be a real valued function, defined on $R-\{-1,1\}$ and given by $f(x)=3\log_e\left|\frac{x-1}{x+1}\right|-\frac{2}{x-1}$. Then in which of the following intervals, function *f*(x) is increasing ? **[JEE(Main)Mar.2021]**

Ans. $(-\infty,-1)\cup\left(\left[\frac{1}{2},\infty\right)-\{1\}\right)$

Sol. $f(x)=3\ln(x-1)-3\ln(x+1)-\frac{2}{(x-1)}$

$$f'(x)=\frac{3}{x-1}-\frac{3}{x+1}+\frac{2}{(x-1)^2}$$

$$f'(x) = \frac{4(2x-1)}{(x-1)^2(x+1)} \geq 0$$



$$x \in (-\infty,-1) \cup \left(\left[\frac{1}{2},\infty\right) - \{1\}\right)$$

8. Let $g(x) = f(x) + f(1-x)$ and $f''(x) > 0, x \in (0,1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{-1}(2\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right) + \tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to :

माना $g(x) = f(x) + f(1-x)$ तथा $f''(x) > 0, x \in (0,1)$ है। यदि g अंतराल $(0, \alpha)$ में ह्रासमान है तथा अंतराल $(\alpha, 1)$ में वर्धमान है, तो $\tan^{-1}(2\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right) + \tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ बराबर है :

Ans. π

Sol. $g'(x) = f'(x) - f'(1-x)$

If $x > 1-x$

$\Rightarrow f'(x) > f'(1-x) \because f'(x)$ is strictly increasing

$\Rightarrow g'(x) > 0 \Rightarrow$ so g(x) is increasing when $x > 1-x \Rightarrow x > \frac{1}{2}$

$\Rightarrow \alpha = \frac{1}{2}$

similarly when $x < 1-x \Rightarrow g(x)$ is decreeing

Now

$$\tan^{-1}(2\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right) + \tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$$

$$= \tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3) = \pi (\because 1+2+3 = 1.2.3)$$

2.3 ALWAYS INCREASING/DECREASING BASED QUESTIONS

1. Find range of a if $f(x) = ax - \sin x$ is always increasing/decreasing

Ans: $\uparrow$ for $a \geq 1$ & $\downarrow$ for $a \leq -1$

Sol. $f(x) = ax - \sin x$ ...(1)

$f'(x) = a - \cos x$

for always increasing

$f'(x) \geq 0$

$a - \cos x \leq 0$

$a \geq [-1, 1]$

a will be greater than maximum value of interval

$a \geq 1$

for always decreasing

$f'(x) \leq 0$

$a \leq [-1, 1]$

a will be less than minimum value of interval

$a \leq -1$

2. If the function $f(x) = (a+2)x^3 - 3ax^2 + 9ax - 1$ is always decreasing $\forall\, x \in R$, find 'a'.

Ans : $(-\infty, -3]$

Sol. $f(x) = (a+2)x^3 - 3ax^2 + 9ax - 1$

$f'(x) = 3(a+2)x^2 - 6ax + 9a$ .....(i)

ATQ $f'(x) \leq 0$

$3(a+2)x^2 - 6ax + 9a < 0$

L.C. $< 0 \Rightarrow a + 2 < 0$

$a < -2$ .....(ii)

and $D \leq 0$

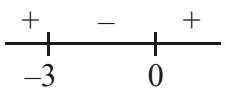
$36a^2 - 4(9a)3(a+2) \leq 0$

$36a[a - 3a - 6] \leq 0$

$36a[-2a - 6] \leq 0$

$-72a(a+3) \leq 0$

$a(a+3) \geq 0$



$a \in (-\infty, -3] \cup [0, \infty)$ ...(3)

from Inequality (2) & (3)

$a \in (-\infty, -3]$

3. Find the range of values of 'a' for which the function $f(x) = x^3 + (2a+3)x^2 + 3(2a+1)x + 5$ is monotonic in R. Hence find the set of values of 'a' for which f(x) in invertible.

Ans. $0 \leq a \leq 3/2$

Sol. $f(x) = x^3 + (2a+3)x^2 + 3(2a+1)x + 5$

$f'(x) = 3x^2 + 2(2a+3)x + 3(2a+1)$

L.C. $= 3 > 0$

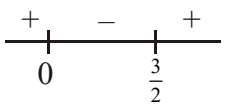
$f'(x) \geq 0 \Rightarrow D \leq 0$

$4(2a+3)^2 - 4 \times 9\,(2a+1) \leq 0$

$4a^2 + 12a + 9 - 18a - 9 \leq 0$

$4a^2 - 6a \leq 0$

$2a(2a-3) \leq 0$



$a \in \left[0, \frac{3}{2}\right]$

4. Prove that $f(x) = \frac{2}{3}x^9 - x^6 + 2x^3 - 3x^2 + 6x - 1$ is always increasing.

Sol. $f(x)=\frac{2}{3}x^9-x^6+2x^3-3x^2+6x-1$ ...(1)

$f'(x)=\frac{2}{3}\cdot 9\cdot x^8-6\cdot x^5+6\cdot x^2-6x+6$

$= 6x^8 - 6x^5 + 6x^2 - 6x + 6$

$f'(x) = 6[x^8 - x^5 + x^2 - x + 1]$

$\because f'(x) > 0 \ \forall \ x \in R$

$\therefore$ f(x) is always increasing.

Hence proved.

5. Prove that $f(x)=\int_{x^2}^{x^3}\frac{dt}{\ln t}$, $(x>0)$ is always an increasing function of x.

Ans. $\uparrow$ in $(0, 1) \cup (1, \infty)$

Sol $f'(x)=\frac{3x^2}{\ln x^3}-\frac{1\cdot 2x}{\ln x^2}=\frac{3x^2}{3\ln x}-\frac{2x}{2\ln x}=\frac{x^2}{\ln x}-\frac{x}{\ln x}=\frac{x^2-x}{\ln x}=\frac{x(x-1)}{\ln x}$

for $(0, 1) \cup (1, \infty)$ $f'(x) > 0 \ \rightarrow$ f is increasing]

6. If $f(x) = (a + 1) \ln x - \frac{x^2}{2}$ is always decreasing then find range of a.

Ans. $a \le -1$

Sol. $f(x)=(a+1)\ell nx-\frac{x^2}{2}$ ...(1)

$f'(x)=\frac{a+1}{x}-x$ ...(2)

ATQ,

$f'(x) \le 0$ (given always decreasing)

$\frac{a+1}{x}-x\le 0$

$\Rightarrow \frac{a+1}{x}\le x$

$\Rightarrow a + 1 \le x^2$ (By cross multiplication due to $x > 0$)

$a + 1 \le [0, \infty)$

$\Rightarrow a + 1 \le 0 \Rightarrow a \le -1$

7. If $f(x) = \sin^4 x + \cos^4 (x) + 2 bx + 3$ is always increasing and do not have any critical point then find range of b.

Ans. $b > 1/2$

Sol $f(x) = \sin^4 x + \cos^4 x + 2bx + 3$

$f'(x) = 4 \sin^3 x \cos x - 4\cos^3 x \sin x + b = - \sin 4x + 2b.$

	$f'(x) > 0$	for all	$x \in R$
$\Leftrightarrow$	$2b > \sin 4x$	for all	$x \in R$
$\Leftrightarrow$	$2b > 1$		

8. If $f(x) = e^{2x} - (a + 1)e^x + 2x$ is always increasing then find range of a.

Ans. $a \le 3$

Sol $f(x) = e^{2x} - (a+1)e^x + 2x$

$f'(x) = 2e^{2x} - (a+1)e^x + 2$

Now, $2e^{2x} - (a+1)e^x + 2 \ge 0$ for all $x \in R$

$\Rightarrow 2\left(e^x + \frac{1}{e^x}\right) - (a+1) \ge 0$ for all $x \in R$

$(a+1) \le 2\left(e^x + \frac{1}{e^x}\right)$ for all $x \in R$

$\Rightarrow a + 1 \le 4 \quad \left(\because \ e^x + \frac{1}{e^x} \text{ has minimum value } 2\right)$

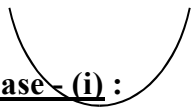
$\Rightarrow a \le 3$

Aliter (Using graph)

$2e^{2x} - (a+1)e^x + 2 \ge 0$ for all $x \in R$

putting $e^x = t \quad ; \ t \in (0, \infty)$

$2t^2 - (a+1)t + 2 \ge 0$ for all $t \in (0, \infty)$



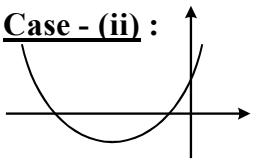
Case - (i) : $D \le 0$

$\Rightarrow (a+1)^2 - 4 \le 0$

$\Rightarrow (a+5)(a-3) \le 0$

$\Rightarrow a \in [-5, 3]$

or



Case - (ii) : both roots are non positive

$D \ge 0 \ \& \ -\frac{b}{2a} < 0 \ \& \ f(0) \ge 0$

$\Rightarrow a \in (-\infty, -5] \cup [3, \infty) \ \& \ \frac{a+1}{4} < 0 \ \& \ 2 \ge 0$

$\Rightarrow a \in (-\infty, -5] \cup [3, \infty) \ \& \ a < -1 \ \& \ a \in R$

$\Rightarrow a \in (-\infty, -5]$

Taking union of (i) and (ii), we get $a \in (-\infty, 3]$.

9. The set of integral value(s) of 'b' for which $f(x) = \sin 2x - 8(b+2)\cos x - (4b^2 + 16b + 6)x$ is monotonic decreasing for $\forall \ x \in R$ and has no critical point, is

Ans $\left(-\infty, -\left(3+\sqrt{3}\right)\right) \cup \left(\sqrt{3}-1, \infty\right)$

[Sol.: $f'(x) = 2\cos 2x + 8(b+2)\sin x - (4b^2 + 16b + 6)$

$= 2(1 - 2\sin^2 x) + 8(b+2)\sin x - (4b^2 + 16b + 6)$

$= -4[\sin^2 x - 2(b+2)\sin x + (b^2 + 4b + 1)]$

for monotonic decreasing and no critical points $f'(x) < 0 \quad \forall \quad x \in R$

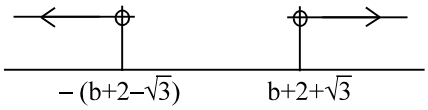
hence $\sin^2 x - 2(b+2)\sin x + b^2 + 4b + 1 > 0$

$[\sin x - (b+2)]^2 - (b+2)^2 + 4b + b^2 + 1 > 0$

$[\sin x - (b+2)]^2 - \left(\sqrt{3}\right)^2 > 0$

$[\sin x - b - 2 - \sqrt{3}][\sin x - b - 2 + \sqrt{3}] > 0$

$[\sin x - (b + 2 + \sqrt{3})][\sin x - (b + 2 - \sqrt{3})] > 0$



Hence $\sin x > b + (2 + \sqrt{3})$

or $b + (2 + \sqrt{3}) < \sin x \quad \forall\, x \in R$

$\therefore \quad b + (2 + \sqrt{3}) < -1$

$b < -(3 + \sqrt{3}) \Rightarrow \mathbf{b} \in \left(-\infty, -\left(3+\sqrt{3}\right)\right)$

|| ly $\sin x < (b + 2 - \sqrt{3})$

or $(b + 2 - \sqrt{3}) > \sin x \quad \forall\, x \in R.$

$b + 2 - \sqrt{3} > 1$

$b > -1 + \sqrt{3}$

$b > \sqrt{3} - 1 \Rightarrow \mathbf{b} \in \left((\sqrt{3}-1), \infty\right)$. Now finally $\left(-\infty, -\left(3+\sqrt{3}\right)\right) \cup \left(\sqrt{3}-1, \infty\right)$]

10. Consider the function,

$f(x) = x^3 - 9x^2 + 15x + 6$ for $1 \le x \le 6$ and $g(x) = \begin{cases} \min . f(t) & \text{for } 1 \le t \le x,\ 1 \le x \le 6 \\ x - 18 & \text{for } x > 6 \end{cases}$ then which of the following hold(s) good? **[O]**

(A) g(x) is differentiable at x = 1 (B) g(x) is discontinuous at x = 6

(C) g(x) is continuous and derivable at x =5 (D) g(x) is monotonic in (1, 5)

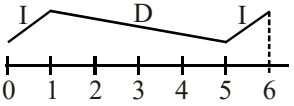
Ans ABCD

[Sol. $f(x) = x^3 - 9x^2 + 15x + 6$

$f(t) = t^3 - 9t^2 + 15t + 6$

$f'(t) = 3t^2 - 18t + 15 = 3[t^2 - 6t + 5] = 3(t-5)(t-1)$

hence f is increasing in (5, 6) and f is decreasing in (1, 5)



now $g(x) = \begin{cases} f(x) = x^3 - 9x^2 + 15x + 6 & 1 \le x < 5 \\ f(5) = -19 & 5 \le x \le 6 \\ x - 18 \quad \text{if} & x > 6 \end{cases}$

$\therefore \quad g(x) = \begin{cases} x^3 - 9x^2 + 15x + 6 & \text{if} \quad 1 \le x < 5 \\ -19 & \text{if} \quad 5 \le x \le 6 \\ x - 18 & \text{if} \quad x > 6 \end{cases}$

Hence g is continuous and differentiable at x = 1

g is continuous and differentiable at x = 5

g is neither continuous nor derivable at x = 6]

11. Given f : [0, ∞) → R be a strictly increasing function such that the functions g (x) = f(x) – 3x and h (x) = f(x) – x^3 are both strictly increasing function. Then prove that function F (x) = f(x) – x^2 – x is always increasing. **[O]**

[Sol. $3F(x) = 3[f(x) - x^2 - x] = \underbrace{2[f(x) - 3x]}_{g(x)} + \underbrace{\left(f(x) - x^3\right)}_{h(x)} + x^3 - 3x^2 + 3x - 1 + 1$

$3F(x) = 2g(x) + h(x) + (x-1)^3 + 1 \quad \Rightarrow \quad F(x)$ is increasing $\forall\ x \in [0, \infty)$

Alternatively: Given f(x) is increasing $\Rightarrow$ $f'(x) > 0$

$g(x) = f(x) - 3x$

$g'(x) = f'(x) - 3 > 0 \Rightarrow f'(x) > 3$

also $h(x) = f(x) - x^3$

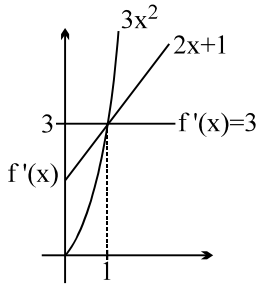
$h'(x) = f'(x) - 3x^2 > 0 \Rightarrow f'(x) > 3x^2$

to prove that $F(x) = f(x) - x^2 - x$ is increasing

i.e $F'(x) = f'(x) - 2x - 1 > 0$

$F'(x) > 0$

$f'(x) > 2x + 1$



for in [0, 1), obviously

$f'(x) > 3 > 2x + 1$ and in (1, 2), $3x^2 > 2x + 1$ hence proved.]

12. Let $f: (0, \infty) \to (0, \infty)$ be a derivable function and F (x) is the primitve of f(x) such that $2(F(x) - f(x)) = f^2(x)$ for any real positive x.then **[O]**

(i) Prove that f is strictly increasing (ii) Prove that $\lim\limits_{x\to\infty} \frac{f(x)}{x} = 1$

Sol. (a) Given $2(F(x) - f(x)) = f^2(x)$ and $\frac{dF}{dx} = f(x)$

$\therefore \quad F(x) = \frac{f^2(x)}{2} + f(x)$; $F'(x) = f(x) \cdot f'(x) + f'(x)$

$\therefore \quad f(x) = f'(x)\,(1 + f(x))$

$\therefore \quad f'(x) = \frac{f(x)}{1+f(x)} = 1 - \frac{1}{1+f(x)} > 0 \quad (\text{as } f(x) > 0)$

Hence f is strictly increasing.

(b) $\lim\limits_{x\to\infty} \frac{f(x)}{x}$

or $\lim\limits_{x\to\infty} f'(x)$ using L'Hospital's Rule

$= \lim\limits_{x\to\infty} \left(1 - \frac{1}{1+f(x)}\right)$ as $x \to \infty$; $f(x) \to \infty$; $\therefore \lim\limits_{x\to\infty} \frac{f(x)}{x} = 1$

13. Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval (a, a + 1). Then, |a| is equal to _____. **[JEE(Main)Feb.2021]**

Ans. 2

Sol. Let $P(x) = 2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$

$P'(x) = 10x^4 + 20x^3 + 30x^2 + 20x + 10$

$P'(x) = 10x^2\left[x^2 + 2x + 3 + \frac{2}{x} + \frac{1}{x^2}\right]$

$= 10x^2\left[x^2 + \frac{1}{x^2} + 2\left(x + \frac{1}{x}\right) + 1\right]$

$= 10x^2\left[\left(x + \frac{1}{x}\right)^2 + 2\left(x + \frac{1}{x}\right) + 1\right]$

$= 10x^2\left[x+\frac{1}{x}+1\right]^2$

$P'(x) = 10\,[x^2 + x + 1]^2 > 0$

P(x) is an odd degree increasing polynomial function

$\Rightarrow$ only one real root

$P(0) = 10$

$P(-1) = -2 + 5 - 10 + 10 - 10 + 10 = 3$

$P(-2) = -64 + 80 - 80 + 40 - 20 + 10 = -34$

$P(-1)\,P(-2) < 0$

Using Intermediate value theorem

$\Rightarrow P(x) = 0$ has at least on root in the $(-2, -1)$

Since $P(x) = 0$ has exactly one real root

$\therefore$ all real roots lie in the interval $(-2, -1)$

$\therefore\ a = -2 \qquad |a| = 2.$

3.1 GREATEST AND LEAST VALUE OF A FUNCTION

For continuous function f(x) greatest and least value exist either at critical points or boundary points.

1. Find the greatest and the least values of the continuous function given below in the indicated intervals.

(i) $f(x) = e^{x^2-4x+3}$ in $[-5, 5]$

(ii) $y = \int_{\frac{5\pi}{4}}^{x}(3\sin t + 4\cos t)dt$ in $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

(iii) $y = \int_{\frac{5\pi}{3}}^{x}(6\cos u - 2\sin u)du$ in $\left[\frac{5\pi}{3}, \frac{7\pi}{4}\right]$

(iv) $f(x) = \cos 3x - 15\cos x + 8$ in $\left[\frac{\pi}{3}, \frac{3\pi}{2}\right]$

Ans. (i) M at $x = -5$ & $f(-5) = e^{48}$; m at $x = 2$ & $f(2) = \frac{1}{e}$]

(ii) $\frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$ at $x = \frac{4\pi}{3}$ which is the least value ; greatest value = zero]

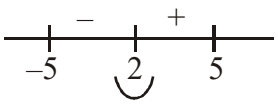
(iii) y is $\uparrow$ hence max. occurs at $\frac{7\pi}{4} = 3\sqrt{3} - 2\sqrt{2} - 1$. Min. Value = 0]

(iv) Max. at $x = \pi$, $y = 22$ & Min. at $x = \frac{\pi}{3}$, $y = -\frac{1}{2}$]

Sol. (i) $f(x) = e^{x^2-4x+3}$

$f'(x) = e^{x^2-4x+3}(2x-4)$

$f'(x) = 0$ at $x = 2$



minima at $x = 2$

$f(-5) = e^{48} \rightarrow$ maximum

$f(2) = \frac{1}{e} \rightarrow$ minimum

$f(5) = e^{8}$

(ii) $y = \int_{\frac{5\pi}{4}}^{x} (3\sin t + 4\cos t)\,dt$ in $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

$$y = \left(3(-\cos t) + 4\sin t\right)_{\frac{5\pi}{4}}^{x}$$

$$f'(x) = \frac{dy}{dx} = 3\sin x + 4\cos x$$

for critical point

$$\left.\begin{array}{c} f'(x) = 0 \\ 3\sin x = -4\cos x \\ \tan x = -\frac{4}{3} \end{array}\right\} x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$$

$\forall x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

$f'(x) < 0$

$f(x)$ is decreasing

$$m = f\left(\frac{4\pi}{3}\right) = \left\{3\left(-\cos\frac{4\pi}{3}\right) + 4\sin\frac{4\pi}{3}\right\} - \left\{-3\cos\frac{5\pi}{4} + 4\sin\frac{5\pi}{4}\right\}$$

$$= \frac{3}{2} - \frac{4\sqrt{3}}{2} + \frac{1}{\sqrt{2}}$$

$$M = f\left(\frac{5\pi}{4}\right) = 0$$

(iii) $y = \left(6\sin u + 2\cos u\right)_{\frac{5\pi}{3}}^{x}$

$$y = 6\sin x + 2\cos x - \left(6\sin\frac{5\pi}{3} + 2\cos\frac{5\pi}{3}\right)$$

$$y = 6\sin x + 2\cos x - \left(-3\sqrt{3} + 1\right) \qquad ...(1)$$

$\frac{dy}{dx} = 6\cos x - 2\sin x$

$f'(x) = 0$

$3\cos x = \sin x$

$3 = \tan x$

$x \notin \left[\frac{5\pi}{3}, \frac{7\pi}{4}\right]$

No critical point

$f'(x) \geq 0$

$\therefore$ f(x) is increasing

$m = f\left(\frac{5\pi}{3}\right) = 0$

$M = f\left(\frac{7\pi}{4}\right) = 6\sin\frac{7\pi}{4} + 2\cos\frac{7\pi}{4} + 3\sqrt{3} - 1$

$= 3\sqrt{3} - 1 - \frac{4}{\sqrt{2}}$

(iv) $f(x) = \cos 3x - 15\cos x + 8$ in $\left[\frac{\pi}{3}, \frac{3\pi}{2}\right]$

$f'(x) = -3\sin 3x + 15\sin x$

$f'(x) = 0 \Rightarrow \sin 3x - 5\sin x = 0$

$3\sin x - 4\sin^3 x - 5\sin x = 0$

$-2\sin x - 4.\sin^3 x = 0$

$-2\sin x\,[1 + 2.\sin^2 x] = 0$

$\sin x = 0$ or $\sin^2 x = -\frac{1}{2}(\text{N.P})$

$\rightarrow x = \pi$

$M = f(\pi) = -1 + 15 + 8 = 22$

$m = f\left(\frac{\pi}{3}\right) = \frac{-1\times 2}{2} - \frac{15}{2} + \frac{8\times 2}{2} = -\frac{1}{2}$

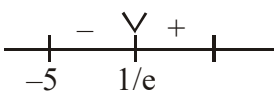
$f\left(\frac{3\pi}{2}\right) = 8$

2 Find minimum value of $y = x^x$ $(x > 0)$

Ans. $(1/e)^{1/e}$

Sol. $f'(x) = \frac{dy}{dx} = x^x(1 + \ell nx)$

$f'(x) = 0 \rightarrow x = \frac{1}{e}$



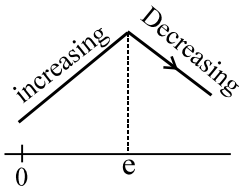
Clearly minima at $x = \frac{1}{e}$

minimum value $= \left(\frac{1}{e}\right)^{\frac{1}{e}}$

3 Use the function $f(x) = x^{\frac{1}{x}}$ $(x > 0)$ to ascertain whether π^e or e^π is greater.

Ans $e^\pi > \pi^e$

Sol $f(x) = x^{\frac{1}{x}}$; $f(x) = e^{\frac{\ln x}{x}} \Rightarrow f'(x) = x^{\frac{1}{x}}\left[\frac{1-\ln x}{x^2}\right]$



for $0 < x < e$, $f'(x) > 0$ and $x > e$, $f'(x) < 0$

$f(x)$ attains the maximum at $x = e$ and max. value $e^{1/e}$

$\therefore$ $f(e) > f(x)$ for all $x > 0 \Rightarrow f(e) = f(\pi)$; $e^{\frac{1}{e}} > \pi^{\frac{1}{\pi}}$; $\therefore$ $e^{\frac{1}{e}} > \pi^{\frac{1}{\pi}}$; $e^\pi > \pi^e$]

4. If f(x) is increasing and g(x) is decreasing for all $x \in R$ find nature of following functions

(i) $h(x) = -f(x)$. (ii) $h(x) = g(x^3)$ (iii) $h(x) = g(x^2)$ (iv) $h(x) = f(x) - g(x)$

(v) $h(x) = \frac{1}{f(x)}$ (vi) $h(x) = \frac{1}{g(x)}$ (vii) $h(x) = g(g(x))$ (viii) $h(x) = g(f(x))$

Sol. (i) $h(x) = -f(x)$

$h'(x) = -f'(x)$

non –ve

$$\left\{\begin{array}{c} \text{if } (x)\uparrow, g(x)\downarrow \\ f'(x) \geq 0, g'(x) \leq 0 \\ f'(x) \geq 0 \\ g'(x) \leq 0 \end{array}\right\}$$

$h'(x) \leq 0$

h(x) is decreasing.

(ii) $h(x) = g(x^3)$

$h'(x) = g'(x^3) \cdot 3x^2$

non +ve (–ve)

$h'(x) \leq 0$

h(x) is decreasing

(iii) $h(x) = g(x^2)$

$h'(x) = g'(x^2) \cdot 2x$

C-I $x < 0$	C-II $x > 0$
(≤ 0) –ve	$g'(x^2) \cdot 2$ (+ve)
≥ 0	≤ 0
$h'(x) \geq 0$	$h'(x) \leq 0$
h(x) is increasing	h(x) is decreasing

(iv) $h(x) = f(x) - g(x)$

$h'(x) = f'(x) - g'(x)$

$\geq 0 \quad \leq 0$

$h'(x) \geq 0$

$h(x)$ = increasing.

(v) $h(x) = \dfrac{1}{f(x)}$

$$h'(x) = -\frac{1}{(f(x))^2} f'(x)$$

$h'(x) \leq 0$

$h(x)$ is decreasing.

(vi) $h(x) = \dfrac{1}{g(x)}$

$$h'(x) = -\frac{1}{(g(x))^2} g'(x)$$

≥ 0

$h'(x) \geq 0$

$h(x)$ is increasing.

(vii) $h(x) = g(g(x))$

$h'(x) = g'(g(x)) \cdot g'(x)$

≥ 0

$h'(x) \geq 0$

$h(x)$ is increasing.

(viii) $h(x) = g(f(x))$

$h'(x) = g'(f(x)) \cdot f'(x)$

≤ 0

$h'(x) \leq 0$

$h(x)$ is decreasing.

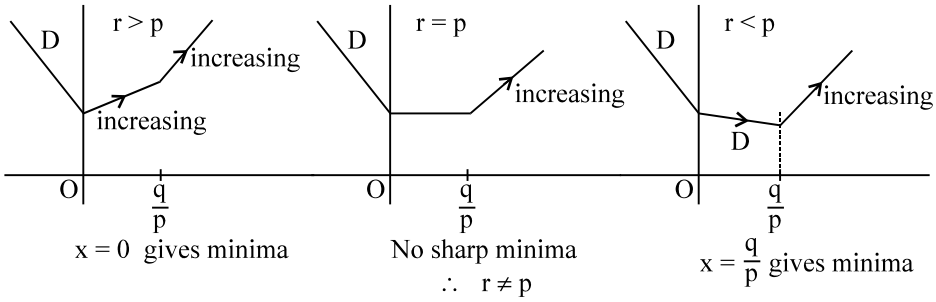
5. If $f(x) = |px - q| + r|x|$, $x \in (-\infty, \infty)$, $p, q, r > 0$, assumes its minimum value only at one point then find the relation between p, q, r.

Ans. $p \neq r$

Sol. Let $x \geq \dfrac{q}{p}$; $f(x) = px - q + rx$; $f'(x) = p + r > 0 \Rightarrow f(x)$ increasing for $x \geq \dfrac{q}{p}$

$0 \leq x < \dfrac{q}{p}$ $\quad f(x) = q - px + rx$; $f'(x) = r - p \Rightarrow$ $\begin{cases} f(x) \text{ is } \uparrow \text{ if } r > p \\ \text{constant} \quad r = p \\ f(x) \text{ is } \downarrow \text{ if } r < p \end{cases}$

$x < 0$, $\quad f(x) = q - px - rx$; $f'(x) = -(p + r) \Rightarrow f(x)$ is decreasing

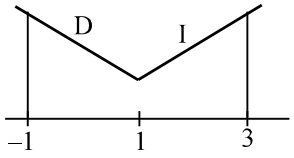


6. Find the image of interval [–1, 3] under the mapping specified by the function $f(x) = 4x^3 - 12x$

Ans. [–8, 72]

[Hint: $f'(x) = 0 \Rightarrow x = \pm 1$

Find $f(-1)$, $f(1)$ and $f(3)$ and interpret.



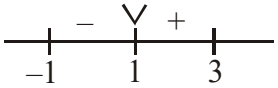
Sol. $f(x) = 4x^3 - 12x$ ...(1)

$f'(x) = 12x^2 - 12$

$= 12(x^2 - 1)$

$= 12(x - 1)(x + 1)$

$f'(x) = 0 \Rightarrow x = \pm 1$



$m = f(1) = 4 - 12 = -8$

$f(-1) = 8$

$M = f(3) = 108 - 36$

$= 72$

Image of [–1, 3] in f(x) is [–8, 72]

7. Let $f(x) = \begin{cases} -x^3 + \dfrac{\left(b^3 - b^2 + b - 1\right)}{\left(b^2 + 3b + 2\right)} & ,0 \le x < 1 \\ 2x - 3 & ,1 \le x \le 3 \end{cases}$

Find all possible real values of b such that f(x) has the smallest value at x = 1.

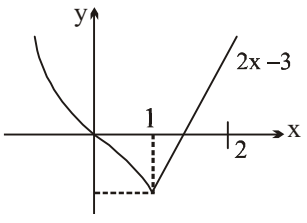
Ans: $b \in (-2, -1) \cup [1, \infty)]$

[Sol. Obviously f is decreasing in $0 \le x < 1$ and increasing in [1, 3]. Also $f(1) = -1$

Note that $f(1) \le f(1 - h)$ in order that f(x) has the smallest value at x = 1

$$\lim_{h \to 0} f(1-h) \ge f(1)$$

$$\lim_{h \to 0} -(1-h)^3 + \frac{(b^2+1)(b-1)}{(b+1)(b+2)} \ge -1$$



or $\dfrac{(b^2+1)(b-1)}{(b+1)(b+2)} \ge 0 \Rightarrow b \in (-2, -1) \cup [1, \infty)]$

8. Find the minimum value of the function $f(x) = 8^x + 8^{-x} - 4(4^x + 4^{-x})$, $\forall\ x \in R$.

Ans. -10

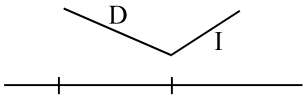
[Sol. $f(x) = (2^x + 2^{-x})^3 - 3(2^x + 2^{-x}) - 4[(2^x + 2^{-x})^2 - 2]$

$2^x + 2^{-x} = t \geq 2,\ x \in R$

$g(t) = t^3 - 3t - 4t^2 + 8 = t^3 - 4t^2 - 3t + 8$

$g'(t) = 3t^2 - 8t - 3$

$= (3t + 1)(t - 3)$



Hence $f_{min} = g(3) = -10$

9. Find all possible values of 'a' for which $f(x) = \log_a(4ax - x^2)$ is monotonically increasing for every $x \in \left[\frac{3}{2}, 2\right]$.

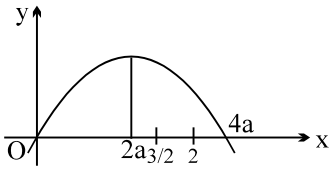
Ans. $\left(\frac{1}{2}, \frac{3}{4}\right] \cup (1, \infty)$

[Sol. Case-I:

If $0 < a < 1$ (obviously 'a' can not be < 0)

then for $f(x)$ to be increasing

$4ax - x^2$ should be decreasing in $\left(\frac{3}{2}, 2\right)$



$\Rightarrow \quad \frac{3}{2} \geq 2a \quad$ and $\quad 2 < 4a$

$\therefore \quad a \leq \frac{3}{4}$ and $a > \frac{1}{2} \Rightarrow \quad a \in \left(\frac{1}{2}, \frac{3}{4}\right]$

Caser-II: If $a > 1$ then for $f(x)$ to be increasing

$4ax - x^2$ increasing in $\left(\frac{3}{2}, 2\right)$

$\therefore \quad 2a \geq 2 \Rightarrow \quad a \geq 1$ but $a \neq 1$; $\quad \therefore \quad a > 1$

$\therefore$ final answer is $\left(\frac{1}{2}, \frac{3}{4}\right] \cup (1, \infty)$ Ans.]

10. In which of the following ordered pair(s) of (a, b) the inequality $a > b$ hold(s) good? **[O]**

(A) (π^e, e^π) (B) $\left((2008)^{2009}, (2009)^{2008}\right)$

(C) $\left(e^{5/2}, \left(\frac{5}{2}\right)^e\right)$ (D) $\left(\left(\frac{7}{4}\right)^{5/3}, \left(\frac{5}{3}\right)^{7/4}\right)$

Ans BCD

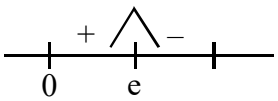
[Hint: Consider $f(x) = x^{1/x}$ which is maximum at $x = e$, intercept the result]

Sol. $f(x) = x^{\frac{1}{x}} \quad ...(1) \quad (x > 0)$

$f'(x) = x^{\frac{1}{x}}\left[-\frac{1}{x^2}\ell nx + \frac{1}{x}\cdot\frac{1}{x}\right]$

$$f'(x) = x^{\frac{1}{x}}\left[\frac{1-\ell nx}{x^2}\right]$$

$f'(x) = 0$



I.O.I. $x \in (0, e)$

I.O.D. $x \in (e, \infty)$

maxima at $x = e$

(A) $e^{\frac{1}{e}} > \pi^{\frac{1}{\pi}}$

apply 'πe' power both sides

$e^{\frac{\pi e}{e}} > \pi^{\frac{\pi e}{\pi}}$

$e^{\pi} > \pi^{e}$

$\Rightarrow a < b$

Therefore A doesn't follow $b < a$.

(B) $(2008)^{\frac{1}{2008}} > (2009)^{\frac{1}{2009}}$

apply power 2009×2008 both sides

$\Rightarrow (2008)^{2009} > (2009)^{2008}$

Clearly $a > b$

(c) $e^{\frac{1}{e}} > \left(\frac{5}{2}\right)^{\frac{2}{5}}$

apply power $\frac{5}{2} \times e$ both sides

$\Rightarrow e^{\frac{5}{2}} > \left(\frac{5}{2}\right)^{e}$

$\Rightarrow a > b$

(d) $\left(\frac{7}{4}\right)^{\frac{4}{7}} > \left(\frac{5}{3}\right)^{\frac{3}{5}}$

apply power $\frac{7}{4} \times \frac{5}{3}$ both sides

$\Rightarrow \left(\frac{7}{4}\right)^{\frac{5}{3}} > \left(\frac{5}{3}\right)^{\frac{7}{4}}$

$\Rightarrow a > b$

11. The maximum value of $f(x) = |x \ln x|$ in $x \in (0, 1)$ is **[O]**

(a) 1/e (b) e (c) 1 (d) none of these

Ans A

Sol $f(x) = |x \ln x|$. for, $x \in (0, 1)$, $f(x) = -x \ln x$

$$f'(x) = -\left[x.\frac{1}{x} + \ln x\right] = -[1 + \ln x] = 0 \Rightarrow x = 1/e$$

$f''(x) = [-1/x] < 0$

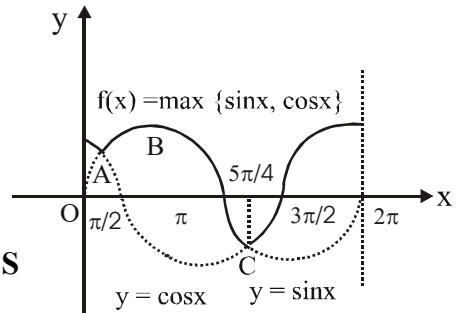
$f(x)$ will be maximum at $x = 1/e$

Maximum value of $f(x) = |-1/e| = 1/e$

Hence (a) is the correct answer.

12. The number of critical points of $f(x) = \max\{\sin x, \cos x\}$ for $x \in (0, 2\pi)$ is **[O]**

Ans 3



Sol Clearly A, B and C are the critical points

3.2 QUESTIONS ON PROVING INEQUALITIES

1. Prove that $x \geq \sin x \ \forall \ x \in \left[0, \frac{\pi}{2}\right]$

Sol. Let $h(x) = x - \sin x$ ...(1)

$h'(x) = 1 - \cos x$

$h'(x) \in [0, 1]$

$h'(x) \geq 0$ in $\left[0, \frac{\pi}{2}\right]$

$h(x)$ is increasing.

$\therefore h(x) \geq h(0)$

$h(x) \geq 0$

from equation (1)

$x - \sin x \geq 0$

$x \geq \sin x$

2. Prove that $\tan x \geq x \ \forall \ x \in \left[0, \frac{\pi}{2}\right)$

Sol. Let $h(x) = \tan x - x$

$h'(x) = \sec^2 x - 1$

$\downarrow$

$[1, \infty)$

$h'(x) \geq 0$

$\Rightarrow h(x)$ is increasing f × n.

$h(x) \geq h(0)$

$\tan x - x \geq 0$

$\tan x \geq x$

3 Prove that $2\sin x + \tan x \geq 3x \quad (\forall\ 0 \leq x < \frac{\pi}{2})$

[Sol. Consider, $f(x) = 2\sin x + \tan x - 3x$

$f'(x) = 2\cos x + \sec^2 x - 3$

$$= \frac{2\cos^3 x - 3\cos^2 x + 1}{\cos^2 x} = \frac{(\cos x - 1)^2 (2\cos x + 1)}{\cos^2 x}$$

Hence $f'(0) > 0 \qquad \forall\, x \in \left(0, \frac{\pi}{2}\right) \quad \Rightarrow \quad f(x)$ is $\uparrow$ in $x \in \left(0, \frac{\pi}{2}\right)$

$\therefore \quad f(x) > f(0)\ \forall\, x$ in $\left(0, \frac{\pi}{2}\right)$; but $f(0) = 0$

$\therefore \quad f(x) \geq 0$; as equaly holds

Hence $f(x) > 0$ in $\left[0, \frac{\pi}{2}\right) \Rightarrow$ the result]

4. Find the set of values of x for which $\ell n(1 + x) \geq \frac{x}{1+x}$

Ans. $(-1, \infty)$

[Hint: $f(x) = \ell n(1 + x) > \frac{x}{1+x}$

$f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} \quad \Rightarrow \quad \uparrow$ for $x > 0$ and $\downarrow\ -1 < x < 0$

$\Rightarrow$ $f(0)$ is the minimum value of $f(x)$ for all $x > -1$]

5. Prove that $\ln(\sec x) \geq \frac{x^2}{2}\ \forall\ x \in \left[0, \frac{\pi}{2}\right)$

Sol. Let

$$g(x) = \ell n(\sec x) - \frac{x^2}{2}$$

$$g'(x) = \frac{1}{\sec x}\sec x \tan x - \frac{2x}{2}$$

$g'(x) = \tan x - x$

$g'(x) \geq 0$

$g(x)$ is increasing

$g(x) \geq g(0)$

$g(x) \geq 0$

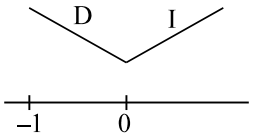
$$\ell n(\sec x) - \frac{x^2}{2} \geq 0$$

$$\ell n(\sec x) \geq \frac{x^2}{2}$$

H.P.

6. $\dfrac{\tan x}{x} > \dfrac{x}{\sin x} \quad \forall \quad 0 < x < \dfrac{\pi}{2}.$

[Sol. We have to show that



$$\frac{\tan x \sin x - x^2}{x \sin x} > 0 \text{ for } 0 < x < \pi/2$$

Since $x \sin x > 0$ for $0 < x < \pi/2$, it is enough to show that

$\tan x \cdot \sin x - x^2 > 0, \ 0 < x < \pi/2$

Let $f(x) = \tan x \sin x - x^2$ for $0 < x < \pi/2$

$f'(x) = \sin x \sec^2 x + \tan x \cos x - 2x = \sin x \sec^2 x + \sin x - 2x$

$f''(x) = \cos x \sec^2 x + \sin x \cdot 2 \sec^2 x \tan x + \cos x - 2$

$= \sec x + \cos x - 2 + 2 \sin x \tan x \sec^2 x$

$= \left(\sqrt{\sec x} - \sqrt{\cos x}\right)^2 + 2 \sin x \tan x \sec^2 x > 0$ for $0 < x < \pi/2$

$\therefore$ f' is strictly increasing in $[0, \pi/2)$. Also $f'(0) = 0$

$\Rightarrow$ $f'(x) > 0$ for $0 < x < \pi/2$

$\Rightarrow$ f is strictly increasing in $[0, \pi/2)$. Also $f(0) = 0$

$\Rightarrow$ $f(x) > 0$ for $0 < x < \pi/2$.

$\Rightarrow$ $\tan x \sin x - x^2 > 0$ for $0 < x < \pi/2$.

$\therefore$ $\dfrac{\tan x \sin x - x^2}{x \sin x} > 0, \quad 0 < x < \pi/2.$

$\Rightarrow$ $\dfrac{\tan x}{x} > \dfrac{x}{\sin x}$ for $0 < x < \dfrac{\pi}{2}$. Henced proved.]

7. Establish the inequality given below by examining the sign of the derivative of an appropriate function:

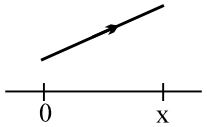
$$\frac{2}{2x+1} < \ell n\left(1 + \frac{1}{x}\right) < \frac{1}{x} \quad \text{for } x > 0$$

[Sol. consider $f(x) = \dfrac{2}{2x+1} - \ell n\left(1 + \dfrac{1}{x}\right)$; $f'(x) = \dfrac{-4}{(2x+1)^2} - \dfrac{1}{1+(1/x)} \cdot \left(-\dfrac{1}{x^2}\right)$

$$= \frac{1}{x(x+1)} - \frac{4}{(2x+1)^2} = \frac{1}{x(x+1)(2x+1)^2}$$

which is always +ve for $x > 0$

hence $f(x)$ is $\uparrow$ for $x > 0$



i.e. $f(x) < \lim\limits_{x \to \infty} f(x)$ **(note carefully)**

but $\lim\limits_{x \to \infty} f(x) = \lim\limits_{x \to \infty} \left\{\dfrac{2}{2x+1} - \ell n\left(1 + \dfrac{1}{x}\right)\right\} = 0$

so $f(x) < 0$ i.e.

$$\frac{2}{2x+1} < \ell n\left(1+\frac{1}{x}\right)$$ Proved

Similarly consider $g(x) = \ell n\left(1+\frac{1}{x}\right) - \frac{1}{x}$; $g'(x) = \frac{1}{x^2} - \frac{1}{x(x+1)} = \frac{1}{x^2(x+1)}$

$g'(x)$ is $\uparrow$ i.e.

so $g(x) < \lim\limits_{x\to\infty} g(x)$ but $\lim\limits_{x\to\infty} f(x) = 0$

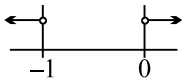
so $g(x) < 0$ hence proved]

8. Prove that $f(x) = \left(1+\frac{1}{x}\right)^x$ is always an increasing function for all x in its domain. State its range and also plot the graph of the function.

Ans $(1, e) \cup (e, \infty)$

[Sol. $f(x) = y = \left(1+\frac{1}{x}\right)^x = e^{x\, ln\left(1+\frac{1}{x}\right)}$

Hence for domain $\frac{x+1}{x} > 0$, $\Rightarrow$ $x \in (-\infty, -1) \cup (0, \infty)$



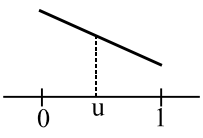
now $\lim\limits_{x\to\pm\infty}\left(1+\frac{1}{x}\right)^x = e$; Also $ln\, y = x\, ln\left(1+\frac{1}{x}\right)$

$$\therefore \quad \lim_{x\to 0^+} \frac{ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}} \left(\frac{\infty}{\infty}\right) = \frac{x}{x+1}\left(\frac{-\frac{1}{x^2}}{-\frac{1}{x^2}}\right) = 0 \Rightarrow y = e^0 = 1 \; ; \; \therefore \lim_{x\to 0^+} f(x) = 1$$

$\lim\limits_{x\to -1^-} f(x) = \infty$ (obviously)

Now $f'(x) = \left(1+\frac{1}{x}\right)^x\left[x\left(\frac{1}{1+x}-\frac{1}{x}\right) + ln\frac{1+x}{x}\right] = \left(1+\frac{1}{x}\right)^x\left[\frac{x}{1+x} - 1 + ln\frac{1+x}{x}\right]$

Let $\frac{x}{1+x} = u$ [for $x > 0$, $u \in (0, 1)$]



consider a function

$g(u) = u - 1 - ln\, u$

$g'(u) = 1 - \frac{1}{u} < 0$ which is negative for $u \in (0, 1)$

Hence $g(u)$ is decreasing in $(0, 1]$

$\therefore$ $g(u) > g(1)$ $\forall$ $u \in (0, 1]$

But $g(1) = 0$

$\therefore$ $g(u) > 0$ $\forall$ $u \in (0, 1)$

$\therefore$ $\frac{x}{1+x} - 1 + ln\frac{1+x}{x} > 0$ $\forall$ $x > 0$

$\therefore$ $f'(x) > 0$

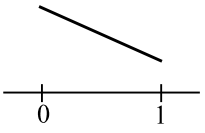
$\therefore$ $f(x)$ is increasing for $x > 0$ **....(1)**

again for $x < -1$

$$f'(x) = \left(1+\frac{1}{x}\right)^x\left[\ln\frac{1+x}{x} + \frac{x}{1+x} - 1\right] \qquad \text{put } \frac{1+x}{x} = u$$

consider $g(u) = \ln u + \frac{1}{u} - 1$ as $x \in (-\infty, -1)$, $u \in (0, 1)$

$$g'(u) = \frac{1}{u} - \frac{1}{u^2} = \frac{u-1}{u^2} < 0 \quad \text{for } u \in (0, 1)$$



$g(u)$ is decreasing in $(0, 1)$

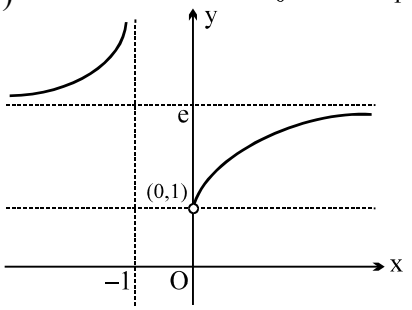
$\therefore$ $g(u) > g(1)$

$g(u) > 0$

$$\ln\frac{1+x}{x} + \frac{x}{1+x} - 1 > 0$$

$\therefore$ $f'(x) > 0$ for $x \in (-\infty, -1)$

$\therefore$ $f(x)$ is increasing in $(-\infty, -1)$ **....(2)**



Graph of $f(x)$

from (1) and (2)

$f(x)$ is increasing for all x in the domain of $f(x)$.

range is $(1, e) \cup (e, \infty)$

9. Find the smallest positive constant A such that $\ln x \le Ax^2$ for all $x > 0$.

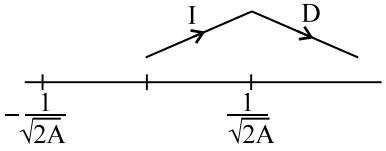
Ans. $\frac{1}{2e}$

[Sol. Note that $A > 0$

Consider the function

$$f(x) = \ln x - Ax^2$$

$$f'(x) = \frac{1}{x} - 2Ax = \frac{1-2Ax^2}{x}$$



$$f'(x) > 0 \text{ for } 0 < x < \frac{1}{\sqrt{2A}}$$

and $f'(x) < 0$ for $x > \frac{1}{\sqrt{2A}}$; $\therefore$ $\ln x < Ax^2$ for $x > 0$

this must also hold good for $x = \frac{1}{\sqrt{2A}}$

$$\therefore \quad \ln\frac{1}{\sqrt{2A}} < A\cdot\frac{1}{2A}; -\ln\sqrt{2A} < \frac{1}{2}; \ln\sqrt{2A} > -\frac{1}{2}$$

$$\sqrt{2A} = e^{-\frac{1}{2}}; A > \frac{1}{2e}; \qquad \therefore \quad \text{least value of } A = \frac{1}{2e} \text{ Ans.]}$$

10. Least natural number a for which $x + ax^{-2} > 2 \ \forall x \in (0,\infty)$ is **[O]**

(a) 1 (b) 2 (c) 5 (d) none of these

Ans B

Sol Let $f(x) = x + ax^{-2}$

$$f'(x) = 1 - 2ax^{-3} = 0 \Rightarrow x = (2a)^{\frac{1}{3}}$$

$$f''(x) = 6ax^{-4} > 0 \ \forall x \in (0,\infty) \text{ (as a is a natural number)}$$

Thus $(2a)^{1/3} + a(2a)^{-2/3} > 2 \Rightarrow a > \frac{32}{27} \Rightarrow$ least natural number a = 2.

Alternative solution

$x + ax^{-2} > 2 \Rightarrow x^2 - 2x^2 + a > 0$

Let $f(x) = x^3 - 2x^2 + a$

Since $f(x) > 0 \ \forall x \in (0,\infty) \Rightarrow \min f(x) > 0$

For minimum f(x), $f'(x) = 3x^2 - 4x = 0 \Rightarrow x = 0,\ 4/3$

$$f(4/3) > 0 \Rightarrow a > \frac{32}{27}.$$

Hence (b) is the correct answer.

11. For $x \in (0, 1)$ prove that $x - \frac{x^3}{3} < \tan^{-1} x < x - \frac{x^3}{6}$ & hence or otherwise find $\lim_{x\to 0}\left[\frac{\tan^{-1} x}{x}\right]$ **[O]**

Ans 0

Sol. Let $f(x) = x - \frac{x^3}{3} - \tan^{-1}x \Rightarrow f'(x) = 1 - x^2 - \frac{1}{1+x^2} = -\frac{x^4}{1+x^2}$

$f'(x) < 0$ for $x \in (0, 1) \Rightarrow$ f(x) is M.D. $\Rightarrow$ $f(x) < f(0)$

$\Rightarrow x - \frac{x^3}{3} - \tan^{-1}x < 0 \Rightarrow x - \frac{x^3}{3} < \tan^{-1}x$ ..... (i)

Similarly $g(x) = x - \frac{x^3}{6} - \tan^{-1}x$

$g'(x) = 1 - \frac{x^2}{2} - \frac{1}{1+x^2} = \frac{x^2(1-x^2)}{2(1+x^2)}$

$g'(x) > 0$ for $x \in (0, 1) \Rightarrow$ g(x) is M.I.

$x - \frac{x^3}{6} - \tan^{-1}x > 0 \Rightarrow x - \frac{x^3}{6} > \tan^{-1}x$ ..... (ii)

from (i) and (ii), we get $x - \frac{x^3}{3} < \tan^{-1}x < x - \frac{x^3}{6}$ Hence Proved

Now, $1 - \frac{x^2}{3} < \frac{\tan^{-1} x}{x} < 1 - \frac{x^2}{6} \Rightarrow \lim_{x\to 0}\frac{\tan^{-1} x}{x} = 1$

but $\lim_{x\to 0}\left[\frac{\tan^{-1} x}{x}\right] = 0$

12. For $x \in \left(0, \frac{\pi}{2}\right)$, prove that $\sin x > x - \frac{x^3}{6}$ **[O]**

Sol. Let $f(x) = \sin x - x + \frac{x^3}{6}$

$f'(x) = \cos x - 1 + \frac{x^2}{2}$

we cannot decide at this point whether $f'(x)$ is positive or negative, hence let us check for monotonic nature of $f'(x)$ $f''(x) = x - \sin x$

Since $f''(x) > 0 \Rightarrow f'(x)$ is M.I. for $x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow f'(x) > f'(0) \Rightarrow f'(x) > 0$

$\Rightarrow f(x)$ is M.I. $\Rightarrow f(x) > f(0)$

$\Rightarrow \sin x - x + \frac{x^3}{6} > 0 \Rightarrow \sin x > x - \frac{x^3}{6}$. Hence proved

4. ROLLE'S THEOREM

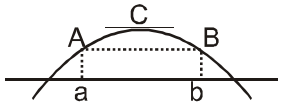
If a function f is satisfying following conditions

(i) f is continuous on [a, b]

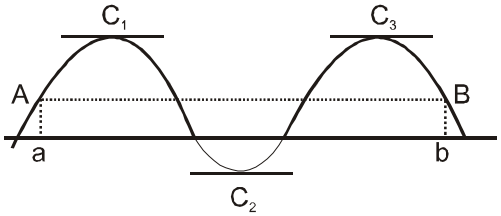
(ii) f is derivable on (a, b) and

(iii) f(a) = f(b),

then there exists at least one real number c between a and b (a < c < b) such that f'(c) = 0



There can be more than one c.

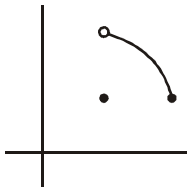


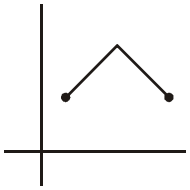
Geometrical Definition

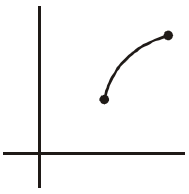
If f(x) is continuous in [a, b], derivable in (a, b) and f(a) = f(b) then there exists at least one point in (a, b) where tangent to the graph will be horizontal.

Note

1. To apply rolle's theorem, all three conditions must be satisfied simultaneously.

(i) 

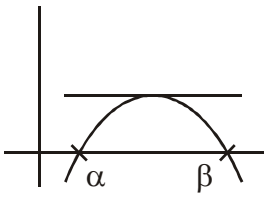
(ii) 

(iii) 

2. For f'(x) = 0 conditions of rolle's theorem are sufficient but not necessary.

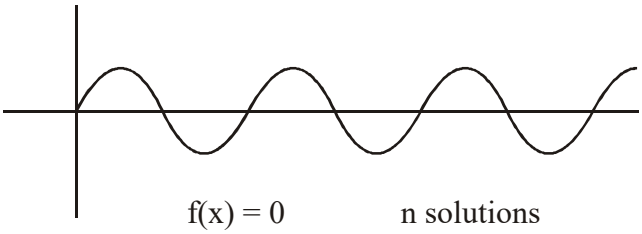
Alternate definition of rolle's theorem.

Between any two consecutive zeroes of derivable function f(x), these exists atleast one solution of equation $f'(x) = 0$.



Note :

(i) For a derivable function f(x), if equation $f(x) = 0$ has n solution then equation $f'(x) = 0$ will have atleast $(n-1)$ solutions.



$f(x) = 0$ n solutions

$f'(x) = 0$ atleast $(n-1)$ solutions

(ii) For a derivable function f(x), if equation $f'(x) = 0$ has m solutions then, equation $f(x) = 0$ will have at most $m+1$ solutions.

1. Verify Rolle's Theorem for and then find c of Rolle's Theorem

(a) $f(x) = x(x+3)e^{-x/2}$ in $[-3, 0]$ (b) $f(x) = x^3 - 3x^2 + 2x + 5$ in $[0, 2]$

(c) $f(x) = 1 - x^{2/3}$ in $[-1, 1]$ (d) $f(x) = |x|^3$ in $[-1, 1]$

Ans. (a) $c = -2$

(b) $c = \dfrac{3 \pm \sqrt{3}}{3}$

(c) Rolles theorem is not applicable

(d) $c = 0$

Sol. (a) $f(x) = x(x+3)e^{-x/2}$ in $[-3, 0]$

f(x) is continue & differentiable in $[-3, 0]$

$$\left.\begin{matrix} f(-3) = 0 \\ f(0) = 0 \end{matrix}\right\}$$

$$f'(x) = 1(x+3)e^{-\frac{x}{2}} + x(1)e^{-\frac{x}{2}} - x(x+3)e^{-\frac{x}{2}} \cdot \left(\frac{1}{2}\right)$$

$$f'(x) = e^{-\frac{x}{2}}\left[2x + 3 - \frac{x^2}{2} - \frac{3x}{2}\right]$$

$$f'(x)=\frac{e^{-\frac{x}{2}}}{2}\left[4x+6-x^2-3x\right]$$

$$f'(x)=\frac{e^{-\frac{x}{2}}}{2}\left[6+x-x^2\right]$$

$$f'(x)=\frac{e^{-\frac{x}{2}}}{2}\left[6+3x-2x-x^2\right]$$

$$f'(x)=\frac{e^{-\frac{x}{2}}}{2}\left[3(x+2)-x(2+x)\right]$$

$$f'(x)=\frac{e^{-\frac{x}{2}}}{2}(x+2)(3-x)$$

$f'(x) = 0$ at $x = -2$

$-2 \in (-3, 0)$

Hence, Rolle's theorem applicable

(b) $f(x) = x^3 - 3x^2 + 2x + 5$ in $[0, 2]$

$f(x)$ is polynomial so, continue & differentiable both.

$f(0) = 5$

$f(2) = 5$

$f'(x) = 3x^2 - 6x + 2$

$f'(c) = 3c^2 + 6c + 2 = 0$

$$c=\frac{-(-6)\pm\sqrt{36-24}}{6}$$

$$c=\frac{6\pm2\sqrt{3}}{6}=\frac{1\pm1}{\sqrt{3}}\begin{cases}1+\frac{1}{\sqrt{3}}\\ 1-\frac{1}{\sqrt{3}}\end{cases}$$

$$c=1\pm\frac{1}{\sqrt{3}}\in(0,2)$$

Hence, Rolle's theorem applicable.

(c) $f(x)=1-x^{\frac{2}{3}}$ in $[-1, 1]$

$f(x)$ is continuous but not differentiable at $x = 0$

$$f'(x)=-\left(\frac{2}{3}\right)\frac{1}{x^{\frac{1}{3}}}=-\frac{2}{3}\cdot\frac{1}{x^{1/3}}$$

Rolle's theorem N.P.

(d) $f(x)=|x|^3=\begin{cases}-x^3 & x<0\\ x^3 & x\geq 0\end{cases}$

Continue at $x=0$

By quick concept

$$f'(x)=\begin{cases}-3x^2 & x<0\\ 3x^2 & x\geq 0\end{cases}$$

also differentiable at $x=0$

$f(-1)=1$

$f(1)=1$

Hence, theorem is applicable.

$f'(x)=0$ at $x=0\in(-1,1)$

$c=0$

2. Let $n\in N$. If the value of c prescribed in Rolle's theorem for the function $f(x)=2x(x-3)^n$ on $[0,3]$ is $3/4$ then $\boldsymbol{n}$ is equal to

Ans 3

[Sol. $f(0)=f(3)$; $f(x)=2x(x-3)^n$

hence $f'(c)=0$

$f'(x)=2[(x-3)^n+nx(x-3)^{n-1}]$

$f'(c)=2[(c-3)^n+nc(c-3)^{n-1}]=0$

or $2(c-3)^{n-1}[c-3+nc]=0$; but $c=3/4$

$\therefore\quad (n+1)\frac{3}{4}=3\ \Rightarrow\ n=3$ **Ans.**]

3. If $2a+3b+6c=0$ then prove that equation $ax^2+bx+c=0$ will have atleast one solution in (0,1).

Sol. Let $f(x)=\frac{ax^3}{3}+\frac{bx^2}{2}+cx$

$f(0)=0$ and $f(1)=\frac{a}{3}+\frac{b}{2}+c=2a+3b+6c=0$

If $f(0)=f(1)$ then $f'(x)=0$ for some value of $x\in(0,1)$

$\Rightarrow ax^2+bx+c=0$ for at least one $x\in(0,1)$

4. For a twice differentiable function P(x), $P(1)=1, P(2)=2, P(3)=3, P(4)=4$.

(i) Prove that equation $P'(x)=1$ will have atleast 3 solutions.

(ii) Prove that equation $P''(x)=0$ will have atleast 2 solutions.

Sol. Given

(i) $P(1)=1, P(2)=2, P(3)=3, P(4)=4$

$P(x)=x$ ...(1)

Let $f(x)=P(x)-x$ which has four zeroes i.e. $x=1,2,3,4$

$f'(x)=P'(x)-1$ ...(2)

Alternate definition of Rolle's Theorem

$f'(x)=0$ has at least 3 solutions.

$P'(x)=1$ has at least 3 solutions.

H.P.

(ii) again differentiating equation (2) w.r.t. x.

$f''(x) = P''(x)$ ...(3)

$f''(x)$ has atleast 2 solutions.

$P''(x)$ have at least 2 solutions.

H.P.

5. Show that between any two roots of the equation $e^x \cos x = 1$ there exists at least one root of $e^x \sin x - 1 = 0$.

Sol Consider $f(x) = \cos x - e^{-x}$ and apply Rolle's theorem in it.

Given

$\cos x = \frac{1}{e^x}$ has 2 roots.

Let $f(x) = \cos x - \frac{1}{e^x}$ has 2 roots.

on differentiating w.r.t. x.

$f'(x) = -\sin x + \frac{1}{e^x}$

$\therefore f'(x) = 0$ has at least one solution.

$-\sin x + \frac{1}{e^x} = 0$

$1 = e^x \sin x$

$e^x \sin x - 1 = 0$

has at least one solution.

H.P.

6. Let $f(x)$ and $g(x)$ be differentiable functions such that $f'(x)\, g(x) \neq f(x)\, g'(x)$ for any real x. Show that between any two real solutions of $f(x) = 0$, there is at least one real solution of $g(x) = 0$. **[O]**

[Sol. Let a, b be the solutions of f(x)=0

suppose g(x) is not equal to zero for any x belonging to [a, b]

now consider h(x)= $f(x)/g(x)$

since g(x) not equal to zero

h(x) is differentiable and continuous in [a, b]

$h(a) = h(b) = 0$ (as $f(a) = 0$ and $f(b) = 0$ but $g(a)$ or $g(b) \neq 0$)

applying rolles theorem for h (x) in [a, b]

$h'(c)=0$ for some c belonging to (a,b)

$f(x)g'(x)= f'(x)g(x)$

this gives the contadiction ...

hence proved]

7. Let P (x) be a polynomial with real coefficients. Let $a, b \in R$, $a < b$, be two consecutive roots of P(x). Show that there exists 'c' such that $a \leq c \leq b$ and $P'(c) + 100\, P(c) = 0$. **[O]**

[Sol. Consider $f(x) = e^{100x} \cdot P(x)$.

Now $f(a) = f(b) = 0$ {as $P(a) = P(b) = 0$ }

Also as P (x) is polynomial $\Rightarrow$ f(x) is continuous and differentiable in [a, b]

$\Rightarrow$ Rolle's theorem can be applied

$\Rightarrow$ $\exists\ c \in (a, b)$ such that $f'(c) = 0$

now $f'(x) = e^{100x}(P'(x) + 100 \cdot P(x))$

$\Rightarrow$ $e^{100c}(P'(c) + 100 \cdot P(c)) = 0$, from (1)

$\Rightarrow$ $P'(c) + 100 \cdot P(c)) = 0$ (as $[e^{100c} \neq 0]$) hence proved.]

8. Consider the function $f(x) = \begin{cases} x \sin\frac{\pi}{x} & \text{for } x > 0 \\ 0 & \text{for } x = 0 \end{cases}$ then the number of points in (0, 1) where the derivative $f'(x)$ vanishes , is

Ans Infinite

[Hint: $f(x)$ vanishes at points where $\sin\frac{\pi}{x} = 0$ i.e. $\frac{\pi}{x} = k\pi$, $k = 1, 2, 3, 4, ...$

hence $x = \frac{1}{k}$. Also $f'(x) = \sin\frac{\pi}{x} - \frac{\pi}{x}\cos\frac{\pi}{x}$ if $x \neq 0$

Since the function has a derivative at any interior point of the interval (0, 1), also continuous in [0,1] and $f(0) = f(1)$

hence Rolle's theorem is applicable to any one of the interval $\left[\frac{1}{2}, 1\right], \left[\frac{1}{3}, \frac{1}{2}\right], \ldots \left[\frac{1}{k+1}, \frac{1}{k}\right]$

hence $\exists$ some c in each of these interval where $f'(c) = 0 \Rightarrow$ infinite points $\Rightarrow$ (D)]

9. Let f be a continuous function on [a, b]. If $F(x) = \left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right)(2x - (a+b))$ then there exist some $c \in$ (a, b) such that **[O]**

(A) $\int_a^c f(t)\,dt = \int_c^b f(t)\,dt$ (B) $\int_a^c f(t)\,dt - \int_c^b f(t)\,dt = f(c)(a + b - 2c)$

(C) $\int_a^c f(t)\,dt - \int_c^b f(t)\,dt = f(c)(2c - (a+b))$ (D) $\int_a^c f(t)\,dt + \int_c^b f(t)\,dt = f(c)(2c - (a+b))$

Ans B

[Sol. Given $F(x) = \left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right)(2x - (a+b))$ ....(1)

as f is continuous hence F (x) is also continuous. Also

put $x = a$

$$F(a) = \left(-\int_a^b f(t)\,dt\right)(a - b) = (b - a)\int_a^b f(t)\,dt$$

and put $x = b$

$$F(b) = \left(\int_a^b f(t)\,dt\right)(b - a)$$

hence $F(a) = F(b)$

hence Rolle's Theorem is applicable to F (x)

$\therefore$ $\exists$ some $c \in (a, b)$ such that $F'(c) = 0$

now $F'(x) = 2\left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right) + (2x-(a+b))[f(x)+f(x)] = 0$

$\therefore \quad F'(c) = \left(\int_a^c f(t)\,dt - \int_c^b f(t)\,dt\right) = f(c)[(a+b)-2c]$ **Ans.**]

10. If f(x) is continuous in [a, b] and differentiable in (a, b) then prove that there exists atleast one $c \in (a,b)$ such that $\frac{f'(c)}{3c^2} = \frac{f(b)-f(a)}{b^3-a^3}$. **[O]**

Sol: We have to prove

$(b^3 - a^3)\ f'(c) - (f(b) - f(a))\ (3c^2) = 0$

Let us assume a function

$F(x) = (b^3 - a^3)\ f(x) - (f(b) - f(a))\ x^3$

which will be continuous in [a, b] and differentiable in (a, b) as f(x) and x^3 both are continuous.

Also $F(a) = b^3 f(a) - a^3 f(b) = F(b)$

So, according to Rolle's theorem, there exists atleast one $c \in (a,b)$ such that, $F'(c) = 0$, which proves the required result.

5. LAGRANGE'S MEAN VALUE THEOREM (LMVT)

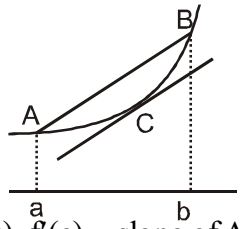
If a function f defined on [a, b] is

(i) continuous on [a, b] and (ii) derivable on (a, b)

then there exists at least one real numbers between a and b $(a < c < b)$ such that $\frac{f(b)-f(a)}{b-a} = f'(c)$

Geometrical Interpretation of LMVT :

The theorem simply states that between two points A and B of the graph of f(x) there exists at least one point where tangent is parallel to chord AB.



C(c, f(c)), f'(c) = slope of AB.

Alternative Statement : If in the statement of LMVT, b is replaced by a + h, then number c between a and b may be written as a + θh, where $0 < \theta < 1$. Thus

$\frac{f(a+h)-f(a)}{h} = f'(a+\theta h)$ or $f(a+h) = f(a) + hf'(a+\theta h),\ 0 < \theta < 1$

Physical Interpretation

Function f(x) is continuous in [a, b] and derivable in (a, b) then these exist atleast a point in (a, b) where instantaneous rate of change of y will be equal to average rate of change of y in the interval.

Q.1 Find 'θ' in the LMVT expressed as $f(a+h) = f(a) + h\,f'(a+\theta h)$ **[O]**

(i) $f(x) = \ell n\, x,\ a = 1,\ h = e - 1$; (ii) $f(x) = 3x^2 - 5x + 12$ in $[0, 1]$; (iii) $f(x) = x^2$ if $a = 1$, and $h = 1$;

Ans. (i) $\theta = \dfrac{e-2}{e-1}$ (ii) $\theta = \dfrac{1}{2}$ (iii) $\theta = \dfrac{1}{2}$

Sol. (i) $f(x) = \ell nx,\ a = 1,\ h = e - 1$

By alternate statement of LMVT

$$f'(a+\theta h) = \frac{f(a+h) - f(a)}{h}$$

$$\frac{1}{1+\theta(e-1)} = \frac{\ell n(e) - \ell n1}{e-1}$$

$$\frac{1}{1+\theta(e-1)} = \frac{1}{e-1}$$

$\Rightarrow 1 + \theta(e - 1) = e - 1$

$$\theta = \frac{e-2}{e-1}$$

(ii) $f(x) = 3x^2 - 5x + 12$ in $[0, 1]$

$a = 0,\ b = 0 + h,\ c = \theta h$

$$f'(\theta h) = \frac{f(h) - f(0)}{h}$$

$$6\theta h - 5 = \frac{3h^2 - 5h + 12 - 12}{h}$$

$6\theta h - 5 = 3h - 5$

$$\theta = \frac{3}{6} = \frac{1}{2}$$

(iii) $f(x) = x^2;\ a = 1,\ h = 1$

$$f'(a+\theta h) = \frac{f(a+h) - f(a)}{h}$$

$$2(1+\theta) = \frac{4-1}{1}$$

$$1 + \theta = \frac{3}{2}$$

$$\theta = \frac{3}{2} - 1 = \frac{1}{2}$$

2. Find c of LMVT

(i) $f(x) = \sqrt{x-1}$ in $[1, 3]$

Ans. $c = \dfrac{3}{2}$

Sol. $f(x)=\sqrt{x-1}$ in $[1,3]$

f(x) is continue in [1,3]

f(x) is differentiable in (1,3)

So, $f'(x)=\frac{1}{2\sqrt{x-1}}$

By LMVT at x = c

$$\frac{f(3)-f(1)}{3-1}=\frac{1}{2\sqrt{e-1}}$$

$$\frac{\sqrt{2}}{2}=\frac{1}{2\sqrt{c-1}}$$

$$c-1=\frac{1}{2} \qquad\qquad c=\frac{3}{2}\in(1,3)$$

(ii) $f(x)=\frac{x}{x+2}$ on [1, 4]

Ans. $c=3\sqrt{2}-2$

Sol. f(x) is continuous in [1, 4]

f(x) is differentiable in (1,4)

$$f'(x)=\frac{2}{(x+2)^2}$$

at LMVT at x = c

$$\frac{f(4)-f(1)}{3}=f'(c)$$

$$\frac{\frac{2}{3}-\frac{1}{3}}{3}=\frac{(c+2)-c}{(c+2)^2}$$

$$\frac{1}{9}=\frac{2}{(c+2)^2}$$

$(c+2)^2=18$

$c+2=\pm 3\sqrt{2}$

$$c=-2\pm 3\sqrt{2}\begin{cases}3\sqrt{2}-2\\ -2-3\sqrt{2}\ (\text{rejected})\end{cases}$$

(iii) If LMVT is known to be applicable for a quadratic polynomial $y=lx^2+mx+n$ in [a, b] then c of LMVT occurs at the midpoint of the interval i.e. $c=\frac{a+b}{2}$.

Sol. $f(x)=\ell x^2+mx+n$ is $[a,b]$

continue & differentiable is [a, b] and (a, b)

$$\frac{\left(\ell b^2+mb+n\right)-\left(\ell a^2+ma+n\right)}{(b-a)}=2\ell c+m$$

$$\frac{\ell\left(b^2-a^2\right)+m(b-a)}{(b-a)}=2\ell c+m$$

$$\Rightarrow \ell(b+a)+m=2\ell c+m$$

$$\Rightarrow c=\frac{a+b}{2}$$

3. Find a point on the curve $f(x)=\sqrt{x-2}$ in [2, 3] when the tangent is parallel to the chord joining the end points.

Ans. (9/4, 1/2)

Sol. $f(x)=\sqrt{x-2}$ ...(1) in [2, 3]

By LMVT

at x = c

$$\frac{f(3)-f(2)}{1}=\frac{1}{2\sqrt{c-2}}$$

$$1=\frac{1}{2\sqrt{c-2}}$$

$$\sqrt{c-2}=\frac{1}{2}$$

$$c-2=\frac{1}{4}$$

$$c=\frac{9}{4}\in(2,3)$$

from equation (1)

$$f(x)=\sqrt{\frac{9}{4}-2}=\sqrt{\frac{1}{4}}=\frac{1}{2}$$

Point $\left(\frac{9}{4},\frac{1}{2}\right)$

4. if a < b, show that a real number 'c' can be found in (a, b) such that $3c^2=a^2+ab+b^2$

Sol Consider $f(x)=x^3$ and use LMVT in [a, b]

Given :

So, let $f(x)=x^3$ ...(1)

By LMVT

at x = c in (a, b)

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$3c^2 = \frac{b^3 - a^3}{b - a}$$

$$3c^2 = \frac{(b - a)(a^2 + ab + b^2)}{(b - a)}$$

H.P.

5. Use LMVT to prove that $\tan x > x$ for $x \in \left(0, \frac{\pi}{2}\right)$

Sol. Let $f(x) = \tan x - x$

By LMVT for $(0, x)$

$$\sec^2 x - 1 = \frac{(\tan x - x) - 0}{x - 0}$$

$$\sec^2 x - 1 = \frac{\tan x}{x} - 1$$

$$\sec^2 x = \frac{\tan x}{x} \qquad ...(1)$$

as $\sec^2 x$ in $\left(0, \frac{\pi}{2}\right)$

$\sec^2 x > 1$

$$\frac{\tan x}{x} > 1$$

$\tan x > x$

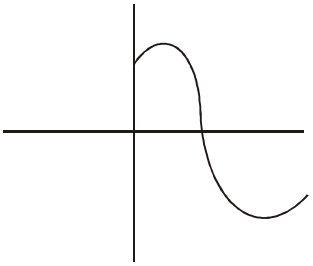
H.P.

6. Use LMVT to prove that $x \geq \sin x$ for $x \geq 0$

Sol. $x \geq \sin x \forall x \geq 0$

Let $g(x) = x - \sin x$

By LMVT in $(0, x)$



$$g'(x) = \frac{g(x) - g(0)}{x - 0}$$

$$g'(x) = \frac{g(x) - g(0)}{x - 0}$$

$$1 - \cos x = \frac{x - \sin x - 0}{x}$$

$$1 - \cos x = 1 - \frac{\sin x}{x}$$

$$\cos x = \frac{\sin x}{x} \qquad \ldots(1)$$

as $\cos x \le 1$

$$\frac{\sin x}{x} \le 1$$

$\sin x \le x$

Hence proved.

7. Use LMVT to prove that $e^x - 1 > x, \forall\, x > 0$

Sol. By LMVT in (0, x)

$$g'(c) = \frac{g(x) - g(0)}{x}$$

$$e^c = \frac{e^x - 1}{x} \forall c \in (0, x)$$

as $e^c > 1$

$$\frac{e^x - 1}{x} > 1$$

$\Rightarrow e^x - 1 > x \qquad$ (as $x > 0$)

8. Using LMVT prove that $|\cos a - \cos b| \le |a - b|$

Sol Consider $f(x) = \cos x$ in [a, b]

$$\left|\frac{\cos b - \cos a}{b - a}\right| = |-\sin c| \le 1 \qquad \Rightarrow \qquad |\cos b - \cos a| \le |b - a|$$

$|\cos a - \cos b| \le |a - b|$ Hence proved.

9. For a function f(x), $f(1) = -1$ and $f'(x) \ge 3$ then find minimum value of f(4)

Ans. 8

Sol. $f(1) = -1$

$f'(x) = 3$

$f(4)|_{\text{minimum}} = ?$

for f(x) in $x \in [1, 4]$ by LMVT

$$\frac{f(4) - f(1)}{4 - 1} = f'(x)$$

$$\frac{f(4) + 1}{3} = f'(x) \qquad \ldots(1)$$

$$f'(x) \ge 3 \Rightarrow \frac{f(4) + 1}{3} \ge 3$$

$f(4) \ge 8$

$f(4)|_{\text{minimum}} = 8$

10. Suppose that *f* and *g* are continuous on [a, b] and differentiable on (a, b). Suppose also that $f(a) = g(a)$ and $f'(x) < g'(x)$ for $a < x < b$. Prove that $f(b) < g(b)$.

Hint: Apply the Mean Value Theorem to the function $h = f - g$

$$\frac{\tan x - 0}{x - 0} = \sec^2 x > 1 \quad \Rightarrow \quad \tan x > x$$

Sol. Given :

$f(a) = g(a)$ ...(1)

also, $f'(x) < g'(x)$ ...(2)

$x \in (a, b)$

Let $h(x) = f(x) - g(x)$ ...(3)

By LMVT in (a, b)

$$h'(x) = \frac{h(b) - h(a)}{b - a}$$

$$f'(x) - g'(x) = \frac{(f(b) - g(b)) - 0}{(b - a)}$$

as $f'(x) - g'(x) < 0$

$$\frac{f(b) - g(b)}{b - a} < 0$$

$\Rightarrow f(b) - g(b) < 0$

$\Rightarrow f(b) < g(b)$

Hence proved

11. If $f(x)$ is continuous on [0, 2], differentiable on (0, 2), $f(0) = 2$, $f(2) = 8$, and $f'(x) \le 3$ for all x in (0, 2), then $f(1)$ has the value equal to

Ans 5

Sol. Applying LMVT in [0, 1] for f and then in [1, 2]

$c_1 \in (0, 1)$, $\frac{f(1) - f(0)}{1 - 0} = f'(c_1) \le 3 \quad \Rightarrow \quad f(1) \le 5$

$c_2 \in (1, 2)$, $\frac{f(2) - f(1)}{2 - 1} = f'(c_2) \le 3 \quad \Rightarrow \quad f(1) \ge 5$

Hence, $f(1) = 5$ Ans.

12. If f(x) and g(x) are continuous on [a, b] and derivable in (a, b) then show that

$$\begin{vmatrix} f(a) & f(b) \\ g(a) & g(b) \end{vmatrix} = (b - a) \begin{vmatrix} f(a) & f'(c) \\ g(a) & g'(c) \end{vmatrix} \text{ where } a < c < b$$

Sol Consider a function

$$F(x) = \begin{vmatrix} f(a) & f(x) \\ g(a) & g(x) \end{vmatrix}$$ which is continuous on [a, b] and differentiable in (a, b).]

Sol. Let $h(x) = \begin{vmatrix} f(a) & f(x) \\ g(a) & g(x) \end{vmatrix}$ ...(1)

By LMVT in (a, b) for h(x)

$$h'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\begin{vmatrix} f(a) & f'(c) \\ g(a) & g'(c) \end{vmatrix} = \frac{\begin{vmatrix} f(a) & f(b) \\ g(a) & g(b) \end{vmatrix} - 0}{(b-a)}$$

$$\Rightarrow \begin{vmatrix} f(a) & f(b) \\ g(a) & g(b) \end{vmatrix} = (b-a)\begin{vmatrix} f(a) & f'(c) \\ g(a) & g'(c) \end{vmatrix}$$

Hence proved.

13. Prove that there exists no such function f (x) such that f (1) = – 1; f (4) = 7 and f' (x) > 3 for all x ∈ R.

[Sol. using LMVT for f in [1, 4]
there must exist some c ∈ (1, 4)

s.t. $f'(c) = \frac{f(4)-f(1)}{4-1} = \frac{7+1}{3} = \frac{8}{3}$

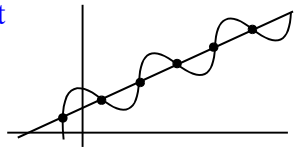
but it is given that f' (x) > 3 ∀ x ∈ R
hence contradiction
∴ no such function exists.

Alternatively: $\int_1^4 f'(x)\,dx > \int_1^4 3\,dx = 9$; f (4) – f (1) > 9 as 8 > 9 → nonsence ⇒ no such f exist]

14. A continuous and differentiable function y = f (x) is such that its graph cuts the line y = px + q at *m* distinct points. Minimum number of points at which f" (x) = 0, is/are

Ans (m – 2)

[Sol. Very obviously if line cuts the graph at *m* points then using LMVT ∃ atleast (m – 1) points where f' (x) = p (slope of line)
∴ using Rolles ∃ atleast (m – 2) points where f" (x) = 0] **[TN, AOD, Rolle's] to be put**



15. If the function f: [0, 16] → R is differentiable. If 0 < α < 1 and 1 < β < 2, then $\int_0^{16} f(t)\,dt$ is equal to

(A) $4\left(\alpha^3 f(\alpha^4) - \beta^3 f(\beta^4)\right)$ (B) $4\left(\alpha^3 f(\alpha^4) + \beta^3 f(\beta^4)\right)$ **[O]**

(C) $4\left(\alpha^4 f(\alpha^3) + \beta^4 f(\beta^3)\right)$ (D) $4\left(\alpha^2 f(\alpha^2) + \beta^2 f(\beta^2)\right)$

Ans B

[Sol. $I = \int_0^{16} f(t)\,dt$

consider $g(x) = \int_0^{x^4} f(t)\,dt$ ⇒ g (0) = 0

LMVT for g in [0, 1] gives, some α ∈ (0, 1) such that $\frac{g(1)-g(0)}{1-0} = g'(\alpha)$ .....(1)

|||ly LMVT in [1, 2] gives some β ∈ (1, 2) such that $\frac{g(2)-g(1)}{2-1} = g'(\beta)$ .....(2)

(1) + (2) $\quad g'(\alpha) + g'(\beta) = g(2) - \underbrace{g(0)}_{\text{zero}}$; but $g'(x) = f(x^4) \cdot 4x^3$

$$4\left(\alpha^3 f(\alpha^4) + \beta^3 f(\beta^4)\right) = \int_0^{16} f(t)\,dt \quad \Rightarrow \quad \textbf{(B)]}$$

16. If $y = f(x)$ is twice differentiable function such that $f(a) = f(b) = 0$, and $f(x) > 0 \ \forall\ x \in (a, b)$, then
(A) $f''(c) < 0$ for some $c \in (a, b)$ (B) $f''(c) > 0 \ \forall\ c \in (a, b)$ **[O]**
(C) $f(c) = 0$ for some $c \in (a, b)$ (D) none

Ans A

Sol Applying LMVT over $f(x)$ for $x \in \left[a, \frac{a+b}{2}\right]$

$$f'(c_1) = \frac{f\left(\frac{a+b}{2}\right) - f(a)}{\left(\frac{b-a}{2}\right)} = \frac{2}{(b-a)} \cdot f\left(\frac{a+b}{2}\right), \ c_1 \in \left(a, \frac{a+b}{2}\right)$$

LMVT in $\left[\frac{a+b}{2}, b\right]$; |||ly $\quad f'(c_2) = -\dfrac{2f\left(\frac{a+b}{2}\right)}{b - \frac{a+b}{2}}, \ c_2 \in \left(\frac{a+b}{2}, b\right)$

Applying LMVT over $y = f'(x)$ in $[c_1, c_2]$

$$f''(x) = \frac{f'(c_2) - f'(c_1)}{c_2 - c_1} = \frac{-\frac{4}{(b-a)} \cdot f\left(\frac{a+b}{2}\right)}{c_2 - c_1} < 0, \text{ where } x \in (c_1, c_2)\ ]$$

17. Suppose *f* and *g* be two differentiable function in [a, b] where $0 < a < b$. If $g(x) \neq 0$ for all x in [a, b]. Prove that there exists 'c' such that $a < c < b$ and $\dfrac{f(b) - f(a)}{g(b) - g(a)} = \dfrac{f'(c)}{g'(c)}$. **[O]**

Use this result to find whether 'c' is in A.M. / G.M. / H.M. / none between 'a' and 'b' for the pair of function given below in [a, b]

(i) $f(x) = \dfrac{1}{x^2}$ and $g(x) = \dfrac{1}{x}$; (ii) $f(x) = e^x$ and $g(x) = e^{-x}$; (iii) $f(x) = \sqrt{x}$ and $g(x) = \dfrac{1}{\sqrt{x}}$

Ans. (i) H.M.; (ii) A.M.; (iii) G.M.]

[Sol. Let $A\,g(x) - f(x)$ in [a, b]

where $A = \dfrac{f(b) - f(a)}{g(b) - g(a)}$

$\phi(a) = A\,g(a) - f(a)$; $\phi(b) = A\,g(b) - f(b)$

now $A\,g(a) - f(a) = A\,g(b) - f(b)$

$f(b) - f(a) = A\left(g(b) - g(a)\right)$

$\Rightarrow \quad \phi(a) = \phi(b)$

$A\,g'(c) - f'(c) = 0; \quad \dfrac{f'(c)}{g'(c)} = A = \dfrac{f(b) - f(a)}{g(b) - g(a)}$]

18. Using Lagrange's mean value theorem, show that $\frac{\beta-\alpha}{1+\beta^2} < \tan^{-1}\beta - \tan^{-1}\alpha < \frac{\beta-\alpha}{1+\alpha^2}$, $\beta > \alpha > 0$. **[O]**

Sol. Let $f(x) = \tan^{-1}x$ ...(1)

and f(x) is continuous in $[\alpha, \beta]$ and differentiable in (α, β) then by LMVT

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1}\beta - \tan^{-1}\alpha}{(\beta-\alpha)} \quad ...(2)$$

as $\alpha < c < \beta$

$\alpha^2 < c^2 < \beta^2$

$1 + \alpha^2 < 1 + c^2 < 1 + \beta^2$

$$\Rightarrow \frac{1}{1+\alpha^2} > \frac{1}{1+c^2} > \frac{1}{1+\beta^2} \quad ...(3)$$

from (2) & (3)

$$\frac{1}{1+\alpha^2} > \frac{\tan^{-1}\beta - \tan^{-1}\alpha}{\beta-\alpha} > \frac{1}{1+\beta^2}$$

$$\Rightarrow \frac{\beta-\alpha}{1+\beta^2} < \tan^{-1}\beta - \tan^{-1}\alpha < \frac{\beta-\alpha}{1+\alpha^2}$$

Hence, proved.

6. PLOTTING THE GRAPH

Q.1 $f(x) = x^{\frac{1}{x}}$.

(i) find domain

(ii) find value/limit of y at boundary point of domain.

(iii) find intervals of monotonocity.

(iv) find points where curve meets x - axis and y - axis.

(v) find value of function at critical points.

(vi) Plot graph

Q.2 $f(x) = x^x$ Q.3 $f(x) = e^{x^2-4x+3}$ Q.4 $f(x) = 4x^3 - 12x$.

Q.5 $f(x) = \frac{x}{\ln x}$ Q.6 $f(x) = x + \ln(1-4x)$ Q.7 $f(x) = \frac{x^2-x+1}{x^2+x+1}$

Q.8 $f(x) = x^2e^{-x}$ Q.9 $f(x) = \frac{1}{x^2-4}$ Q.10 $f(x) = \left(1+\frac{1}{x}\right)^x$

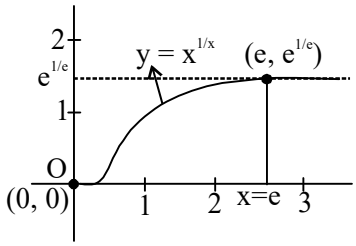
Q.11 $f(x) = \frac{(x+2)(x-1)}{x(x+1)}$ Q.12 $f(x) = \frac{|x-1|}{|x|-1}$

Ans.

1. (i) $(0, \infty)$ (ii) $x \to 0^+ x^{\frac{1}{x}} = 0$ (iii) $(e,\infty)\downarrow,(0,e)\uparrow$ (iv) Never meets

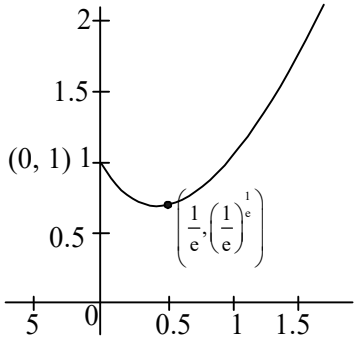
(v) at $x = e$, $y = e^{\frac{1}{e}}$

(vi) $y = x^{1/x}$



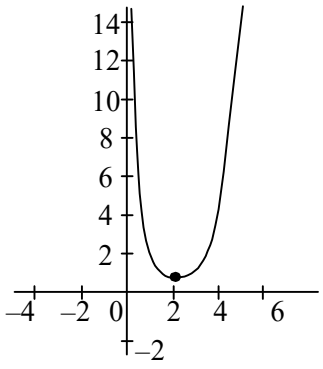
2. (i) $(0, \infty)$ (ii) $\lim\limits_{x\to 0^+} x^x = 1$ (iii) $(0,e)\downarrow,(e,\infty)\uparrow$ (iv) Never (v) $x = e, e^e$

(vi) $y = x^x$



3. (i) R (ii) $\lim\limits_{x\to\infty} f(x) = \infty$ (iii) $(2,\infty)\uparrow,(-\infty,2)\downarrow$ (iv) $x = 0, y = e^3$ (v) $x = 2,\ y = \dfrac{1}{e}$

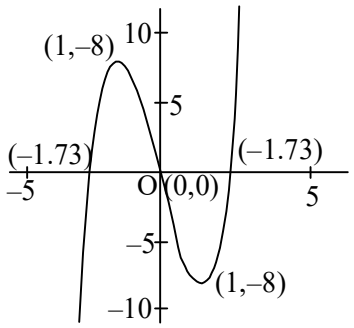
(vi) $y = e^{x^2-4x+3}$



4. (i) R (ii) $\lim\limits_{x\to\infty} f(x) = \infty$ (iii) $(-\infty,-1)\cup(1,\infty)\uparrow$, $(-1,1)\downarrow$

(iv) $x = 0, y = 0$, at $x = \pm\sqrt{3}$, $y = 0$ (v) at $x = 1, y = -8$, at $x = -1, y = 8$

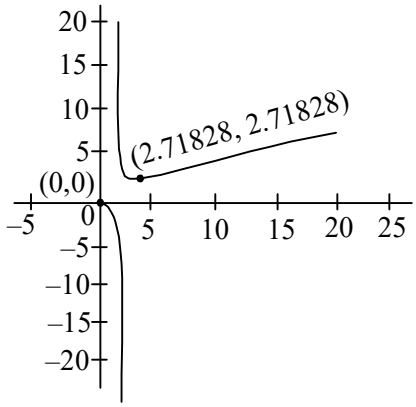
(vi) $y = 4x^3 - 12x$



5. (i) $(0,1) \cup (1,\infty)$ (ii) $\lim\limits_{x \to 1^+} = \infty$, $\lim\limits_{x \to 1^-} = -\infty$ (iii) $(e,\infty)\uparrow, (0,1) \cup (1,e)\downarrow$

(iv) never (v) at $x = e$, $y = e$

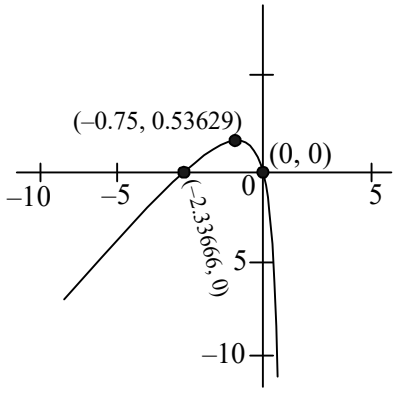
(vi) $y = \dfrac{x}{\ln x}$



6. (i) $x < \dfrac{1}{4}$ (ii) $\lim\limits_{x \to \frac{1}{4}^-} = -\infty$ (iii) $\left(-\infty, -\dfrac{3}{4}\right)\uparrow, \left(-\dfrac{3}{4}, \dfrac{1}{4}\right)\downarrow$

(iv) at $x = 0$, $y = 0$ (v) at $x = -\dfrac{3}{4}$, $y = \ln 4 - \dfrac{3}{4}$

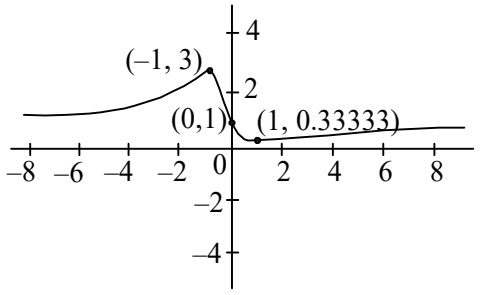
(vi) $y = x + \ln(1 - 4x)$



7. (i) R (ii) $\lim_{x\to\infty} = 1$, $\lim_{x\to-\infty} = 1$ (iii) $(-\infty,-1)\cup(1,\infty)\uparrow, (-1,1)\downarrow$

(iv) at $x = 0$, $y = 1$ (v) at $x = 1$, $y = \frac{1}{3}$, at $x = -1$, $y = 3$

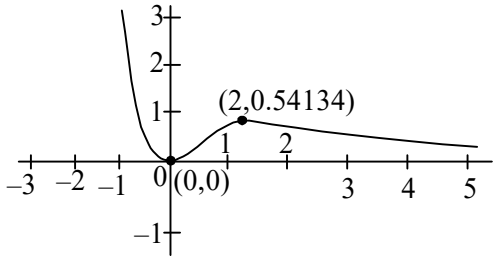
(vi) $y = \frac{x^2 - x + 1}{x^2 + x + 1}$



8. (i) R (ii) $\lim_{x\to\infty} = 0$, $\lim_{x\to-\infty} = \infty$ (iii) $(0,2)\uparrow, (-\infty,0)\cup(2,\infty)\downarrow$

(iv) at $x = 0$, $y = 0$ (v) at $x = 2$, $y = \frac{4}{e^2}$

(vi) $f(x) = x^2 e^{-x}$

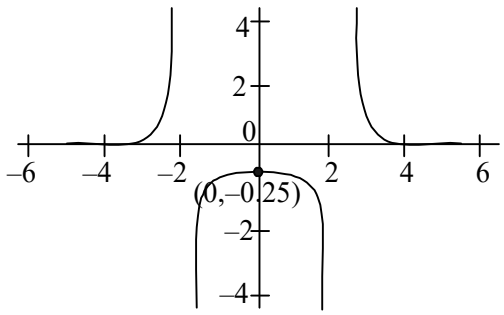


9. (i) $R - \{-2,2\}$ (ii) $\lim_{x\to2^+} = \infty$, $\lim_{x\to2^-} = -\infty$, $\lim_{x\to-2^+} = -\infty$, $\lim_{x\to-2^-} = \infty$

(iii) $(0,2)\cup(2,\infty)\downarrow, (-\infty,-2)\cup(-2,0)\uparrow$ (iv) at $x = 0$, $y = -\frac{1}{4}$

(v) at $x = 0$, $y = -\frac{1}{4}$

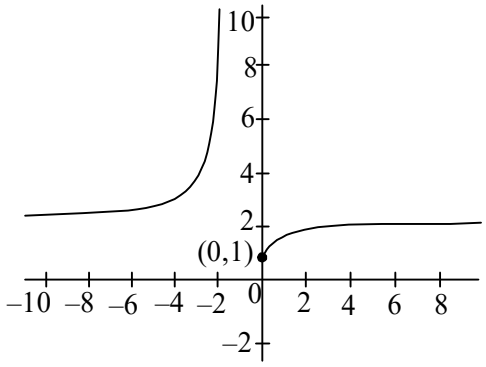
(vi) $y = \frac{1}{x^2 - 4}$



10. (i) $(-\infty,-1)\cup(0,\infty)$ (ii) $\lim\limits_{x\to\infty} = e$, at $x=-1$, $y=\infty$, at $x=0^+$, $y=1$

(iii) $\uparrow$ in domain

(vi) $f(x)=\left(1+\frac{1}{x}\right)^x$

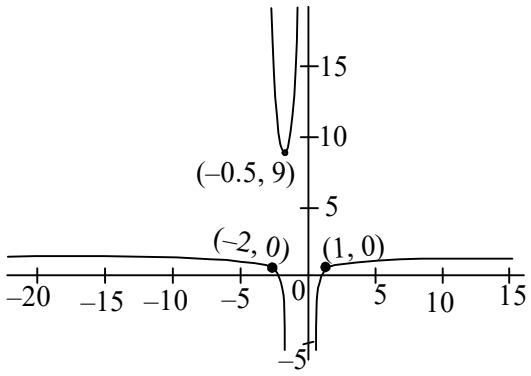


11. (i) $R-\{0,-1\}$ (ii) $\lim\limits_{x\to0^+}=-\infty$, $\lim\limits_{x\to0^-}=\infty$, $\lim\limits_{x\to-1^+}=-\infty$, $\lim\limits_{x\to-1^-}=\infty$

(iii) $\left(-\frac{1}{2},\infty\right)\uparrow,\left(-\infty,-\frac{1}{2}\right)\downarrow$ (iv) at $x=1$, $y=0$, at $x=-2$, $y=0$

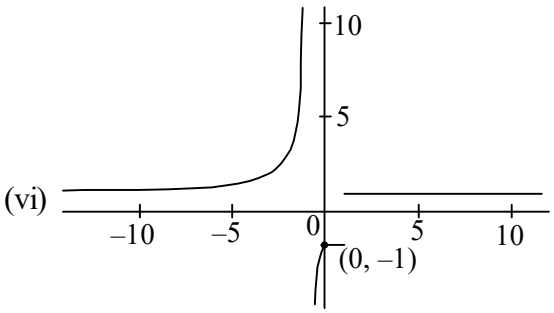
(v) at $x=-\frac{1}{2}$, $y=9$

(vi) $y=\frac{(x+2)(x-1)}{x(x+1)}$



12. (i) $R-\{-1,1\}$

(ii) $\lim\limits_{x\to1^+}=1, \lim\limits_{x\to1^-}=-1, \lim\limits_{x\to-1^+}=-\infty, \lim\limits_{x\to-1^-}=\infty,$



Q.13 Prove that equation $\frac{x^3+1}{x^2+1} = 5$, have no root in [0, 2]. **[O]**

Sol $x^3 + 1 = 5x^2 + 5 \quad \Rightarrow \quad x^3 - 5x - 4 = 0$

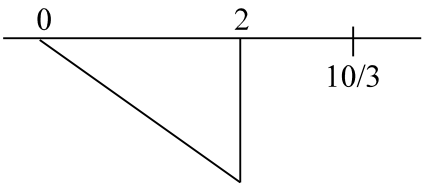
Let $f(x) = x^3 - 5x - 4$

$f'(x) = 3x^2 - 10x = x(3x - 10)$

hence f is decreasing in (0, 10/3)

$\therefore \quad f(2) < f(0) < 0$

hence no root in [0, 2] **Hence proved**]



Q.14 Find number of solutions of equation $\frac{1}{(1+x)^3} = 3x - \sin x$ **[O]**

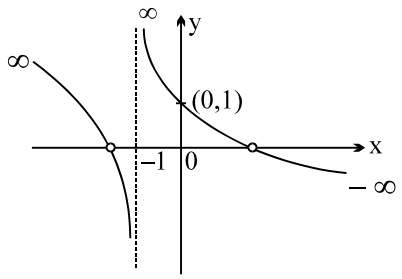
Ans 2

[Sol. $f'(x) = -\frac{3}{(x+1)^4} - 3 + \cos x < 0$

hence f(x) is always decreasing, Also as $x \to \infty$, $f(x) \to -\infty$ and as $x \to -\infty$, $f(x) \to +\infty$

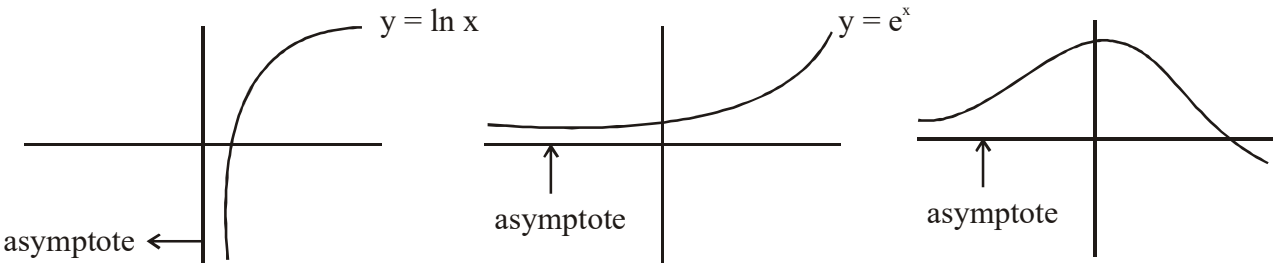
hence one positive and one negative root

Graph is as shown

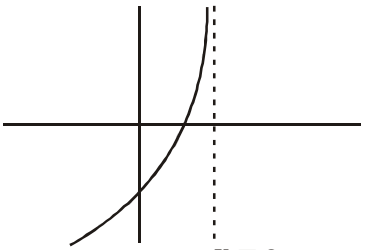


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7. ASYMPTOTES

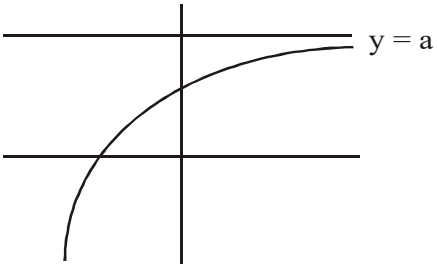


(i) Vertical asymptote :



$\lim_{x \to a} f(x) = \infty$

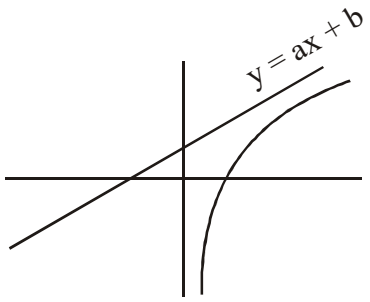
(ii) Horizontal asymptote :



$\lim_{x \to \infty} f(x) = a$

$\Rightarrow \lim_{x \to \infty} f'(x) = 0$

(iii) Inclined asymptote :



$\lim_{x \to \infty} (f(x) - (ax + b) = 0$

$\lim_{x \to \infty} f'(x) = a$

$\Rightarrow$ **Asymptotes for rational functions :**

(i) Vertical asymptote

Ex.: $f(x) = \dfrac{x^2 - x + 1}{(x+1)(x-2)}$

$x = -1, x = 2$ are asymptotes.

(ii) Horizontal asymptote

Degree of Numerator is not greater than degree of Denominator.

$$f(x) = \frac{a_0 x^n + a_1 x^{n-1} + ...}{b_0 x^m + b_1 x^{m-1} + ...}$$

$n = m$	$n < m$
$y = \dfrac{a_0}{b_0}$	$y = 0$

(iii) Inclined asymptote

Degree of N^r will be 1 more than degree of D^r.

$$f(x) = \frac{P(x)}{Q(x)}$$

$$f(x) = (ax + b) + \frac{r(x)}{Q(x)}$$

$\Rightarrow$ $y = ax + b$ will be asymptotes.

Que. $y = \dfrac{x^2 - 1}{x}$

$y = x - \dfrac{1}{x}$

$\Rightarrow y = x$ will be asymptote.

Que. $y = \dfrac{x^3 + x^2 + x - 3}{x^2 + 1}$

$$y = \frac{\left(x^2+1\right)\left(x+1\right) - 4}{x^2+1}$$

$$y = x + 1 - \frac{4}{x^2+1}$$

$\Rightarrow y = x + 1$ will be asymptote.

MAXIMA & MINIMA

1.1 MAXIMA AND MINIMA AT A POINT

1. Maxima :

A function f(x) is said to have a maximum at $x = a$ if

$f(a) > f(a+h)$ and $f(a) > f(a-h)$ where h is sufficiently small positive quantity

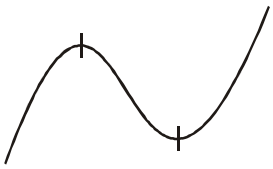
2. Minima :

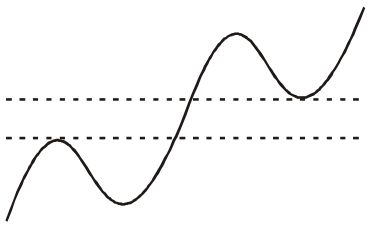
A function f(x) is said to have a minimum at $x = b$ if

$f(a) < f(a+h)$ and $f(a) < f(a-h)$ where h is sufficiently small positive quantity

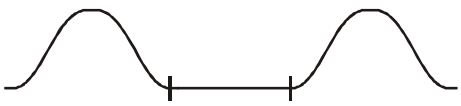
1.2 IMPORTANT POINTS

1. Maxima and Minima are points in domain.
2. To define maxima and minima function need not to be continuous.
3. Maxima, maximum, local maxima, local maximum, relative maxima, relative maximum are synonyms.
4. Minima, minimum, local minima, local minimum, relative minima, relative minimum are synonyms.
5. Greatest value, global maxima, global maximum, absolute maxima, absolute maximum are synonyms.
6. Least value, global minima, global minimum, absolute minima, absolute minimum are synonyms.
7. Value of function at maxima may not be the greatest value.
8. Value of function at minima may not be the least value.
9. Value at maxima can be less than value at minima.





10. Maxima and minima are together known as point of extrema/extremum/extremal points/ turning points.
11. For a continuous function maxima and minima occur alternately (provided function is not constant are interval).



12. Maxima and minima at boundary points : In [a, b]

(i)



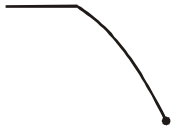
$x = a$ Maxima

$x = b$ Maxima

(ii) 

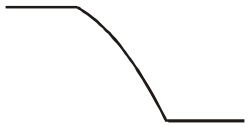
x = a Maxima

x = b minima

(iii) 

x = a Neither Maxima Nor Minima

x = b minima

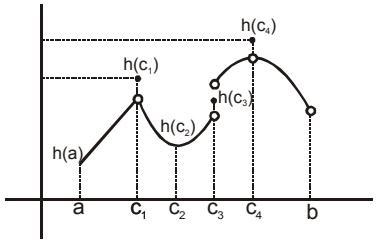
(iv) 

x = a

} Neither Maxima Nor Minima

x = b

Consider the graph of y = h(x), x ∈ [a, b)

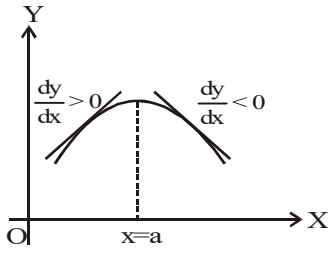


$x = c_1$, $x = c_4$ are points of local maxima, with maximum values $h(c_1)$, $h(c_4)$ respectively.
$x = a$, $x = c_2$ are points of local minima, with minimum values $h(a)$, $h(c_2)$ respectively.
$x = c_3$ is neither a point of maxima nor a minima.
Global maximum is $h(c_4)$

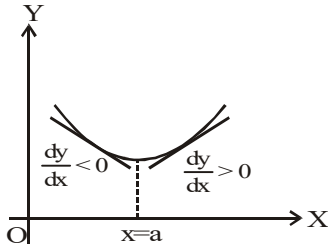
Global minimum is h(a)

2.1 FIRST DERIVATIVE APPROACH

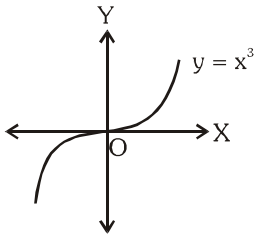
1. If in nbd of x = a, f'(x) changes sign from positive to negative then x = a is said to be a point **local maxima.**



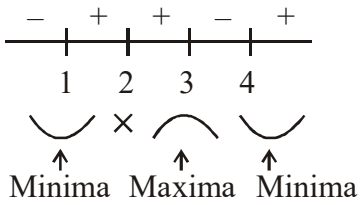
2. If in nbd of x = a, f'(x) changes sign from negative to positive then x = a is said to be a point **local minima.**



3. If in nbd of x = a, f′(x) does not change its sign then x = a will be neither maxima nor minima.



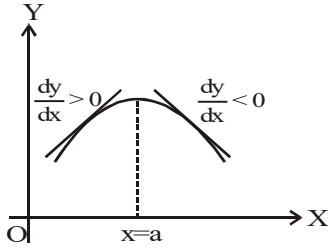
Ex. : $f'(x) = (x-1)(x-2)^2(x-3)(x-4)$



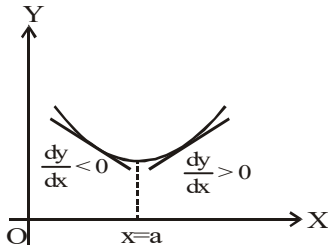
2.2 SECOND DERIVATIVE APPROACH

If f(x) is continuous and differentiable at x = a where f'(a) = 0 and f''(a) also exists then for ascertaining maxima/ minima at x = a, 2nd derivative test can be used -

(i) If f''(a) > 0 $\Rightarrow$ x = a is a point of local minima



(ii) If f''(a) < 0 $\Rightarrow$ x = a is a point of local maxima



(iii) If $f''(a) = 0 \Rightarrow$ second derivative test fails.

To identify maxima/minima at this point either first derivative test or higher derivative test can be used.

1. Find extremum points

(i) $f(x) = x^2 e^{-x}$ (ii) $y(x) = \dfrac{x}{\ln x}$

(iii) $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x$ in $[0, \pi]$ (iv) $y(x) = 4x^3 - 12x$

(v) $f(x) = |x|$ (vi) $f(x) = |x^2 - 1|$ (vii) $y(x) = x^4 - 4x^3 + 10$

Ans (i) M : x = 2, m : x = 0 (ii) m : x = e (iii) M : x = 0, $2\pi/3$ m : x = $\pi/2$, π

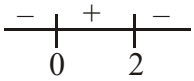
(iv) M : x = –1, m : x = 1 (v) m = x = 0 (vi) M : x = 0, m : x = 1, –1

(vii) m : x = 3

Sol. (i) $f(x) = x^2 e^{-x}$

$f'(x) = 2^x e^{-x} - x^2 e^{-x}$

$f'(x) = e^{-x} \cdot x(2 - x)$

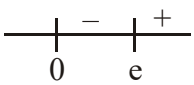


Maxima at x = 2

minima at x = 0

(ii) $f(x) = \dfrac{x}{\ell nx}$

$f'(x) = \dfrac{\ell nx - 1}{(\ell nx)^2}$, $(x > 0)$

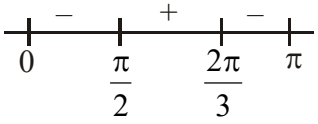


minima at x = e

(iii) $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x$

$f'(h) = -12\cos^3 x \sin x - 30\cos^2 x \sin x - 12\cos x \sin x$

$= -6 \sin x \cos x (2\cos x + 1)(\cos x + 2)$



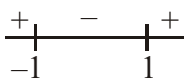
Maxima at $x = 0, \dfrac{2\pi}{3}$

Mionima at $x = \frac{\pi}{2}, \pi$

(iv) $f(x) = 4x^3 - 12x$

$f'(x) = 12x^2 - 12$

$= 12(x - 1)(x + 1)$



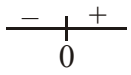
Maxima at $x = 1$

minima at $x = -1$

(v) $f(x) = |x|$

$$f(x) = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

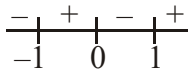


Minima at $x = 0$

(vi) $f(x) = |x^2 - 1|$

$$f(x) = \begin{cases} x^2 - 1 & x > 1 \\ 1 - x^2 & -1 \leq x \leq 1 \\ x^2 - 1 & x < -1 \end{cases}$$

$$f'(x) = \begin{cases} 2x & x > 1 \\ -2x & -1 < x < 1 \\ 2x - 0 & x < -1 \end{cases}$$



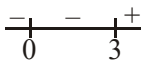
Maxima at $x = 0$

Minima at $x = 1, -1$

(vii) $f(x) = x^4 - 4x^3 + 10$

$f'(x) = 4x^3 - 12x^2$

$f'(x) = 4x^2(x - 3)$



Minima at $x = 3$

2. For what numbers a and b does the function $f(x) = axe^{bx^2}$ have the maximum value $f(2) = 1$

Ans $a = \frac{\sqrt{e}}{2}$, $b = -\frac{1}{8}$

Sol. $f(x) = axe^{bx^2}$; $f(2) = 1$ and $f'(2) = 0; 2ae^{4b} = 1$ ....(1)

also $f'(x) = a[\, xe^{bx^2} \cdot 2bx + e^{bx^2} \,] = axe^{bx^2}\,[2bx^2 + 1]$

$f'(2) = ae^{4b}(8b + 1) = 0;$ $a = 0$ or $b = -1/8$ but $a \neq 0;$ $a = e/2$

hence $a = \frac{\sqrt{e}}{2}$ and $b = -\frac{1}{8}$]

3. Let f(x) be a polynomial of degree 6 in x, in which the coefficient of x^6 is unity and it has extrema at $x = -1$ and $x = 1$. If $\lim\limits_{x\to 0}\frac{f(x)}{x^3} = 1$, then 5.f(2) is equal to _____. **[JEE(Main)Feb.2021]**

माना x में एक बहुपद f(x) की घात 6 है, तथा पद x^6 का गुणांक है और $x = -1$ तथा $x = 1$ इसके चरम बिन्दु हैं। यदि $\lim\limits_{x\to 0}\frac{f(x)}{x^3} = 1$ है, तो 5.f(2) बराबर है : **[JEE(Main)Feb.2021]**

Ans. 144

Sol. $f(x) = x^6 + bx^5 + cx^4 + x^3 + ex^2 + fx + g$

$\lim\limits_{x\to 0}\frac{f(x)}{x^3} \quad \Rightarrow \quad d = 1$

$e = f = g = 0$

$f(x) = x^6 + bx^5 + cx^4 + x^3$

$f'(1) = f'(-1) = 0$

$f'(x) = 6x^5 + 5bx^4 + 4cx^3 + 3x^2$

$f'(1) = 6 + 5b + 4c + 3 = 0 \quad ... (1)$

$f'(-1) = -6 + 5b - 4c + 3 = 0 \quad ... (2)$

By (1) and (2)

$$b = -\frac{3}{5}$$

$$c = -\frac{3}{2}$$

$$f(x) = x^6 - \frac{3}{5}x^5 - \frac{3}{2}x^4 + x^3$$

$$f(2) = \frac{144}{5}$$

$$5f(2) = 5 \times \frac{144}{5} = 144$$

2.3 n^{th} DERIVATIVE TEST

Let f(x) have derivatives up to nth order

If $f'(c) = f''(c) = \ldots\ldots\ldots = f^{n-1}(c) = 0$ and $f^n(c) \neq 0$ then we have following possibilities

(i) n is even, $f^{n)}(c) < 0 \Rightarrow x = c$ is point of maxima
(ii) n is even, $f^{n)}(c) > 0 \Rightarrow x = c$ is point of minima.
(iii) n is odd, $f^{n)}(c) < 0 \Rightarrow f(x)$ is decreasing about $x = c$
(iv) n is odd, $f^{n)} > 0 \Rightarrow f(x)$ is increasing about $x = c$.

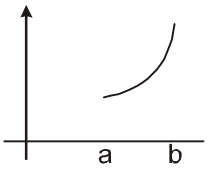
Q.1 Find points of local maxima or minima of $f(x) = x^5 - 5x^4 + 5x^3 - 1$

Sol. $f(x) = x^5 - 5x^4 + 5x^3 - 1$
$f'(x) = 5x^2 (x - 1) (x - 3)$
$f'(x) = 0 \quad \Rightarrow \quad x = 0, 1, 3$
$f''(x) = 10x (2x^2 - 6x + 3)$
Now, $f''(1) < 0 \quad \Rightarrow$ Maxima at $x = 1$
$f''(3) > 0 \quad \Rightarrow$ Minima at $x = 3$
and, $f''(0) = 0 \Rightarrow$ IInd derivative test fails
so, $f'''(x) = 30 (2x^2 - 4x + 1)$
$f'''(0) = 30$
$\Rightarrow$ Neither maxima nor minima at $x = 0$.

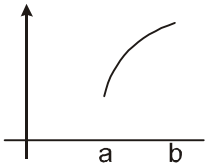
3.1 SIGNIFICANCE OF SIGN OF SECOND ORDER DERIVATIVE

A function f(x) is concave in (a, b) if tangent drawn at every point $(x_0, (f(x_0))$, for $x_0 \in (a, b)$ lie below the curve. f(x) is convex in (a,b) if tangent drawn at each point $(x_0, f(x_0))$, $x_0 \in (a, b)$ lie above the curve.

Results : 1. If $f''(x) > 0 \ \forall \ x \in (a, b)$, then the curve $y = f(x)$ is concave in (a, b)



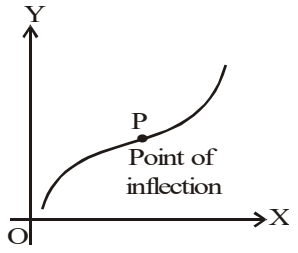
2. If $f''(x) < 0 \ \forall \ x \in (a, b)$ then the curve $y = f(x)$ is convex in (a, b)



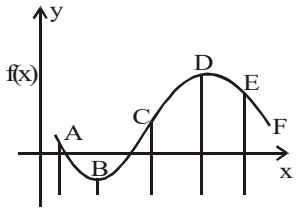
3.2 POINT OF INFLECTION

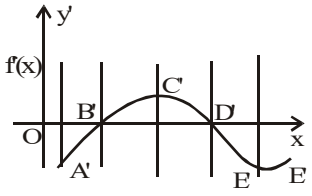
A point where the graph of a function has a tangent line and where the concavity changes is called a point of inflection.

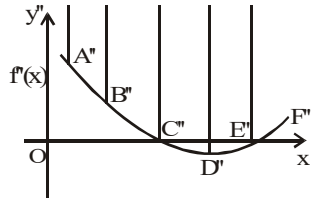
If function $y = f(x)$ is double differentiable then the point at which $\frac{d^2y}{dx^2} = 0$ & changes its sign is the point of inflection.



Note : If at any point $\frac{d^2y}{dx^2}$ does not exist but sign of $\frac{d^2y}{dx^2}$ changes about this point then it is also called point of inflection. e.g. for $y = x^{1/3}$, $x = 0$ is point of inflection. For a given function f(x) graphs of f'(x) and f''(x) can be drawn as shown in the adjacent figure. Here point C & E are point of inflection.





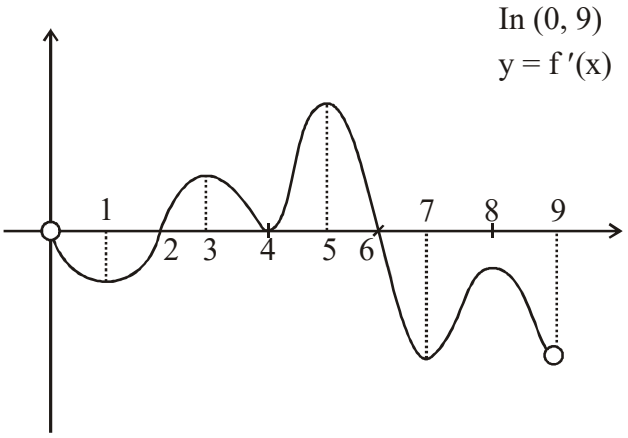


Note :

(i) At point of inflection, f ″(x) will be zero or not defined.

(ii) Point of inflection of y = f(x) will be extremum points for y = f ′(x).

Q.1



For y = f(x), find :

(i) Critical point (ii) Intervals of monotonicity (iii) Points of extremum

(iv) Nature of concavity (v) Point of inflection

Ans (i) x = 2, 4, 6 (ii) ↑ in (2, 6); ↓ in (0, 2) ∪ (6, 9) (iii) x = 2 Minima; x = 6 Maxima

(iv) concave up in (1, 3) ∪ (4, 5) ∪ (7, 8); concave down in (0, 1) ∪ (3, 4) ∪ (5, 7) ∪ (8, 9)

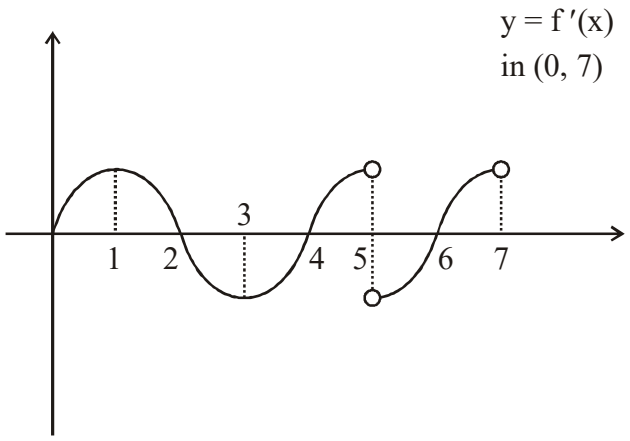
(v) x = 1, 3, 4, 5 7, 8.

Note

⇒ Point of inflection of f(x) are extremum points of f ′(x).

⇒ When f ′(x) is ↑, then graph of f(x) will be concave upwards and when f ′(x) is ↓ then graph of f(x) will be concave downwards.

Q.2



For y = f(x), find :

(i) Critical point (ii) Intervals of monotonicity (iii) Points of extremum

(iv) Nature of concavity (v) Point of inflection

Sol. (i) 2,4,6,5

(ii) ↑ in (0,2) ∪ (4,5) ∪ (6,7)

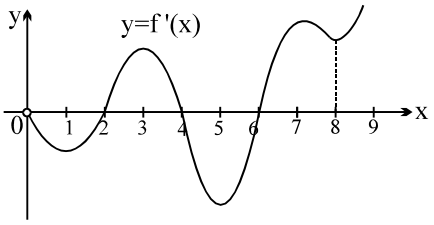
(iii) Minima = 4, 6

Maxima ⇒ x = 2, 5

(iv) Concave up in (0, 1) ∪ (3, 5) ∪ (5, 7)

(v) x = 1,3,5

3. The graph of the first derivative *f* ' of a function f is shown. **[O]**



(a) On what intervals is *f* increasing? Explain.

(b) At what values of x does *f* have a local maximum or minimum?

(c) On what intervals is *f* concave upwards or concave downwards?

(d) What are the x-coordinates of the inflection points of *f*?

[Ans. (a) increasing in (2, 4) ∪ (6, ∞) and decreasing in (0, 2) ∪ (4, 6)

(b) x = 2 (minima), 4 (Maxima) and 6 (minima)

(c) concave up in (1, 3) ∪ (5, 7) ∪ (8, ∞); concave down in (0, 1) ∪ (3, 5) ∪ (7, 8)

(d) x = 1, 3, 5, 7, 8]

4. Sketch a possible graph of a differentiable function *f* that satisfies the following conditions: **[O]**

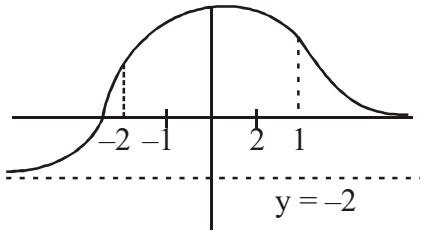
(i) f ' (x) > 0 on (– ∞, 1),

f ' (x) < 0 on (1, ∞)

(ii) $f''(x) > 0$ on $(-\infty, -2)$
and $(2, \infty)$, $f''(x) < 0$ on $(-2, 2)$

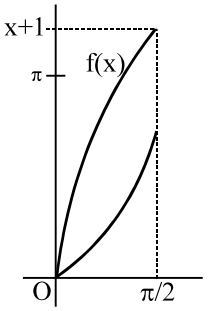
(iii) $\lim\limits_{x \to -\infty} f(x) = -2$, $\lim\limits_{x \to \infty} f(x) = 0$

[Sol. Condition (i) tells us that f is increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$.
Condition (ii) says that f is concave upward on $(-\infty, -2)$ and $(2, \infty)$, and concave downward on $(-2, 2)$.
From condition (iii) we know that the graph of f has two horizontal asymptotes: $y = -2$ and $y = 0$.
We first draw the horizontal asymptote $y = -2$ and a dashed line. We then draw the graph of f approaching this asymptote at the far left, increasing to its maximum point at $x = 1$ and decreasing toward the x-axis at the far right. We also make sure that the graph has inflection points when $x = -2$ and 2. Notice that we made the curve bend upward for $x < -2$ and $x > 2$, and bend downward when x is between -2 and 2.]



5. Prove the inequality : $\sin x + 2x \geq \dfrac{3x(x+1)}{\pi} \quad \forall \; x \in \left[0, \dfrac{\pi}{2}\right]$

[Sol. $f(x) = \sin x + 2x$; $g(x) = \dfrac{3(x^2 + x)}{\pi}$



$f'(x) = \cos x + 2 \; (\uparrow)$ $\qquad g'(x) = \dfrac{3}{\pi}(2x+1)$

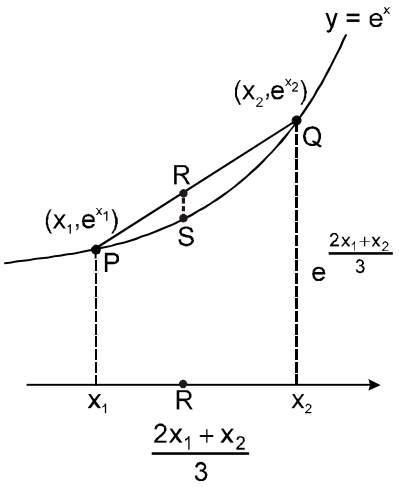
$f''(x) = -\sin x < 0$ $\qquad g''(x) = \dfrac{3}{\pi}(2) > 0$

$\Rightarrow$ concave down and increasing $\qquad \Rightarrow$ concave up and increasing

$f\left(\dfrac{\pi}{2}\right) = \pi + 1 \qquad g\left(\dfrac{\pi}{2}\right) = 3\dfrac{\pi}{2\pi}\left(\dfrac{\pi}{2}+1\right) = \dfrac{3}{2}\left(\dfrac{\pi}{2}+1\right) = \dfrac{3\pi}{4} + \dfrac{3}{2} = \pi + \dfrac{3}{4} - \dfrac{\pi}{4}$]

6. Prove that for any two numbers x_1 & x_2, $\dfrac{2e^{x_1} + e^{x_2}}{3} > e^{\frac{2x_1 + x_2}{3}}$

Sol. Assume $f(x) = e^x$ and let x_1 & x_2 be two points on the curve $y = e^x$.
Let R be another point which divides $\overline{PQ}$ in ratio 1 : 2.



y coordinate of point R is $\frac{2e^{x_1}+e^{x_2}}{3}$ and y coordinate of point S is $e^{\frac{2x_1+x_2}{3}}$. Since $f(x) = e^x$ is concave up, the point R will always be above the point S.

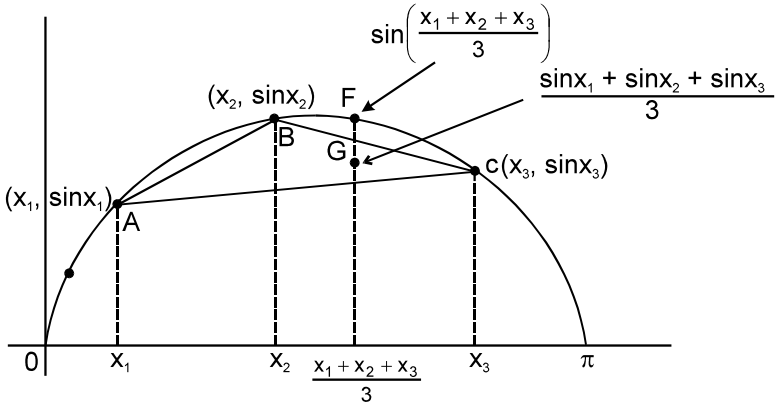
$\Rightarrow \quad \frac{2e^{x_1}+e^{x_2}}{3} > e^{\frac{2x_1+x_2}{3}}$

Alternate : Above inequality could also be easily proved using AM and GM.

7. If $0 < x_1 < x_2 < x_3 < \pi$ then prove that $\sin\left(\frac{x_1+x_2+x_3}{3}\right) > \frac{\sin x_1+\sin x_2+\sin x_3}{3}$. Henceprove that :

if A, B, C are angles of a triangle then maximum value of $\sin A + \sin B + \sin C$ is $\frac{3\sqrt{3}}{2}$. **[O]**

Sol.



Point A, B, C form a triangle.

y coordinate of centroid G is $\frac{\sin x_1+\sin x_2+\sin x_3}{3}$ and y coordinate of point F is

$\sin\left(\frac{x_1+x_2+x_3}{3}\right)$. Hence $\sin\left(\frac{x_1+x_2+x_3}{3}\right) \geq \frac{\sin x_1+\sin x_2+\sin x_3}{3}$.

If $A + B + C = \pi$, then

$\sin\left(\frac{A+B+C}{3}\right) \geq \frac{\sin A+\sin B+\sin C}{3}$

$\Rightarrow \quad \sin\frac{\pi}{3} \geq \frac{\sin A + \sin B + \sin C}{3}$

$\Rightarrow \quad \frac{3\sqrt{3}}{2} \geq \sin A + \sin B + \sin C$

$\Rightarrow$ maximum value of $(\sin A + \sin B + \sin C) = \frac{3\sqrt{3}}{2}$

4.1 USEFUL FORMULAE OF MENSURATION TO REMEMBER :

1. Volume of a cuboid = ℓbh.
2. Surface area of cuboid = $2(\ell b + bh + h\ell)$.
3. Volume of cube = a^3
4. Surface area of cube = $6a^2$
5. Volume of a cone = $\frac{1}{3}\pi r^2 h$.
6. Curved surface area of cone = $\pi r\ell$ (ℓ = slant height)
7. Curved surface area of a cylinder = $2\pi rh$.
8. Total surface area of a cylinder = $2\pi rh + 2\pi r^2$.
9. Volume of a sphere = $\frac{4}{3}\pi r^3$.
10. Surface area of a sphere = $4\pi r^2$.
11. Area of a circular sector = $\frac{1}{2}r^2\theta$, when θ is in radians.
12. Volume of a prism = (area of the base) × (height).
13. Lateral surface area of a prism = (perimeter of the base) × (height).
14. Total surface area of a prism = (lateral surface area) + 2 (area of the base)
(Note that lateral surfaces of a prism are all rectangle).
15. Volume of a pyramid = $\frac{1}{3}$ (area of the base) × (height).
16. Curved surface area of a pyramid = $\frac{1}{2}$ (perimeter of the base) × (slant height).
(Note that slant surfaces of a pyramid are triangles).

4.2 GEOMETRICAL PROBLEMS

1. When possible, draw a figure to illustrate the problem & label those parts that are important in the problem. Constants & variables should be clearly distinguished.

2. Write an equation for the quantity that is to be maximized or minimized. If this quantity is denoted by 'y', it must be expressed in terms of a single independent variable x. This may require some algebraic manipulations.

3. If y = f(x) is a quantity to be maximum or minimum, apply concept of maxima and minima to get required result

Questions

1. A rectangle is inscribed is a semicircle of radius 'r'. Find maximum area of rectangle.

Ans. r^2

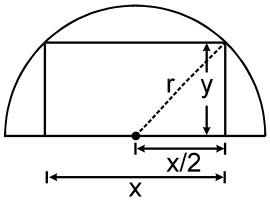
Sol Let sides of rectangle be x and y (as shown in figure).

$\Rightarrow \quad A = xy.$

Here x and y are not independent variables and are related by Pythogorus theorem with r.

$\frac{x^2}{4} + y^2 = r^2 \quad \Rightarrow \quad y = \sqrt{r^2 - \frac{x^2}{4}}$

$\Rightarrow \quad A(x) = x\sqrt{r^2 - \frac{x^2}{4}}$



$\Rightarrow \quad A(x) = \sqrt{x^2r^2 - \frac{x^4}{4}}$

Let $f(x) = r^2x^2 - \frac{x^4}{4}$; $x \in (0, r)$

A(x) is maximum when f(x) is maximum

Hence $f'(x) = x(2r^2 - x^2) = 0 \Rightarrow x = r\sqrt{2}$

also $f'(r\sqrt{2^+}) < 0$ and $f'(r\sqrt{2^-}) > 0$

confirming at f(x) is maximum when $x = r\sqrt{2}$ & $y = \frac{r}{\sqrt{2}}$.

2. Of all the open right circular cylinders with a given volume V, Find ratio of height to radius of the cylinder having least surface area.

Ans $h = r$

Sol. $\text{Area} = 2\pi rh + \pi r^2$

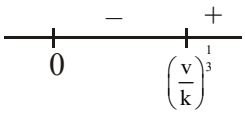
$\pi r^2 h = v$

$$h = \frac{v}{\pi r^2}$$

$$A = \text{Area} = 2\pi r\frac{v}{\pi r^2} + \pi r^2$$

$$\frac{dA}{dr} = -\frac{2v}{r^2} + 2\pi r$$

$$= \frac{2\left(\pi r^3 - v\right)}{r^2}$$



$$r = \left(\frac{v}{\pi}\right)^{\frac{1}{3}}$$

$$h = \left(\frac{v}{k}\right)^{\frac{1}{3}}$$

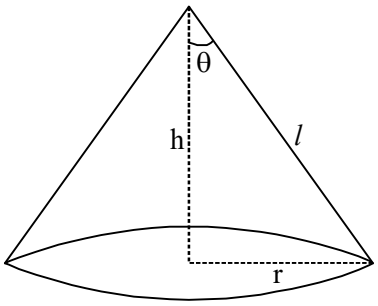
$$\frac{r}{h} = 1$$

$h = r$

3. For a given slant height prove that volume of conical tent is maximum if semi vertical angle $\theta = \tan^{-1}\sqrt{2}$

Ans.

Sol.



$$h^2 + r^2 = \ell^2$$

$$\frac{\ell^2}{3} + r^2 = \ell^2$$

$$r = \frac{\sqrt{2}}{\sqrt{3}}\ell$$

$$v = \frac{1}{3}\pi r^2 h$$

$$v = \frac{1}{3}\pi\left(\ell^2 - h^2\right)h$$

$$\frac{dv}{dh} = \frac{\pi}{3}\left(\ell^2 - h^2 + h\left(-2h\right)\right)$$

$$\frac{dv}{dh} = \frac{\pi}{3}\left(\ell - \sqrt{3}h\right)\left(\ell + \sqrt{3}h\right)$$

$$h = \frac{\ell}{\sqrt{3}}$$

$$\tan\theta = \frac{r}{h} = \frac{\sqrt{2}\ell}{\sqrt{3}} \times \frac{\sqrt{3}}{\ell}$$

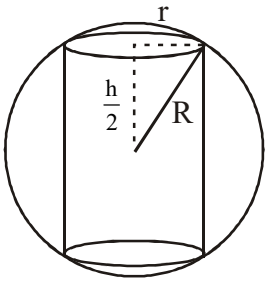
$$\tan\theta = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\sqrt{2}\right)$$

4. A cylinder is inscribed in a sphere of a radius R if volume of cylinder is maximum then find its height.

Ans. $\frac{2R}{\sqrt{3}}$

Sol.



$V = \pi r^2 h$ $\qquad r^2 + \frac{h^2}{4} = R^2$

$$V = \pi\left(R^2 - \frac{h^2}{4}\right)h$$

$$\frac{dV}{dh} = \pi\left(R^2 - \frac{h^2}{4} + h\left(-\frac{h}{2}\right)\right)$$

$$= \pi\left(R^2 - \frac{3h^2}{4}\right)$$

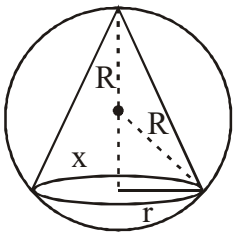
$$R^2 = \frac{3}{4}h^2$$

$$h = \frac{2}{\sqrt{3}}R$$

5. Find the altitude of the right cone of maximum volume that can be inscribed in a sphere of radius R.

Ans. $4R/3$

Sol.



$x^2 + r^2 = R^2$

$$V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi\left(R^2 - x^2\right)\left(R + x\right)$$

$$\frac{dV}{dx} = \frac{\pi}{3}\left[\left(R + x\right)\left(-2x\right) + \left(R^2 - x^2\right)\right] = \frac{\pi}{3}\left[-2Rx + R^2 - 3x^2\right] = 0$$

$R^2 = 3x^2 + 2Rx$

$3x^2 + 2Rx - R^2 = 0$

$$x = \frac{-2R \pm 4R}{6} = \frac{R}{3}$$

$$x = \frac{R}{3}$$

Height = Altitude = $R + x$

$$= R + \frac{R}{3} = \frac{4R}{3}$$

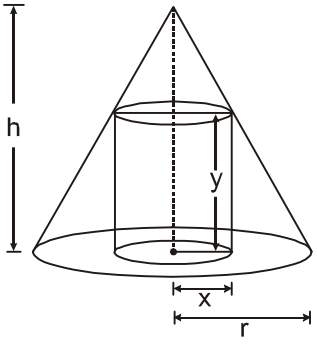
6. Find the altitude of the right circular cylinder of maximum volume that can be inscribed in a given right circular cone of height 'h'.

Ans. $h/3$

Sol Let x be the radius of cylinder and y be its height $v = \pi x^2 y$

x, y can be related by using similar triangles (as shown in figure).

$\frac{y}{r-x} = \frac{h}{r} \quad \Rightarrow \quad y = \frac{h}{r}(r-x)$



$\Rightarrow v(x) = \pi x^2 \frac{h}{r}(r-x) \qquad x \in (0, r)$

$\Rightarrow v(x) = \frac{\pi h}{r}(rx^2 - x^3)$

$v'(x) = \frac{\pi h}{r} x(2r - 3x)$

$v'\left(\frac{2r}{3}\right) = 0 \quad \text{and} \quad v''\left(\frac{2r}{3}\right) < 0$

Thus volume is maximum at $x = \left(\frac{2r}{3}\right)$ and $y = \frac{h}{3}$.

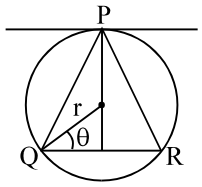
7. Find the coordinates of the point P on the curve $\frac{x^2}{8} + \frac{y^2}{18} = 1$ in the 1st quadrant so that the area of the triangle formed by the tangent at P and the coordinate axes is minimum.

Ans. (2, 3)]

Sol. $(x = 2\sqrt{2}\cos\theta, y = 3\sqrt{2}\sin\theta)$]

8. A point P is given on the circumference of a circle of radius *r*. Chords QR are parallel to the tangent at P. Determine the maximum possible area of the triangle PQR if r = 2.

Sol. Ar. of $\Delta PQR = \frac{2r\cos\theta(r + r\sin\theta)}{2}$ [Ans: $\left(\frac{3\sqrt{3}}{4}\right)\cdot r^2$]



$A(\theta) = r^2\cos\theta(1 + \sin\theta)$

$A'(\theta) = r^2[\cos\theta \cdot \cos\theta - (1 + \sin\theta)\sin\theta]$

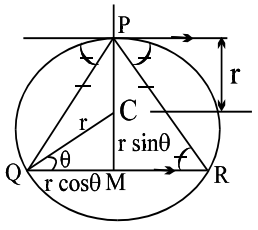
for maximum or minimum $A'(\theta) = 0$

$\cos^2\theta - \sin\theta(1 + \sin\theta) = 0 \quad = \quad 1 - \sin^2\theta - \sin\theta - \sin^2\theta = 0$

$= 2\sin^2\theta + \sin\theta - 1 = 0 \quad = \quad (\sin\theta + 1)(2\sin\theta - 1) = 0$

$\sin\theta = -1$ (not possible)

$\sin\theta = 1/2$ hence, $\theta = \pi/6$ or $5\pi/6$



$\frac{d^2A}{d\theta^2} = -2\cos\theta\sin\theta - \cos\theta - 2\sin\theta\cos\theta$

$\left.\frac{d^2A}{d\theta^2}\right]_{\theta=\pi/6} = -\text{ve} \quad \Rightarrow \quad A \text{ is maximum}$

$A_{max} = 2r^2 \frac{\sqrt{3}}{2}\left(1 + \frac{1}{2}\right)\frac{1}{2} \quad \Rightarrow \quad A_{max} = \frac{3\sqrt{3}}{4}r^2$]

9. For a train/steamer the cost of fuel varies as the square of its speed (in km/hr.) and the cost is Rs 24/hr when the speed is 12 km/hr. If other expenses amounts to Rs. 96 /hr, find the most economical speed and the cost of journey for 100 km.

Ans. 24 km/hr; Rs. 800/-

Sol. $F \propto v^2$ $\qquad$ $\text{total time} = \dfrac{\text{dist}}{\text{speed}}$

$F = KV^2$

$24 = K(144)$ $\qquad$ $\text{total time} = \dfrac{100}{v}$

$K = \dfrac{1}{6}$ $\qquad$ total cost = (per hour cost) (total time)

$F = \dfrac{v^2}{6}$ $\qquad$ $C = \left(\dfrac{v^2}{6} + 96\right)\dfrac{100}{v}$

$$\frac{dc}{dv} = \frac{-100}{v^2}\left(\frac{v^2}{6} + 96\right) + \frac{100}{v}\left(\frac{v}{3}\right)$$

$$= \frac{-100}{6} - \frac{100 \times 96}{v^2} + \frac{100}{3}$$

$$= 100\left(\frac{v^2 - 576}{6v^2}\right)$$

$$\frac{dc}{dv} = \frac{100}{6v^2}(v - 24)(v + 24)$$

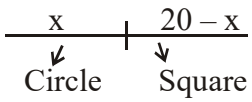
$V = 24$ km/hr

$$C = \left(\frac{576}{6} + 96\right)\frac{100}{24} = 800$$

10. A wire of length 20 cm is cut into two pieces. One piece converted into a circle and the other into a square. Where the wire is to be cut from so that the sum total of the areas of two plane figures is
(a) minimum **(b)** maximum.

Ans

Sol.



$2\pi r = x$ $\qquad$ $4a = 20 - x$

$r = \dfrac{x}{2x}$ $\qquad$ $a = \dfrac{20 - x}{4}$

$$A(x) = \frac{\pi x^2}{4\pi^2} + \left(\frac{20 - x}{4}\right)^2 = \frac{x^2}{4\pi} + \left(\frac{20 - x}{4}\right)^2$$

$$\frac{dA}{dx} = \frac{x}{2\pi} + 2\left(\frac{20 - x}{4}\right)\left(-\frac{1}{4}\right) = 0$$

$$\frac{x}{2\pi}=\frac{20-x}{8}$$

$8x = 40\pi - 2\pi x$

$$x=\frac{20\pi}{4+\pi}$$

$x \in [0, 20]$

So, maximum, min. Check at boundary points, where $f(x) = 0$

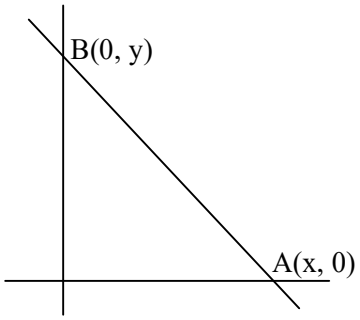
$A(0) = 25$ $\qquad A(20)=\frac{100}{\pi}$ $\qquad A\left(\frac{20x}{4+\pi}\right)=\frac{100\pi}{(\pi+4)^2}+\frac{400\pi^2}{(\pi+4)^2}$

Maxima $\qquad$ Minima

11. Find the equation of a line through (1, 8) cutting the positive semi axes at A and B if
    (i) the area of ΔOAB is minimum
    (ii) its intercept between the coordinate axes is minimum.
    (iii) sum of its intercept on the coordinates axes is minimum.

Ans. (i) $8x + y = 16$, (ii) $2x + y = 10$; min. intercept $5\sqrt{5}$, (iii) $2\sqrt{2}x + y = 8 + 2\sqrt{2}$]

Sol. (i)



$$y-y_1=\frac{-y_1}{x_1}(x-x_1)$$

$$8-y_1=\frac{-y_1}{x_1}(1-x_1)$$

$2x_1y_1 = 8x_1 + y_1$

$$\text{Area}=\frac{1}{2}x_1y_1$$

$$A=\frac{1}{2}y_1\left(\frac{y_1}{2y_1-8}\right)$$

$$\frac{dA}{dy}=0$$

$$\frac{1}{2}\left(\frac{(2y_1-8)(2y_1)-y_1^2(2)}{(2y_1-8)}\right)=0$$

$y_1(2y_1 - 8) = y_1^2$
$2y_1 - 8 = y_1$
$y_1 = 8 \qquad x_1 = 1$
equation of line $y + 8x = 16$

(ii) $I = \sqrt{x_1^2 + y_1^2}$

$$I^2 = x_1^2 + \left(\frac{8x_1}{2x_1 - 1}\right)^2$$

$$\frac{dI^2}{dx_1} = 0$$

$$2x_1 + 2\left(\frac{8x_1}{2x_1 - 1}\right)\left(\frac{8(2x_1 - 1) - 8x_1(2)}{(2x_1 - 1)^2}\right) = 0$$

$$x_1 = \frac{2 \times 8x_1 \times 8}{(2x_1 - 1)^3}$$

$(2x_1 - 1)^3 = 64$

$$x_1 = \frac{5}{2} \qquad y_1 = 5$$

$$y - 5 = -2\left(x - \frac{5}{2}\right)$$

$y + 2x = 10$

(iii) Sum $= x_1 + y_1$

$$S = x_1 + \frac{8x_1}{(2x_1 - 1)}$$

$$\frac{ds}{dx_1} = 0$$

$$1 + \frac{16x_1 - 8 - 16x_1}{(2x_1 - 1)^2} = 0$$

$(2x_1 - 1)^2 = 8$

$$x_1 = \frac{2\sqrt{2} + 1}{2} \quad \text{or} \qquad x_1 = \frac{1 - 2\sqrt{2}}{2} \text{ (Rejected)}$$

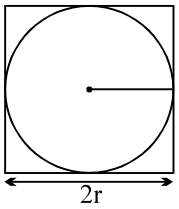
$y_1 = 4 + \sqrt{2}$

$$y-\left(4+\sqrt{2}\right)=\frac{-\left(4+\sqrt{2}\right)}{\frac{1+2\sqrt{2}}{2}}\left(x-\frac{1+2\sqrt{2}}{2}\right)$$

$$y+2\sqrt{2}x=8+2\sqrt{2}$$

12. You are designing a 1728 cm^3 closed right circular cylindrical cans whose manufacture will take waste into account. There is no waste in cutting the aluminium sheet for the curved surface, but the tops and bottom of radius "*r*" will be cut from squares that measure "2r" units on a side. Find the total quantity of aluminium (in square cm) for the manufacture of a most economical can. **[O]**

Ans. 864



[Sol. $V = \pi r^2 h$, $V = 1728$

$S = 2\pi rh + 8r^2$

$$S(r) = 8r^2 + 2\pi r \cdot \frac{V}{\pi r^2} = 8r^2 + \frac{2V}{r}$$

$$S'(r) = 16r - \frac{2V}{r^2} = 0$$

$$r^3 = \frac{2V}{16} = \frac{V}{8} = \frac{1728}{8} = 216$$

$$S(r)|_{min} = 8 \cdot 36 + \frac{2 \cdot 1728}{6}$$

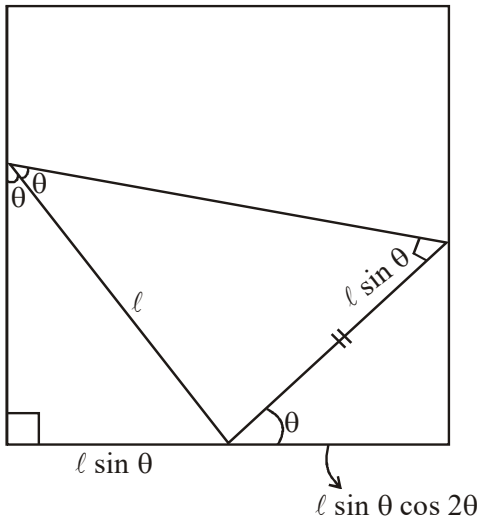
$= 288 + 576 = 864$ **Ans.**]

(Maxima) : $(h = 2d)$

13. One corner of a rectangular sheet of paper, width 1m, is folded over so on just reach the opposite edge of the sheet. Find the minimum length of crease. **[O]**

Ans. $\frac{3\sqrt{3}}{4}$

Sol.



$\ell \sin\theta + \ell \sin\theta \cos 2\theta = 1$

$$\ell = \frac{1}{\sin\theta(1+\cos 2\theta)}$$

$$\ell = \frac{1}{\sin\theta(2\cos^2\theta)}$$

$$\ell = \frac{1}{\sin 2\theta \cos\theta}$$

$y = \sin 2\theta \cos 3\theta$

$$\frac{dy}{d\theta} = \cos\theta(2\cos 2\theta) + \sin 2\theta(-\sin\theta) = 0$$

$\tan\theta = 2\cot 2\theta$

$\tan\theta \tan 2\theta = 2$

$$\tan\theta \frac{(2\tan\theta)}{1-\tan^2\theta} = 2$$

$2\tan^2\theta = 2 - 2\tan^2\theta$

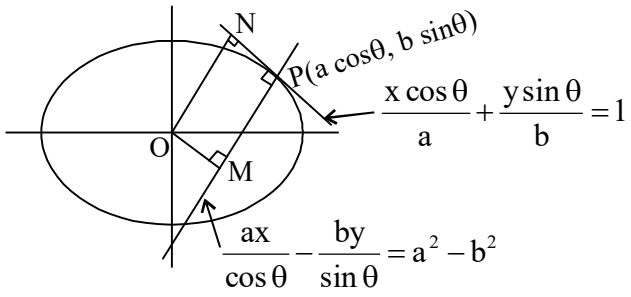
$$\tan^2\theta = \frac{1}{2} \qquad \tan\theta = \frac{1}{\sqrt{2}}$$

$$\sin\theta = \frac{1}{\sqrt{3}} \qquad \cos\theta = \sqrt{\frac{2}{3}}$$

14 Find the co-ordinates of all the points P on the ellipse $(x^2/a^2) + (y^2/b^2) = 1$ for which the area of the triangle PON is maximum, where O denotes the origin and N the foot of the perpendicular from O to the tangent at P.

Ans. $\pm \frac{a^2}{\sqrt{a^2+b^2}}, \pm \frac{b^2}{\sqrt{a^2+b^2}}$

Sol.



$$\text{Area} = \frac{1}{2}(ON)(PN)$$

$$A = \frac{1}{2}(ON)(OM)$$

$$ON = \frac{ab}{\sqrt{b^2\cos^2\theta + a^2\sin^2\theta}} \qquad OM = \frac{(a^2-b^2)\sin\theta\cos\theta}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

$$A = \frac{1}{2}\frac{\left(a^2-b^2\right)\sin\theta\cos\theta\times ab}{b^2\cos^2\theta + a^2\sin^2\theta} = \frac{ab\left(a^2-b^2\right)}{2}\left(\frac{1}{b^2\cot\theta + a^2\tan\theta}\right)$$

Let minimize $y = b^2\cot\theta + a^2\tan\theta$

$$\frac{dy}{d\theta} = b^2\left(-\operatorname{cosec}^2\theta\right) + a^2\left(\sec^2\theta\right) = 0$$

$$\frac{a^2}{\cos^2\theta} = \frac{b^2}{\sin^2\theta}$$

$$\tan^2\theta = \frac{b^2}{a^2} \Rightarrow \tan\theta = \pm\frac{b}{a}$$

$$\cos\theta = \frac{\pm a}{\sqrt{a^2+b^2}} \qquad \sin\theta = \frac{\pm b}{\sqrt{a^2+b^2}}$$

$$P\left(\pm\frac{a^2}{\sqrt{a^2+b^2}}, \pm\frac{b^2}{\sqrt{a^2+b^2}}\right)$$

15. The corridors of width a & b meet at right angles . Show that the length of the longest pipe that can be passed round the corner horizontally is, $(a^{2/3}+b^{2/3})^{3/2}$. **[O]**

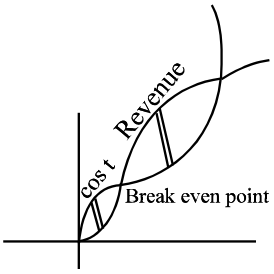
16 The cost function at American Gadget is $C(x) = x^3 - 6x^2 + 15x$ (x in thousands of units and $x > 0$)
The production level at which average cost is minimum is **[O]**

Ans 3

[Hint: $C(x)$ = cost of producing x items > 0

$$\text{average cost} = \frac{C(x)}{x} = x^2 - 6x + 15 = f(x)$$

$f'(x) = 2x - 3 = 0 \quad \Rightarrow x = 3$]



17. Manufacturer can sell x items at a price of Rs. $\left(5-\frac{x}{100}\right)$ each. The cost price of x items is Rs $\left(\frac{x}{5}+500\right)$. Find the number of items he should sell to earn maximum profit. **[O]**

Ans. $x = 240$

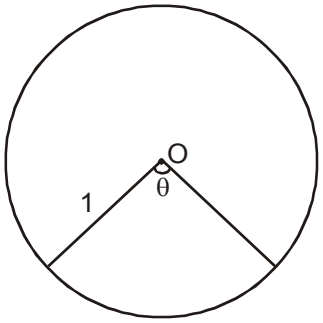
Sol.

18. A conical vessel is to be prepared out of a circular sheet of metal of unit radius. In order that the vessel has maximum volume, the sectorial area that must be removed from the sheet is A_1 and the area of the given sheet is A_2.

If $\frac{A_2}{A_1} = m + \sqrt{n}$, where $m, n \in N$, then $m + n$ is equal to. **[O]**

Ans. 9

Sol. $$\frac{A_2}{A_1} = \frac{\pi r^2}{\left(\frac{2\pi-\theta}{2\pi}\right).\pi r^2} = \frac{2\pi}{2\pi-\theta}$$



$$V = \frac{\pi}{3}\left(\frac{\theta}{2\pi}\right)^2 \sqrt{1^2 - \left(\frac{\theta}{2\pi}\right)^2}$$

$$\frac{dv}{d\theta} = 0 \Rightarrow \theta = \sqrt{\frac{8}{3}}\pi$$

$$\frac{A_2}{A_1} = \frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}} = 3 + \sqrt{6}$$

5. ANALYSIS OF CUBIC EQUATION

$f(x) = ax^3 + bx^2 + cx + d$

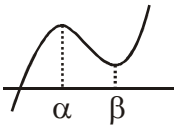
$f'(x) = 3ax^2 + 2bx + c$ — α, β

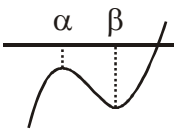
f is monotonic ($D \le 0$)

- 3 real and coincident roots
  $f(\alpha) = 0$
  $f'(\alpha) = 0$
  $f''(\alpha) = 0$
- One real and two imaginary roots otherwise

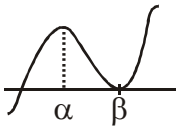
f is non monotonic ($D > 0$)

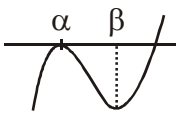
- One real and two imaginary roots
  $f(\alpha)\cdot f(\beta) > 0$



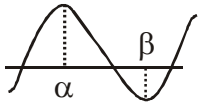


- Two coincident and one distinct roots
  $f(\alpha)\cdot f(\beta) = 0$





- Three real and distinct roots
  $f(\alpha)\cdot f(\beta) < 0$



1. If the cubic $y = x^3 + px + q$ has 3 distinct real roots then prove that $4p^3 + 27q^2 < 0$.

Sol. $y = x^3 + px + q$

$$\frac{dy}{dx} = 3x^2 + p = 0$$

$$x = -\sqrt{\frac{-p}{3}} \quad \text{or} \quad x = \sqrt{\frac{-p}{3}}$$

$$f\left(-\sqrt{\frac{-p}{3}}\right) + \left(\sqrt{\frac{-p}{3}}\right) < 0$$

Let $\sqrt{\frac{-p}{3}} = t$

$f(-t)\, f(t) < 0$

$(-t^3 - pt + q)(t^3 + pt + q) < 0$

$\{q + (pt + t^3)\}\ \{q - (pt + t^3)\} < 0$

$q^2 - t^2 (p + t^2)^2 < 0$

$$q^2 + \frac{p}{3}\left(\frac{2p}{3}\right)^2 < 0$$

$27q^2 + 4p^3 < 0$ H.P.

2. For a cubic $f(x) = \frac{ax^3}{3} + (a+2)x^2 + (a-1)x + 2$, $a > 0$, Find the value of 'a' for which it **[O]**

(a) + ve point of maximum (b) – ve point of minimum

(c) + ve point of minimum (d) – ve point of maximum

(e) – ve point of inflection (f) + ve point of inflection

Sol (a) both root of f'(x)=0 must be + ve and distinct($a > 0$)

(b) both root of f'(x)=0 must be – ve and distinct($a > 0$)

(c) at least one root of f'(x)=0 must be + ve and roots must be distinct

(d) at least one root of f'(x)=0 must be – ve and roots must be distinct

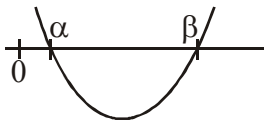
(e) abscissa corresponding to $\frac{d^2y}{dx^2} = 0$ must be – ve

(f) abscissa corresponding to $\frac{d^2y}{dx^2} = 0$ must be + ve

Sol. (a) +ve point of maximum

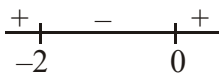
$f'(x) = ax2 + 2(a+2)x + (a-1) = 0 \begin{cases} \alpha \\ \beta \end{cases}$

Both roots are greater than zero of $f'(x)$



(i) $-\frac{b}{2a} > 0$ (ii) $D > 0$ (iii) $f'(0) > 0$

(i) $-\frac{(a+2)}{a} > 0 \Rightarrow \frac{a+2}{a} < 0$



$a \in (-2, 0)$ ...(i)

(ii) $D > 0$

$4(a+2)^2 - 4a(a-1) > 0$

$a^2 + 4a + 4 - a^2 + a > 0$

$5a + 4 > 0$

$$a > -\frac{4}{5} \Rightarrow a \in \left(-\frac{4}{5}, \infty\right)$$ ...(ii)

(iii) $f'(0) > 0$

$a - 1 > 0$

$a > 1$

$a \in (1, \infty)$

Final answer

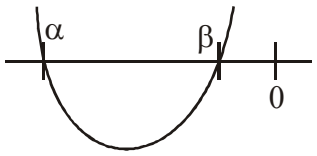
$(1) \cap (2) \cap (3)$

$a \in \phi$

(b) –ve point of minima

$f'(x) = ax^2 + 2(a + 2)x + (a - 1),\ a > 0$

Both roots are less than zero of $f'(x)$.

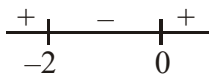


(i) $-\dfrac{b}{2a} < 0$

(ii) $D > 0$

(iii) $f'(0) = 0$

(i) $\dfrac{-(a+2)}{a} < 0 \Rightarrow \dfrac{a+2}{a} > 0$



$a \in (-\infty, -2) \cup (0, \infty)$ ...(1)

(ii) $D > 0$

$a \in \left(-\dfrac{4}{5}, \infty\right)$ ...(2)

(iii) $f'(0) > 0$

$a > 1$

$a \in (1, \infty)$ ...(3)

Final answer

$(1) \cap (2) \cap (3)$

$a \in (1, \infty)$

(c) $f'(x) = ax^2 + 2(a + 2)x + (a - 1)$

at least one root of $f'(x) = 0$ must be +ve and roots must be distinct.

Case I :

Both roots are +ve.

$a \in \phi$

Case II :

Exactly one root is +ve and other is –ve

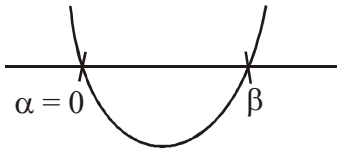
$f'(0) < 0$

$a < 1$

$a \in (-\infty, 1)$

Case III :

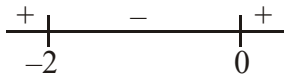
One root is zero and other is +ve.



(i) $f'(0) = 0$

$a = 1$

(ii) $-\frac{(a+2)}{a} > 0 \Rightarrow \frac{a+2}{a} < 0$



$a \in (-2, 0)$

(iii) $D > 0$

$\left(-\frac{4}{5}, \infty\right)$

$a \in \phi$

Final answer $\Rightarrow$ Case I $\cup$ Case II $\cup$ Case III

$a \in (-\infty, 1)$

(d) –ve point of maximum

$f'(x) = ax^2 + 2(a + 2)x + (a - 1)$, $a > 0$

at least one root of $f'(x) = 0$ must be –ve and roots must be distinct.

Case I : Both roots are –ve

$a \in (1, \infty)$

Case II : One root is +ve and other is –ve.

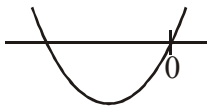
$a \in (-\infty, 1)$

Case III : One root is –ve and other is zero.

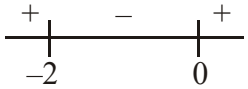
(i) $f'(0) = 0$

(ii) $\frac{-b}{2a} < 0$

(iii) $D > 0$



(i) $a = 1$ (ii) $-\frac{(a+2)}{a} < 0$ (iii) $\frac{a+2}{a} > 0$



(iii) $D > 0$ $a = 1$

$a \in \left(-\frac{4}{5}, \infty\right)$

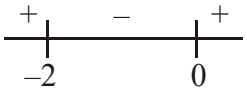
Final answer $a \in R$

(e) –ve point of inflection

abscissa corresponding to $\frac{d^2y}{dx^2} = 0$ must be –ve

$$-\frac{b}{2a} < 0$$

$$-\frac{(a+2)}{a} < 0$$



$$\frac{a+2}{a} > 0$$

$a \in (-\infty, -2) \cup (0, \infty)$

(f) +ve point of inflection

$$-\frac{6}{2a} > 0$$

$$-\frac{(a+2)}{a} > 0 \Rightarrow \frac{a+2}{a} < 0$$

$a \in (-2, 0)$